

Manual

Special Collection Functions for MES Batches, ERP Batches, Serial Numbers AIP-LCS 8.2

Version 1.1.23049

Last changed on: 01.09.2020

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1 Special Data Collection Functions for Batches and Serial Numbers

Purpose

The AIP features provided in this function package make it possible to enter batch-related data directly in production using shop floor terminals or data acquisition PCs.

Integration

Data entered via the AIP terminal can be displayed and/or evaluated in different MOC applications. Collected data can be uploaded via interfaces.

Features

Besides batch data, HYDRA shop floor clients (AIP) also collect and generate serial numbers to provide for complete documentation and traceability. The function package additionally provides further special features for palletizing, packaging, assembling/finishing and weighing of components.

Functions to collect and use serial numbers:

- Predefined series or serial numbers are transferred in HYDRA standard format via the HYDRA
- ERP interface
- Serial numbers are entered or generated via a user-friendly posting function at the BDE terminal
- Option to assign components also identified by serial numbers
- Classification into yield/ scrap quantities with reasons
- Automatic generation of goods receipts/ goods issues
- Optional verification of serial numbers already defined for the order
- Entry of order-related series and serial numbers in order to identify processed parts
- Validation check for already posted series or serial numbers
- Display of available and already used series or serial numbers at workstation PCs (MES Operation Center)
- Upload of posted series and serial numbers in HYDRA standard format via the HYDRA ERP interface

Functions for palletizing / packaging / assembling/finishing:

- Mapping of specific data collection processes for palletizing and packaging at packing stations

- Generation of handling units at the BDE terminal
- Function to edit handling units (e.g. assigning batches and removing batches from handling units)

Functions to weigh components:

- Display of the components to be weighed
- Management of multiple charges for each weighing order
- Presentation of target / actual quantities for each component and charge
- Weight recording via the interface connected to the scale
- Warning if tolerance quantities are exceeded or not reached
- Function to correct quantities and/or to change charge quantities for components
- Validation check for weighed batches

Additional licenses may be needed in order to use the functions listed above. Adding and coordinating the specific requirements and implementing them are considered a customized HYDRA service.

2 AIP2 Operation

2.1 Special Control and Display Elements on the AIP2

Tables

Uniform selection lists are used in AIP 8.2 posting dialogs:

Filter	
Status	Status
1	PRODUCTION
2	SETUP
3	ORDER SHORTAGE
4	OUT OF MATERIAL
5	OUT OF STAFF
6	START UP
22	COLOUR CHANGE
97	ELECTRICAL FAILURE
98	MECHANICAL FAILURE
99	GENERAL DISTURBANCE

Page 1 • Page 2

If information is available for more than one page, the page numbers are displayed below the table. The current page is highlighted in bold letters. If the user clicks/touches a page, the display directly changes to this page.

If more pages are available than the page numbers displayed, the following buttons can be displayed on the left or right hand side depending on the context (available as of SP10/2016):

- : If you click this button, the system jumps to the first page of the next page navigation. This means: If *Page 1 ... Page 9* were displayed for the page navigation, the system jumps to *Page 10*.
- : If you click this button, the system jumps to the first page of the next page navigation. This means: If *Page 10 ... Page 18* were displayed for the page navigation, the system jumps to *Page 9*.
- : If you click this button, the system directly jumps to *Page 1*.
- : If you click this button, the system directly jumps to the last page.

You can select an operation using the mouse, touch screen, keyboard (arrow keys: '↑' or '↓'), scanner or by entering it manually.

The content of tables or lists depends on the respective context. Example: When you log on an operation, those operations are available that are included in the sequencing list or planned for the respective workplace or group. When you interrupt an operation, only running operations are available for selection.



Scrolling page by page (up or down) in the table.



Scrolling to the left or right. Only those buttons are activated that make sense for the current situation (context sensitive). This figure shows that scrolling to the left has been deactivated.

Filter

Optionally you can display a “table filter” (customization). This is an automatic filter that, once it has been entered, directly affects the table without having to update it. This process is realized through full-text search for (defined) columns. The search is *case-insensitive*.

Virtual keyboard

Using the virtual keyboard, you can enter data manually via touch screen or a connected mouse. The virtual keyboard is displayed automatically as soon as the focus is on an input field. The keyboard layout, which is installed and activated in the Windows language settings, specifies the layout of the virtual keyboard.



Moving the virtual keyboard



Hiding the keyboard for 10 seconds



Switching between the alphanumeric and numeric keyboard



Selecting the keyboard layout (language)



Changing the scaling/size of the keyboard



To move the keyboard, you must configure the driver accordingly (configuration in the control panel of the terminal/PC)!

If you do not want to display the virtual keyboard in general, you must enter the parameter `-t` in the entry `parameters=` of the configuration file `ctaip.ini`.

Date display

AIP supports a country-specific date format in dynamic dialogs. The option "short date" has to be selected in the "regional settings" of the Windows "control panel" of the terminal/PC. Please note:

- Years are always four characters long.
- Months and days are always 2 characters long.
- Allowed separators are: '-' (minus), '/' (backslash) and '.' (dot).
- Blanks must not be included in the "short date" format, i.e. the <BLANK> separator is not allowed.
- The date separator "." (dot) is only allowed in connection with the DD.MM.YYYY format.
- The date format, which might possibly be configured in dynamic dialogs, is ignored.

Examples

- | | |
|-----------------------|------------|
| • English(USA) | MM/DD/YYYY |
| • Danish | MM-DD-YYYY |
| • Customer-specific 1 | YYYY-MM-DD |
| • Customer-specific 2 | YYYY/MM/DD |

- Customer-specific 3 MM/YYYY/DD

Note

If the date format used is other than the permitted formats, a note appears when the program is started and the date format is set to MM/DD/YYYY.

In the status bar, the year format is shortened and displayed only with two characters.

2.2 General description of the posting process on the AIP2

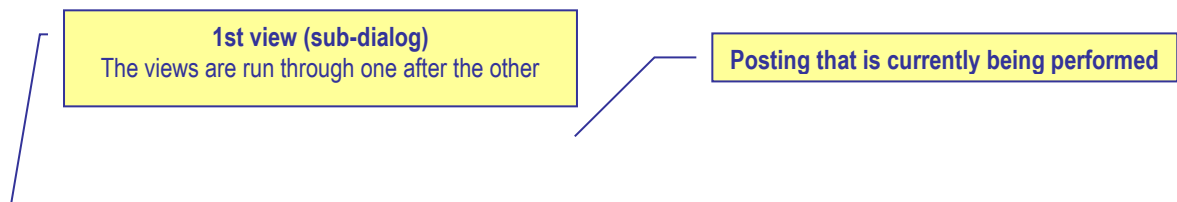
Many AIP posting dialogs are divided into several views (sub-dialogs). These views (sub-dialogs) cover the entire screen so that only one dialog is visible at a time. In a “workflow concept” the user is navigated through the posting dialog step by step. In the following, this process is described using the example *Interrupt operation*. Other posting dialogs are operated in the same way.

The action *Interrupt operation* is performed. To start this action, you click the button *Interrupt* when you have selected an operation:



Interrupt

The dialog *Interrupt operation* opens and the first view (sub-dialog) is displayed. The header displays the function that is currently being executed (here: *Interrupt operation*).



In the first dialog *Enter quantities*, the user can enter the produced yield or scrap quantities. Subject to the active input field, the virtual keyboard is shown or hidden automatically.

Quantities can be entered using the virtual or real keyboard. The user can go to the next field using the tabulator key (which can also be found on the virtual keyboard). When the user has entered all values in the first view, the next view (sub-dialog) can be opened by clicking **Next**.

The **Cancel** button is displayed in all sub-dialogs. Click this button to cancel/close the entire process at any time.

To open the next view (**Select status** in the example), click the **Next** button or another tab (in our example: **Select status** or **Confirmation**). Please note in this context, that no view can be skipped when they are navigated upwards (view 1 → view 2 → view 3). This means: When you are in the first view (*enter quantities*) and you click the third view (*confirmation*), the second view (*select status*) will be displayed first.

Vice versa, when navigating downwards (e.g. from the *confirm* view to the *enter quantities* view), each view can directly be opened by clicking the required tab. In this case, views can actually be skipped. Using the **Back** button, views are opened one after the other (upwards).

As long as the dialog has not been confirmed, the data entered can be changed at any time by scrolling back and forth.

Interrupt operation S61003050100 InjectionMolding at workplace 60610

1 Enter quantities 2 **Choose status** 3 Confirmation

Workplace 60610 ENG21

Actual status OUT OF MATERIAL since 02.10.2015 15:40:10

Filter

Status	Status
1	PRODUCTION
2	SETUP
3	ORDER SHORTAGE
4	OUT OF MATERIAL
5	OUT OF STAFF
6	START UP
15	MAINTENANCE
20	REPAIR
22	COLOUR CHANGE
24	TECHNICAL FAILURE
97	ELECTRICAL FAILURE
98	MECHANICAL FAILURE

Page 1 • Page 2

New status 4 OUT OF MATERIAL Estimated duration 0 Minutes

Comment

Cancel Back Go on

04.12.15 15:15:00

In the second view *Select status*, you select the workplace status that is set, when the operation has been interrupted. You can select the status from the status list displayed. This list can be restricted using the *Filter* field. Once the required values have been entered, the next view/sub-dialog can be opened by clicking **Next** (in our example it is the last view).

Interrupt operation S61003050100 InjectionMolding at workplace 60610

1 Enter quantities 2 Choose status 3 Confirmation

Operation 0100 InjectionMolding

Prod.Order	S6100305	Target qty.	90000 PCS
Article	SAR-FM0011-K-5013	Yield	1442 PCS
	Mirror housing right, blu	Scrap	33 PCS
		Completion	1%

Workplace 60610 ENG21

New status OUT OF MATERIAL

Entered quantities

Yield	5
Scrap	0

Staff Number

Buttons: Cancel, Back, Interrupt OP

Footer: mpcdv, UK flag, 04.12.15 15:15:45

The sub-dialog *Confirmation* shows a summary of all values entered in the dialog. If the user agrees with the entered data, the **Interrupt operation** dialog can be confirmed, once the badge number has been entered. Then the dialog is sent to the server and posted.

If an input field of the dialog is not completed properly (e.g. a mandatory field is empty), the field is highlighted in red in the respective view and gets the focus. The user can then directly correct the value.



If a workflow dialog is opened, you can click the ESC key to directly exit the dialog. This exit is also possible, if the virtual keyboard is displayed. As a consequence, you cannot use the ESC key to close the virtual keyboard.

3 Main View with Tiles

With the AIP2, the user can switch between the tile design optimized for touchscreens and the list format. By default, the tile layout is shown, which is described in the sections that follow.



To ensure proper processing and posting, terminals with "MDE" operation mode must not be switched off during times without shift.

3.1 Main view – header and footer

Header



The AIP logo is displayed on the top left of the screen, which may be replaced with a customer logo after configuration.

Possible messages are displayed to the right of it (e.g. if a dialog is opened for more than five minutes).

A separate window opens to display error messages that occur during data collection (e.g. validity checks).

Main views

You can assign a maximum of 16 workplaces or machines to the AIP2 terminal. The different workplaces are listed in the order that they were assigned to the terminal on the client. .

In the main view of the AIP2, you can use the button "< Overview" to switch to the icon view of workplaces. In the terminal configuration of the client you can specify whether you want to use the icon view. The sections that follow describe the main view and the icon view.

Footer



CTAIP V8.2.0.0 Server:win2008-7:7 TNR:888

09.09.15 14:47:56

The MPDV logo can be found at the bottom left of the AIP2 terminal. Double clicking the logo opens the info dialog where you can start further administration functions. This dialog closes automatically after approx. 5 seconds.

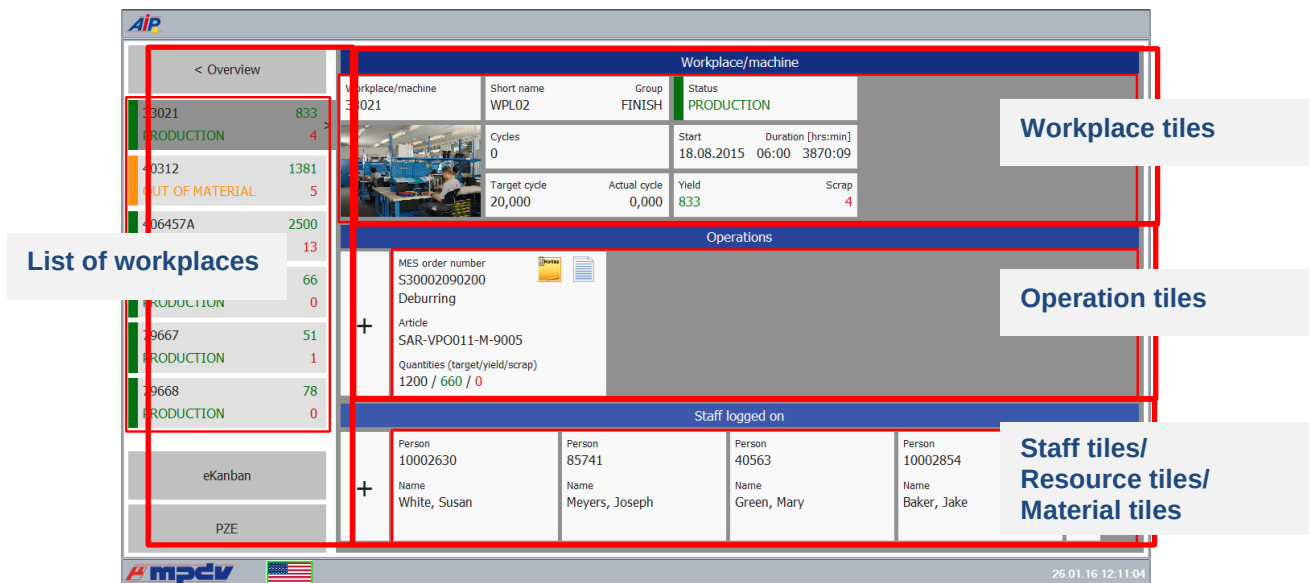
In the middle, further information is displayed: the current terminal status, AIP2 version number, date of the build, IP address of the server and the terminal number.

The current date and time are displayed to the right.

AIP2 statuses

	Network connection has been established. The terminal is ONLINE. Server communication is enabled. All saved data records have been transferred.
	The terminal is sending data to the server.
	No network connection or no connection to the server. The terminal is OFFLINE. Server communication is interrupted. Online functions, such as the display of information, are disabled. But you can still record certain postings. These postings are transferred to the server, once data connection has been re-established.
	Data is being received. The terminal reads files from the server or writes data to the server.
	The terminal is sending saved data records to the server.
	DEMO mode The terminal is in DEMO mode, i.e. server communication is disabled.

3.2 Main view with "tiles"



Please note: The actual display can be different to the above illustration.

Subject to the configurations made, the main view with tiles consists of two or three rows of tiles. While the first two rows of tiles (workplace and operation tiles) are always displayed, it is up to the user whether or not the third row of tiles is shown (optional display). In the configuration of workplaces you can configure for each workplace separately if you want to show the third row of tiles.

Additionally, there is the list of workplaces to the left. Here, you can select the workplaces for which details are displayed on the right-hand side.

List of workplaces

The workplace list shows all workplaces/machines assigned to the terminal. If many workplaces are assigned, swipe to get to the workplaces displayed further down.

This information is shown for the workplaces:

Machine/workplace number

Shows the machine and/or workplace number.

Status

The status is displayed for each machine in color on the left-hand side and also the status text is colored. Coloring is as follows:

- green: production
- yellow: assigned status
- red: not assigned

If the production lock is enabled, an exclamation mark is displayed in the same color as the status.

Quantities

On the right-hand side, the first figure shows the produced yield in green and the red figure shows the produced scrap.



If you have enabled the *Compensate manual quantities* option (e.g. set off scrap against yield) and the machine list also shows shift-related quantities, they will not be updated immediately. The application only updates the quantities, once the lists have been reloaded.

Unit for yield and scrap

If no operation is logged on, the primary quantity unit from the workplace/machine configuration is displayed as unit for yield and scrap. If an operation is logged on, the primary quantity unit of the operation is displayed.

Workplace tiles

The workplace tiles provide the following details:

Workplace/machine

No workplace or machine number is displayed.

Short name / group

The short name of the workplace or machine and the machine group are displayed.

Status

The status is displayed in color and as status text on the left-hand side. Coloring is as follows:

- green: production
- yellow: assigned status
- red: not assigned

If the production lock is enabled, an exclamation mark is displayed in the same color as the status.

Machine image

Shows the picture of the machine stored in the configuration of workplaces.

Clocks

Shows the recorded machine cycles of the current shift.

Start / Duration [hrs:min]

Point in time since the status has been available and the resulting duration at the current point in time.

With BDE workplaces¹, this point in time refers to the last manual status change. With MDE workplaces, it refers to the time when the last status change was identified (for machine connections). It also refers to the point in time when the status was last changed manually or to the time of the last shift change.

Target/actual cycle

Current target and actual cycle of the workplace.

The largest target cycle of all operations logged on to the workplace is shown. The largest target cycle is transferred to the MDE for monitoring.

If the target cycle is smaller than the minimum cycle time, the target cycle is still shown.

If an operation is logged off or interrupted, the largest target cycle of the remaining operations is identified and displayed. After logoff or interruption of the last operation at the workplace, the last target cycle set is still displayed.

If no operation is logged on, the target cycle specified in the machine list is displayed. Thus, even after a restart, the terminal can get the target cycle that last applied.

Yield / Scrap

Yield and scrap quantities of the current shift produced at the machine/workplace.

KPIs: OEE, utilization efficiency, scrap ratio



This function is only available if you enable the extension [aipkpi](#).

Shows the KPIs OEE, utilization efficiency and scrap ratio. The application calculates the KPIs at cyclic intervals (scheduler job "MDE keyfigure calculation"). The KPIs always refer to the current shift. The application calculates the KPIs based on the formulas that are also used for the OEE report and/or the efficiency report on the MOC. The application shows the KPIs with two decimal places. To the right of the KPI, the AIP2 GUI highlights in color if limit values are exceeded or not reached. If you have not defined limit values, the application shows the KPI in gray, otherwise in the color you defined for exceeding/not reaching limit values. For further information on the configuration, refer to the document [MDE KPI Configuration.pdf](#).

¹ An MDE workplace is a workplace that is assigned to a terminal, which runs in the "MDE" operation mode. Otherwise, it is a BDE workplace.



The AIP calculates and updates data at cyclic intervals. This may result in deviations between the collected values and the displayed KPIs.

Linked functions

If you click the workplace status, the dialog for changing the status opens. If you click one of the other tiles, the dialog opens where you can start the functions available for the selected workplace.



The buttons displayed depend on the selected workplace. The button *Lock production status*, for example, is only available for MDE machines.

List of operations logged on

The middle area on the right-hand side shows the logged on operations as tiles. The following data is shown:

MES order number

Order number and operation number of the operation logged on. The combination of these two numbers is the *MES order number*.

Article

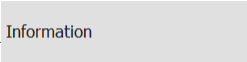
Article defined for the operation.

Quantities (target / yield / scrap)

Shows the target quantity defined for the operation, the produced yield and the scrap. The yield and scrap quantities integrate the counter readings of the available machine connections.

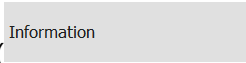


This icon is displayed if at least one note has been recorded for this operation in the graphic planning board of the client that must be shown on the terminal. To display the note(s), click the icon or click the

operation and select the *Information* button ().



This icon is displayed if at least one long text is stored for this operation. Click this icon to display the

long texts or click the operation and select the button *Information* ().

If you click an operation, a screen opens that shows the workplace/operation data. Via this screen, you can also select the operation-related functions.

Operation tiles

In addition to the fields already described, the operation tiles also show the following data:

Comments

This tile shows the user fields 53 and 54 (alphanumeric, 20 characters) of the operation. To edit these fields, you must store a respective user field key for the operation, which includes these two fields.

Completion in %

The bar shows the proportion of "yield", which has been produced until now, compared to the "target quantity".

Since logon (target / yield / deviation)

The production quantity to be expected since the OP has been logged on (depending on the cycle time, partitioning and the time when no production lock has been set for the machine). If the terminal program has been restarted after the OP logon, no value can be calculated.

Calculation:

Target Since Logon = Net Running Time[sec] * Partitioning/Target Cycle[sec/stroke]

Net Running Time: Time since logon while the production lock has not been set. This calculation does not integrate the breaks specified in the shift model or the status times posted to RPA 12 (resource performance account).

Deviation (in percent) between the calculated target quantity since logon and the quantity which has actually been produced "since logon".

Calculation: Deviation[%] = 100% * (Yield Since Logon - Target Quantity Since Logon) / Target Quantity Since Logon.

As of CTAIP 8.2.1.32:

Up to now, the target quantity since OP logon was only calculated if the workplace/machine was assigned to a terminal with operation mode "MDE processing". As of CTAIP 8.2.1.32, the target quantity since OP logon is also calculated if the terminal is configured with operation mode "BDE processing".

If the terminal has been restarted after an operation logon, the target quantity cannot be calculated correctly. To improve transparency, an "*" (asterisk) is shown behind the target quantity (since OP logon) in this case. The asterisk indicates that the target quantity now displayed no longer refers to the time of the operation logon, but to the time of the terminal restart.

While the workplace/machine status 999 is displayed, the target quantity since OP logon is "---". If the status 999 is again changed within a "free shift", the target quantity since OP logon is calculated using the point in time of the OP logon, of the shift start or of the terminal start.

To disable the calculation of the target quantity since OP logon on the terminal, you can use the configuration CalcTargetYieldSinceLogon=0 in the hytnrcfg.ini. The quantity is then displayed using "--".



With workplaces/machines that are configured as "Machining centers", the calculation of the target quantity since OP logon is generally disabled because this calculation contradicts the principles of the machining center.

Planned duration

The field *Planned duration* displays the target processing time of the operation in format [h:mm].

Partitioning

Calculate the displayed partitioning as follows:

TLG _M	Partitioning of the workplace/machine (TLG in mnrlst)
DIV _M	Pulse factor of workplace/machine (IMPFAKT in mnrlst)
TLG _{AG}	Partitioning of the operation (TLG in anrlst)

The application shows the calculated partitioning without decimal places, provided it is an integer value.

In case the partitioning or pulse factor of a machine or an order is 0, calculation is based on the value 1.

Displaying the "3rd list"

The third list is optional. You can configure the third list in the configuration of workplaces. The following lists can be displayed:

- List of staff logged on to the currently selected workplace (BDE)
- List of resources logged on to the currently selected workplace (WRM)
- List of materials/input batches (MPL/TRT) logged on to the currently selected workplace
- List of output batches produced in the currently selected operation (MPL/TRT)

In case you have enabled several lists, you can switch between these lists in the header line that is located above the third list. Activated lists can be selected one after the other.

Maintenance status

If you have purchased the license for the maintenance calendar, the maintenance status is displayed using a yellow or a red field showing a wrench. The color displayed depends on the required maintenance activity.

Calling functions

The functions available are assigned to the relevant objects. Example: The functions *Log person off* and *Log all staff off* are displayed if you click on a person logged on.

3.3 Icon view of workplaces

You can enable this view in the configuration of terminals via the client. Then open this view by clicking the button "< Overview" in the main view. This view shows workplaces in a clear structure and with an image. It shows important information on the single workstations:

Workplace/machine 33021	Workplace/machine 40312	Workplace/machine 406457A	Workplace/machine 406457B	Workplace/machine 79667	Workplace/machine 79668
Status PRODUCTION	Status OUT OF MATERIAL	Status PRODUCTION	Status PRODUCTION	Status PRODUCTION	Status PRODUCTION
Yield 833	Yield 1381	Yield 2500	Yield 66	Yield 51	Yield 78
Scrap 4	Scrap 5	Scrap 13	Scrap 0	Scrap 1	Scrap 0
					

Please note: The actual display can be different to the above illustration.

A colored bar to the left of the image indicates the current status of workplaces/machines:

- green: production
- yellow: assigned status
- red: not assigned

Each tile includes the workplace/machine number, the status (text) of the workplace, the yield and scrap quantity and the image of the workplace.

A colored background with a caliper and/or wrench to the right of the image indicates if an inspection or maintenance is due for the workstation.



If you enable the extension [aipkpi](#), the application also shows the KPIs OEE, utilization efficiency and scrap ratio.

The application calculates the KPIs at cyclic intervals (scheduler job "MDE keyfigure calculation"). The KPIs always refer to the current shift. The application calculates the KPIs based on the formulas that are also used for the OEE report and/or the efficiency report on the MOC. The application shows the KPIs with two decimal places. To the right of the KPI, the AIP2 GUI highlights in color if limit values are exceeded or not reached.



The AIP calculates and updates data at cyclic intervals. This may result in deviations between the collected values and the displayed KPIs.

If you click on a tile, the previously described main view is displayed and the workplace is automatically selected. From there, you can perform the postings for the selected workplace.

Use the option "< Overview" to exit the main view and to return to the icon view.



As part of the advanced configuration options, you can customize the layout of display lists, the displayed data fields and functions. For technical reasons, however, you cannot change the sort sequence of display lists in the main view of the terminal.

As of AIP 8.2.2.28, you can automatically change from the main view with tiles to the icon view of workplaces after a configured time. The configuration `AUTOMATIC-CHANGE-TO-START-DISPLAY` is described in the document [AIP2_Configuration_hytnrcfg.pdf](#).

4 Basic Screen as List View

The new tile design can be disabled for the AIP2 terminal. This chapter describes the basic screen with disabled tiles.

In general, the AIP2 has been designed for entries to be made via touch screen. The corresponding functions can be started, selected or executed by touching the buttons or using the displayed virtual keyboard. Selection lists are provided in many cases, as an alternative to manual entries. Required entries can easily be selected from these lists.

Barcodes can be imported/entered in the current dialog using barcode readers, handheld scanners, or swipe card readers. Subject to the barcode prefix, certain data (e.g. operation data) can directly be assigned to the corresponding input field, without having to focus this input field explicitly.

It goes without saying that mouse and keyboard may also be used.



To ensure proper processing and posting, terminals with "MDE" operation mode must not be switched off during times without shift.

4.1 Basic screens – header and footer

Header



The AIP logo is displayed top left of the screen, which may be replaced with a customer logo after corresponding configuration.

Possible messages are displayed to the right of it (e.g. if a dialog is opened for more than five minutes).

A separate window opens to display error messages that occur during data collection (e.g. validity checks).

Basic screens

A maximum of 16 workplaces or machines can be assigned to the AIP2 terminal. The single workplaces can be found within the list area in the order assigned to the terminal via the client. .

As regards the basic screen of the AIP2 terminal, the user can choose between a tabular view, field-related view and an icon view. This can be configured via the configuration of terminals in the client. The single basic screens are described in the sections that follow.

Footer



The MPDV logo can be found at the bottom left of the AIP2 terminal. Double clicking the logo opens the info dialog where further administration functions can be started. This dialog closes automatically after approx. 5 seconds.

Further information is displayed in the center : the current terminal status, AIP2 version number, date of the build, IP address of the server as well as the terminal number.

The current date and time are displayed to the right.

Terminal status

	Network connection has been established The terminal is ONLINE. Server communication is enabled. All saved data records have been transferred.
	The terminal is sending data to the server.
	No network connection or no connection to the server. The terminal is OFFLINE. Server communication is interrupted. Online functions, such as the display of information, are disabled. But certain postings can be recorded anyway. These postings are transferred to the server, once data connection has been established.
	Data are being received. The terminal reads files from the server or writes data to the server.
	The terminal is sending stored data records to the server.
	DEMO mode The terminal is in the DEMO mode, i.e. server communication is disabled.

4.2 Basic screen “tabular view“

1st list
Workplaces assigned to the terminal

2nd list
List of registered operations

3rd list (optional)
e.g. list of registered staff

The screenshot shows the AIP (Advanced Production Interface) tabular view. It consists of three main sections:

- Machines/workplaces:** A table with columns: Machine/Workplace, Status, Status since, and Note. It lists four entries: 33021 (Manual Workplace Deburring, PRODUCTION, 18.08.15 06:00), 40312 (Manual Workplace Deburring, OUT OF MATERIAL, 18.08.15 20:55), 406457A (Mounting Place 1, PRODUCTION, 18.08.15 18:40), and 406457B (Mounting Place 2, PRODUCTION, 18.08.15 15:44). The status column is color-coded: green for PRODUCTION and yellow for OUT OF MATERIAL.
- Operations on workplace:** A table with columns: Article, Order, Operation, Target qty., Yield, Scrap, N, and T. It lists one entry: SAR-VPO011-M-9005 (S3000209, 0200 Deburring, 1200 PCS, 660, 0).
- Registered persons:** A table with columns: Person, Name, First name, and Advance logon. It lists four entries: 10002630 (White, Susan), 85741 (Meyers, Joseph), 40563 (Green, Mary), and 10002854 (Baker, Jake).

At the bottom, there are buttons for 'Staff work time logging' and 'Staff logging', and a footer showing the date and time: 23.11.15 16:15:15.

Subject to the configurations made, the tabular basic screen consists of two or three tables. While the first two tables are always displayed, it is up to the user whether or not the third table is shown (optional).

“Machines/workplaces” table

The upper table shows the workplaces assigned to the terminal. The following columns are displayed.

Machine/workplace

The machine or workplace number as well as a description are displayed.

Status

The status is highlighted in color and the status text is shown. Coloring is as follows:

- green: production
- yellow: assigned status
- red: not assigned

Status since

Point in time since the status is available.

For ADE workplaces² the point in time refers to the last manual status change. For MDE workplaces it refers to the time when the last status change was identified (for machine connections). It also refers to the point in time when the status was changed manually most recently or to the time of the last shift change.

Please note:

It is indicated here if the "lock production status" function is enabled for the machine/workplace.

Below the first list there is a row including the function buttons mainly relating to machines/workplaces. These functions are described in more detail in the sections that follow.



By way of "customizing" services it is possible to adapt the layout of the display lists, displayed data fields, sort sequences, etc. according to the customer's requirements. For technical reasons, however, the sort sequence of display lists may not be changed in the basic screen of terminals. The software does not allow it.

Provided that the "compensate manual quantities" option (e.g. set off scrap against yield) is enabled and the machine list also shows shift-related quantities (no default setting), they will not be updated immediately. Quantities are only updated once the lists have been reloaded.

"Operations at workplace" table

The second table shows the operations currently logged on to the selected workplace. The following columns are displayed:

Article

Article defined for the operation

Order and operation

Order number and operation number of the registered operation. Together they build the *MES order number*.

Target quantity

Target quantity defined for the operation.

² We talk of an MDE workplace if this workplace is assigned to a terminal, which runs in the "MDE" operation mode. In any other case, it is an BDE workplace.

Yield

Yield already produced for this operation. The counters of possible machine connections are considered as well.

Scrap

Scrap quantity already produced for this operation. The counters of possible machine connections are taken into account as well.

N

It is indicated here if a note visible on the terminal has been recorded for this operation in the graphic planning board of the client. The note(s) is/are displayed by clicking the OP info button ().

T

If a long text is defined for this operation it is indicated here. The long text is displayed using the OP info dialog (button ).

Below the second list there is a row that mainly includes function buttons relating to operations.

"3rd list" table

The third list is optional and may be configured. Information displayed in this list depends on the workplace configuration.

The following lists can be displayed:

- List of staff logged on to the currently selected workplace (BDE)
- List of resources logged on to the currently selected workplace (WRM)
- Materials/input batches logged on to the currently selected workplace (MPL/TRT)
- List of output batches produced in the currently selected operation (MPL/TRT)

The buttons below the third list (to the left) allow switching between these lists.

Please note

The staff logged on displayed in the third list is identical to the list displayed in the dialog "F5 staff logged on...". Selecting a person in the third list does *not* affect the selection of the operation in the list of OPs running at the workplace. Therefore, it neither affects pre-assignment of the operation in the corresponding posting dialogs.

Toolbar in the basic screen

A toolbar, which may be customized, is assigned to each list included the basic screen. This makes the purpose of a function clear to the user. The “partial upload/confirmation” function can be found below the list of registered operations.

In fact, the toolbar may include several “tabs”, which can be made visible by scrolling to the right/left at the right/left end of the toolbar. A posting dialog (e.g. *change partitioning*) can be opened by clicking the corresponding button.

Please note

The displayed buttons depend on the context defined by the respectively selected workplace. Thus, the displayed buttons may vary when selecting another workplace/machine.

4.3 Basic screen "machine overview"

If the “change view” button is clicked in the basic screen, the view changes to the following presentation:

Toolbar of the assigned machines

Workplace/Machine	33021	Status	PRODUCTION	
Short description	WPL02	Start	06:00	18.08.2015
Group	FINISH	Duration [hh:min]	2338:16	
Part.	1	Yield	833	PCS
Target cycle [sec/cycle]	20	Scrap	4	PCS
Actual cycle [sec/clock]	0	Cycles		
		Note		

Machine information

Operation	S30002090200	Target qty.	1200	PCS
Article	SAR-VPO011-M-9005	Yield	660	PCS
Article Designation	Mirror housing right,	Scrap	0	PCS
Comment 1				PCS
Comment 2		Yield since logon	0	PCS
Planned duration	06:40	Completion	55%	

Order information

Buttons: Modify status, Log on operation, Partial confirmation, Interrupt operation, Log off operation

Footer: AIP logo, US Flag, 23.11.15 16:16:51

This presentation gives detailed information on a single machine, whereas the above toolbar still provides an overview of all assigned machines and workplaces.

The presentation consists of three sections:

Toolbar of the assigned machines

The color indicates the current status of all assigned workplaces. Coloring is as follows:

- green: production
- yellow: assigned status
- red: not assigned

It is possible to switch between machines by pressing a machine icon (requires a touch screen). Using the keyboard, the active machine can be selected by the arrow keys. To do this, the toolbar must be active.

Workplace/machine information

This display area shows information relating to workplaces/machines and shifts about the currently selected workplace.

Order information

This display area shows information on the registered order/OP. If several orders/OPs are logged on to the workplace, then extra arrow buttons are displayed. It is possible to switch between individual orders/OPs using these arrow buttons.

Notes on selected fields of the machine overview

Unit for yield and scrap

Provided that no operation is logged on, the primary quantity unit from the workplace/machine configuration is displayed as unit for yield and scrap. If an operation is logged on the primary quantity unit of the operation is displayed.

Partitioning

The displayed partitioning is calculated as follows:

$$\text{Partitioning} = \frac{TLG_M}{DIV_M} * \left[\frac{TLG_{OP1}}{DIV_{OP1}} + \frac{TLG_{OP2}}{DIV_{OP2}} + \dots \right]$$

TLG _M	Partitioning of the machine (TLG in mnr.lst)
DIV _M	Pulse factor of the machine (IMPFAKT in mnr.lst)
TLG _{OPi}	Partitioning of the individual operation (TLG in anr.lst)

The resulting partitioning is displayed without decimal places, provided it is an integer value. Otherwise, 3 decimal places are shown.

In case partitioning or pulse factor of a machine or an order is 0, calculation is based on the value 1.

Having logged off all OPs, the machine continues working with the partitioning of the machine.

Target cycle

The largest target cycle of all operations running at the machine is always displayed in the machine overview of the terminal. If this OP is logged off the largest target cycle of the remaining OPs will be displayed.

In case no OP is logged on, the target cycle from the machine list is displayed. Thus, even after a restart, the terminal can get the target cycle that applied at last.

The largest target cycle is also transferred to MDE for monitoring.

Comment 1, comment 2

These two fields show the user fields 53 and 54 (alphanumeric with 20 characters) of the operation. To be able to edit these fields, a corresponding user field key containing these two fields must be defined for the operation.

Target since logon

The production quantity to be expected since the OP has been logged on (depending on the cycle time, partitioning and the time while the production status was not locked for the machine). No value can be calculated, in case the terminal program has been restarted since the OP was logged on.

Calculation:

$\text{TargetSinceLogon} = \text{NetRunningTime}[\text{sec}] * \text{Partitioning} / \text{TargetCycle}[\text{sec/stroke}]$

NetRunningTime: Time since logon while the production lock has not been set. This calculation does not take into account any breaks defined in the shift model or status times posted on RPA 12 (resource performance account).

Deviation [%]

Deviation (in percent) between the expected target quantity since logon and the quantity which has actually been produced "since logon".

Calculation: $\text{Deviation}[\%] = 100\% * (\text{YieldSinceLogon} - \text{TargetSinceLogon}) / \text{TargetSinceLogon}$

Completion

The bar represents the proportion of "yield", which has been produced until now, compared to the "target quantity".

Machine icon:

Provided that the WRM-WTK license has been purchased, the machine icon may be replaced with a picture showing a yellow or red oilcan. It all depends on the required maintenance activity:



4.4 “Machines as icons” basic display

This view can be configured as the default display in the configuration of terminals at the client. It has the advantage that the user can tell from a distance whether or not all machines are in the “Production” status.

All MDE machines have their own buttons colored according to the corresponding status:

- green: Production
- yellow: Assigned status
- red: Not assigned

The button includes details on the workplace/machine number, the registered operation, the yield and scrap quantities as well as the status (text) of the workplace/machine.

If a button is touched, the “machine overview” basic display is shown for this workplace. From there, postings for the selected workplace can be performed using the standard buttons.

By clicking the “symbol” button (if configured) the view changes from the “machine overview” to the “icon view of machines”.

5 Barcode Input with Prefix

A barcode is interpreted as prefix barcode if the third character is a dot. In this case the first two characters identify the barcode type. The actual barcode starts with the fourth character.

ID+ Prefix	Example	Comment --> processing
-----	-----	---- General ---
00.	00.ABC123	Data not defined, → 00. will be deleted and "ABC123" will be passed on to standard processing
01.	01.OK 01.ESC	Action barcode → Dialog cancelled or ended with OK button or Esc button.
-----	-----	---- HYDRA-ADE + HYDRA-LLE + HYDRA-MDE ---
10.		(combined) Order/sequence/OP number → acronym <ANR>
11.		Order (header) → acronym <AUNR>
12.		Sequence → acronym <AFOLG>
13.		OP → acronym <AGNR>
14.		Sub-order number -> Acronym <UAGNR>
22.		Split no. → acronym <SPLNR>
15.		Upload/confirmation number → Acronym <RMNR>
16.	16.EXTRUDER-7 16,200	Machine → Acronym <MNR> → Passed on to dialog with MNR=EXTRUDER-7 or MNR=200
17.	17.1 17.1001	Machine status → Acronym <MST> → Passed on to dialog with MST=1 or MST=1001
18.	18.1 18.1001	Scrap reason → Acronym <EGG:AUS> → Passed on to dialog with EGG:AUS =1 or EGG:AUS=1001
19.	19.1 19.1001	Deviation reason → Acronym < EGG:GUT > → Passed on to dialog with EGG:GUT =1 or EGG:GUT=1001
20.	20.1 20.MF	Operator position → Acronym <BPOS> → Passed on to dialog with BPOS =1 or BPOS = MF
21.		Wage and premium indicators → Acronym <LPKZ>
-----	-----	----HYDRA-WRM + HYDRA-DNC + HYDRA-PDV + HYDRA-MPL ---
40.	40,100 40.MONTAGE	Destination → Acronym <ZLO> → Passed on to dialog with ZLO=100 or ZLO= MONTAGE
41.	41.KARTON 41.KISTE	Transport unit → Acronym <TPE> → Passed on to dialog with TPE = KARTON or TPE = KISTE
42.		Batch number → Acronym <CNR>→→
43.		Throughput batch number → Acronym <DLL>→→
44.		Alternative batch number → Acronym <CNR:ALT1>→→
45.		Alternative batch number → Acronym <CNR:ALT2>
46.		Alternative batch number → Acronym <CNR:ALT3>
47.		Alternative batch number → Acronym <CNR:ALT4>
48.		Alternative batch number → Acronym <CNR:ALT5>
49.		Alternative batch number → Acronym <CNR:ALT6>
-----	-----	---- Mainly for the HYDRA-PZE module ---

ID+ Prefix	Example	Comment --> processing
50.		Badge number → Acronym <KNR>
51.		Personnel number → Acronym <PNR>
52.	52.EDV 52.VERTRIEB	Cost center → Acronym <KST> → Passed on to dialog with KST=EDV or KST=VERTRIEB
53.	53.1 53.1001	Absence reason → Acronym <FGR> → Passed on to dialog with FGR=1 or FGR=1001
-----	-----	---- Customer-specific barcodes ---
90.		
...		

In case barcodes are required, which actually have a dot at the third place (e.g. if the machine/workplace number has a dot as third character), it is possible to define an alternative indicator for barcode prefixes in the HyTnrCfg.ini terminal configuration, e.g.

[Terminal->USR 0]

BarcodePrefixChar=\$



If another prefix is actually required, the respective barcode font in use must be able to represent this prefix.

Examples for barcodes

Barcode printing: Font "Codedreineun" and prefix "."

Prefix	Barcode	Raw data
10.	*10.123456780100* *10._ABCD12340100*	ANR = 123456780100 ANR=_ABCD12340100

Prefix	Barcode	Raw data
11.	*11.12345678*	AUNR = 12345678
12.	*12.01*	AFOLG = 01
13.	*13.0100*	AGNR = 0100
14.	*14.0000*	UAGNR = 0000
15.	*15.123456789012345*	RMNR = 123465789012345
22.	*22.02*	SPLNR = 02

Prefix	Barcode	Raw data
16.	*16.123456*	MNR = 123456
17.	*17.1122*	MST = 1122
18.	*18.1234*	EGG:AUS = 1234
19.	*19.123456789*	EGG:GUT = 132456789
20.	*20.13*	BPOS = 13
21.	*21.1221*	LPKZ = 1221

Prefix	Barcode	Raw data
50.	*50.1337*	KNR = 1337

5.1 Configuration of customized barcode prefixes

Section [barcode]	
BarKenn90=SAPCNR BarKenn91=EGR:GUT	The barcode prefixes 90...99 can be assigned here according to the customer's requirements. This means, if a barcode with the relevant prefix is used, it will be transferred to the dialog along with the assigned ID. Then the barcode has the following structure: <Prefix>.<Net barcode> e.g.: "90.12345" → SAPCNR=12345

Firmly assigned barcode prefixes:

10:ANR 53:FGR

11:AUNR

12:AFOLG

13:AGNR

14:UAGNR

15:RMNR

16:MNR

17:MST

18:AUSGRD

19:AGGGRD

20:BPOS

21:LPKZ

22:SPLNR

40:ZLO

41:TPE

42:CNR

43:DLL

44:CNR:ALT1

45:CNR:ALT2

46:CNR:ALT3

47:CNR:ALT4

48:CNR:ALT5

49:CNR:ALT6

50:KNR

51:PNR

52:KST

6 AIP2 -Local Configuration

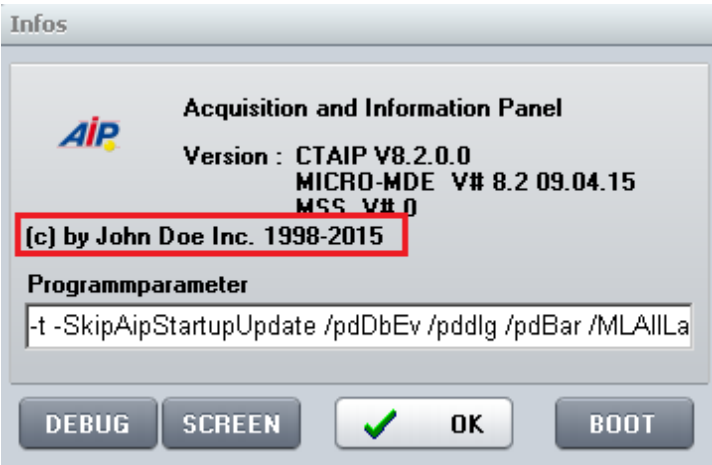
6.1 Local Configuration ctaip.ini

The most important hardware and system settings are defined for each terminal in the CTAIP.INI file of the C:\MPDV\AIP2 directory.



Changes to the configuration file ctaip.ini are only enabled after rebooting the terminal software.

Layout configuration

Entry	Comment
Section [system]	
CompanyInfo=(c) by John Doe Inc. 1998-2015	Overwrites the copyright text in the "About" dialog 
parameters= ... -t +t	The virtual keyboard can be disabled/enabled, irrespective of the terminal type.
DEMO_ALL=ON	The information in the lower bar are normally hidden in demo mode. This option, however, causes all pieces of information to be displayed even in demo mode (version number and date, server, terminal number, online lights).
parameters= ... - SkipDynDlgExtraction...	Internal option preventing the extraction of dialog fields/buttons when updating dialog data. (Only applies if WF directory ".\spool\swf.\$\$\$\" is available)
parameters= ... - SkipAipStartupUpdate...	Internal option preventing the configuration files (*.ini,*.cfg,*.cfl) of the module DLLs (packets*.dll) and application DLLs (functions*.dll) from being synchronized when starting the terminal.
VirtScreenSize=640	All windows are started with the indicated resolution

Entry	Comment
VirtScreenRatio=16:9	The display ratio remains if the configuration VirtScreenSize is used to reduce the window. The width-to-height (aspect) ratio can be changed using VirtScreenRatio. Consequently, the width-to-height ratio 16:9 can be tested with a 4:3 monitor and vice versa. The value can be configured as "16:9" or as floating point value (e.g. 1,77777). Note: Information screens are specifically scaled if monitor screens in portrait format (e.g. 9:16) are used. This special scaling can be disabled in hytnrcfg.ini: [Tnr Konfiguration 0]->PortraitAlignment=off
Section [SKIN]	Configurations for the terminal's skin (see ctaipplay.ini) The online configuration of the terminal can be reached by the info dialog (click MPDV icon). ALT+F1 opens the control elements for the skin. Please note: Only in the design/GUI of AIP 8.1 and/or the dynamic dialogs
Saturation=0	Skin configuration / Default = 0
Hue=0	Skin configuration / Default = 0
Name=mpdv	Skin to be used / Default = mpdv
Active=false	Skins can only disabled in ctaip.ini! Default = true

Entry	Comment
Section [system]	
Usr=21	Distinct terminal number
Hostname=192.9.200.24	Internet address of the server
Offlinetimeout=600	The interval after which online access should be attempted the next time. The interval is specified in seconds
Showcursor=on	Show or hide mouse pointer in terminal application: on: mouse pointer active off: mouse pointer inactive
Loadfile=ctnet\win\aip2.txt	Configuration file to download the application from the server. The paths is relative to the installation directory of the system.
Watchdog=on	ON: Watchdog is activated OFF: Watchdog is not activated

Entry	Comment
Demo=off	'on': Offline demo mode; always off in the production environment!
parameters=-t	The -t parameter switches off the virtual keyboard.
TMOUT_C=xxx	Timeout for CONNECT to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
TMOUT_S=xxx	Timeout for SEND to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
TMOUT_R=xxx	Timeout for RECEIVE of the server If not specified, default = 120 seconds
TMOUT_F=xxx	Timeout for FILESERVER operations to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
Section [barcode]	Configuration of customized barcode prefixes.
BarKenn90=MNR4 ... BarKenn99=ANR3	BarKenn90 > defines the prefix (here: 90); The ID from the dialog (= acronym) is assigned.
Section [comports]	
com1=0 com2=MSS com3=BAR Com3=LEGIC Com4=RFLESER	Assignment of serial interfaces to connected devices: MSS – machine interface BAR, LEGIC, RFLESER – various reading devices
Section [MSS-INIT]	
MSS_DIALOG=10	If the terminal is switched off longer than 15 minutes, a dialog is displayed on terminal restart. The user must then decide whether the counter pulses, which were recorded when the terminal was closed, are posted or discarded. The dialog closes automatically with "Yes" after an entered time has elapsed; in this case the counting impulses are posted. This value configures the time in seconds the dialog is open. If the terminal is switched off for less than 15 minutes, no dialog is opened; the counting pulses recorded in the switch-off phase are accepted and posted without confirmation. Please note: The value can also be configured in hytnrcfg.ini. Entries in the hytnrcfg.ini file take priority.

Entry	Comment
MSS_FILEAGE_MIN=5 MSS_FILEAGE_OVERTIME=delete	Additional configuration for MSS_DIALOG If the backup file for counting pulses of the terminal is older than 5 minutes, no dialog is opened and the back up file deleted. Quantities recorded at the time when the terminal was closed are not used/posted. Please note: The value can also be configured in hytnrcfg.ini. Entries in the hytnrcfg.ini file take priority.
Section [ext. software]	
Button=Editor WindowName=Editor SearchParts=On	Configuration of the button in the top line: A previously started program can be brought to the foreground at the push of a button. Button: button caption WindowName: Name of the program (e.g. from the taskbar). SearchParts=On: Parts of WindowName are sufficient SearchParts=Off: WindowName must be entered completely. The option "SearchParts=On" is recommended for programs such as MSWord that change the title bar subject to the document that is currently being loaded.
ProgFileName=c:\Programme\wincmd\Wincmd32.exe	The program that is started if the program mentioned above cannot be called to the foreground.
AutoStart=on	This option starts the program (ProgFileName) when starting the terminal program.
Section [DLL]	
BusDLL=PCC.EXE	PCC.EXE must be entered here if the terminal is configured with the option "operated as HYDRA-MDE terminal". If necessary, the AIP2 enters this value independently upon starting the program.
Section [PDV]	
PDVProtokoll=ON	Enables PDV logging (prot_pdv.txt) (by default=OFF)
PDVIOPODir=c:\IOPSim\	Path for DNC
Section [CAQ]	
;Supported barcodes	
FieldWNRBarcodeOnly=Y	If this entry is set the tool number may only be entered using a scanner.
FieldNestBarcodeOnly=Y	If this entry is set the cavity number may only be entered using a scanner.
FieldNummBarcodeOnly=Y	If this entry is set the number may only be entered using a scanner.
FieldKNRBarcodeOnly=Y	If this entry is set the badge number may only be entered using a scanner.

Entry	Comment
BarcodeWNR=	This field specifies which acronym is entered into the tool number field by the scanner.
BarcodeNest=	This field specifies which acronym is entered into the cavity number field by the scanner.
BarcodeNumm=	This field specifies which acronym is entered into the number field by the scanner.

6.2 PNG – Files / Bitmaps

The use of PNC files is recommended by MPDV. By default PNG files have a size of 24 x 24 px.

6.2.1 File pict.zip

The file "pict.zip" is updated by the installation tool "inst32.exe" while downloading and includes all default PNG files.

The default PNG files can be overwritten in the file pict_cust.zip. Several PNG files have the extension ".small.png" (e.g. aip.small.png). These PNG files are used with a screen resolution of 640x480.

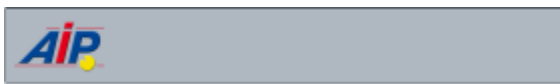
6.2.2 File pict_cust.zip

The file "pict_cust.zip" is loaded from the server directory (e.g. \<serverDir>\1\custom) when starting the program (as is the case for the hycust.mld).

Customized PNG files may be stored in this file and loaded by the AIP2 terminal. Default PNG files may also be "overwritten".

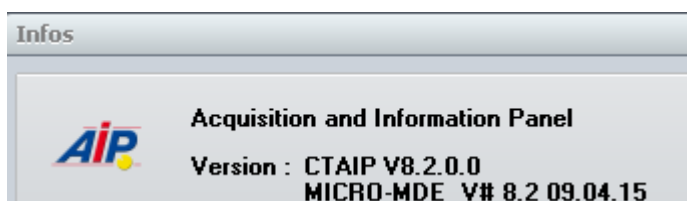
Please note: file sizes are not adjusted.

Customize header



The AIP icon displayed in the header can be replaced by storing a separate AIP.png file in the pict_cust.zip file.

This AIP icon will also be replaced in the "About" dialog.

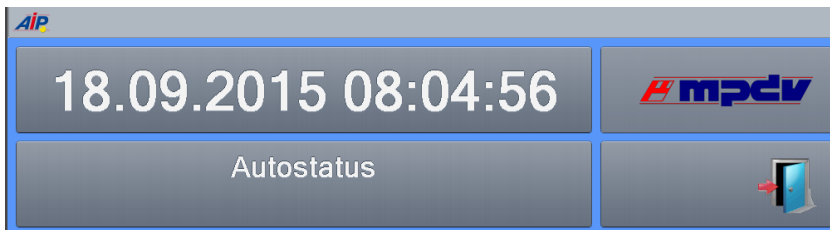


Customize footer



The MPDV icon displayed in the footer can be replaced by storing a separate company.png file in the pict_cust.zip file.

Customize PZE dialog



The MPDV icon displayed in the PZE dialog can be replaced by storing a separate pze_mpdv.png file in the pict_cust.zip file. In case the PZE terminal is operated with a screen resolution of 640x480, a customized pze_mpdv.small.png file has to be integrated in the pict_cust.zip file.

6.3 Multilingualism (*.mld files)

The below-mentioned files are required for the translation of the application. The table shows the priorities for the translation, ownership and the relevant storage locations.

Priority	File	Server directory	Owner	Description
1	ctaipkd.mld	./ctnet/win/aip2/custom	Customer	Customer-specific translations
2	hycust.mld	./1/custom/	MPDV	Customized translations by MPDV. The file is used by the AIP2, CTAIP and CTWIN terminal.
3	ctaip.mld	./ctnet/win/aip2	MPDV	Standard translation file

7 AIP2 - Central Configuration File hytnrcfg.ini

You can use this file as central place to store different configurations for all or for separate terminals.

For each section, a general version is available

[section 0].

The entries included in this section can be overwritten by entries in a terminal-specific section

[section <TNR-USER>]

<TNR-USER> = UserNo = terminal number + 2000 e.g. 2010,2101,..) for exactly one terminal/UserNo



The file hytnrcfg.ini is loaded from the server on every terminal start.

Section / Entry	Comment
	→→→
[Tnr Konfiguration 0]	
FollowExternStatus=on	Transfer of machine statuses when reloading machine list. Useful if status change is set by PDM or another terminal
[Terminal->Installation 0]	
InstallFonts=on	If set to "off", fonts are not installed during restart. ON=DEFAULT
OnlyInstallFontsAfterDownload=false	If "InstallFonts=on": If true, then fonts are only installed directly after a download. If false, then fonts are installed every time the terminal is restarted. (false = DEFAULT)
[Terminal->USR 0]	

Section / Entry	Comment
AttachedApplication=First	Displaying documents of OP info: With this configuration, the system first checks whether or not an application is linked in Windows that matches the file extension of the document. This application is then used to display the document. If no link is available, the viewers configured in ctaip.ini (→ [ext. software]) and internal viewers are used. If an extension is completely unknown, the system tries to display the document as text . Different settings are possible: First → search for linked application first AfterUserViewer → If a UserViewer is configured, this one overrides the linked application (also applies to ExcelViewer, WordViewer and PowerpointViewer) Last → Only if no ctaip.ini assignment is found for the file extension, then the system searches for a linked application (default). Off → The system does not search for a linked application.
HTTPBrowser=standard	Display of documents (via OP info): If documents are configured with a path of schema "http", the file is not downloaded to the terminal, but the link is transferred to a browser. The default browser for the terminal is htmview3.exe, as this one can be operated by touchscreen. If this entry is set, the default browser configured in Windows is used.
SupressErrorMessage=70012	Suppress message "material is not planned"
MSS_DIALOG=10	If the terminal is switched off longer than 15 minutes, a dialog is displayed on terminal restart. The user must then decide whether the counter pulses, which were recorded when the terminal was closed, are posted or discarded. After a configurable period of time, the dialog closes automatically with "Yes" (Yes, posting of pulses). This value configures the time in seconds the dialog is open.
MSS_FILEAGE_MIN=5 MSS_FILEAGE_OVERTIME=delete	If the backup file for counter pulses on the terminal is older than 5 minutes, then no dialog is opened and the backup file deleted. Quantities recorded at the time when the terminal was closed are not used/posted.
[SignatureRecording->User 0]	
ManualBadgeInput=true	This configuration specifies whether or not the field <i>User</i> can be edited on the terminal (by default: no editing) true → activates keyboard input for field <i>User</i> on the terminal

Section / Entry	Comment
Transparency=255	The display of the signature dialog can also be transparent. 255 → Signature dialog is 0 % transparent (not transparent) 1 → Signature dialog is 99% transparent (maximum transparency) (Default = 155)
ShowPosition=TR	You can change the place of the signature dialog: TL Top – Left TM Top – Middle TR Top – Right ML Middle – Left MM Middle – Middle (Default) MR Middle – Right BL Bottom – Left BM Bottom – Middle BR Bottom – Right
USE_SERVICE_ACCOUNT=1	0 (default) SSO: ServiceAccount is not used (requirement: the terminal must be started with the domain "user" (SSO)). Note: ServiceAccount=1 can only be used if all users are in the "root" domain. SubDomain users are not supported.
SIGNATURE_1_USER_TYPE=REPORTING_USER_READONLY	REPORTING_USER_READONLY The user identification using the Windows user is activated. The Windows user is then preassigned in field <i>User</i>. The <i>User</i> field is read-only. Requirement: The "SSO" option must be enabled for all reporting users. Otherwise, successful authentication is not possible. REPORTING_USER_CHANGEABLE The user identification using the Windows user is activated. The Windows user is then preassigned in field <i>User</i>. The <i>User</i> field can be edited. Requirement: The "SSO" option must be enabled for all reporting users. Otherwise, successful authentication is not possible.

Section / Entry	Comment
SIGNATURE_1_LOGON_TYPE=HYDRA	"" / Not set / "EMPTY" There is also an alternative login procedure. HYDRA The user identification using the Windows user is locked. You must enter a user for identification. Requirement: All reporting persons must have been created as users and the option "SSO" must not be enabled for the users. Otherwise, successful authentication is not possible. ACTIVEDIRECTORY The TAB for the user identification via <i>User</i> is locked. The Windows user must be used for identification purposes. Requirement: The "SSO" option must be enabled for all reporting users. Otherwise, successful authentication is not possible. MIXED_BUT_UNIQUE The setting of option "SSO" specifies whether the user login or the Windows login is used. "SSO" enabled → Windows only "SSO" disabled → user only
SIGNATURE_2_LOGON_TYPE=HYDRA	Identical to SIGNATURE_1_LOGON_TYPE (see above)
ExtendedSignatureRecording=true	Used for signatures on the terminal with quality data collection.
[MDE/Blade Configuration 0]	
CONVERT-TO-ANSI- FILE=<list1 list2>	Configuration of the files that are provided from the AIP to the MDEB2 blade in ANSI format when a combined operation is available. The following lists are transferred by default if the entry is not available. counters.lst schicht.lst mnr.lst mstat.lst anr.lst pnr.lst If you want to transfer further lists, you must specify the standard lists and the additional lists.

7.1 Layout configuration

Entry	Comment
Section and/or [terminal configuration 0] [terminal configuration 2XXX];	(general configuration) (2XXX terminal-specific configuration)
AUTO-CONFIRM-UHR-ERROR-MESSAGE=TRUE	This setting specifies that in case of an error that occurred reading the clock (e.g. when activated after standby

Entry	Comment
	mode), the time is transferred without confirmation dialog and the terminal time is later synchronized with the server time via PDM command.
SUPPRESS-MAXIMUM-NUMBER-OF-MACHINES-WARNING=ON	Suppresses the warning after restart of terminal if more than 32 machines are assigned to the terminal (static/dynamic). (Default=OFF)
CalcTargetYieldSinceLogon=2	CalcTargetYieldSinceLogon=1 The duration is calculated from the total runtime since login (all statuses) minus the configured shift breaks. CalcTargetYieldSinceLogon=2 The duration is calculated from the total runtime since login (all statuses). Defined breaks are not used and are not deducted. (default value)
Section [QRD-PRINTER->TICKET 0] ;(general configuration) [QRD-PRINTER->TICKET 2xxx] ;(2XXX configuration for a specific terminal)	
COMPLETE-ABSENCE-OF-LOCAL-MNR-DATA-FOR-EVENT=< Events >	Reloads the machine row for the configured <Events>, if it is not available locally ➔ This configuration might be required for a group workplace without machine assignment.
COMPLETE-ABSENCE-OF-LOCAL-ANR-DATA-FOR-EVENT=< Events >	Reloads the order row for the configured <Events>, if it is not available locally ➔ This option has been implemented to access order data in the master data, e.g. when logging on orders.
COMPLETE-...-EVENT=< Events > COMPLETE-...-EVENT=#ALL# COMPLETE-...-EVENT=A_AN A_P_AN	Explanation on the configuration of <Events> ➔ Using <#ALL#> the row (ANR/MNR) that is not available is reloaded for any event. ➔ <A_AN A_P_AN> restricts reloading of information to specified events. The ID <DLGFAM> is preferred to the ID <DLG> in order to identify the <Event>.
Section [AIP2 Initialization 0]	
XML-GUI=OFF	Disables the new AIP2 design and uses the AIP 8.1 design.
CTWIN-STYLE=ON	Activates the GUI that is similar to CTWIN on the AIP2. The two button bars are shown below the two lists just like on the AIP 8.1.
CTWIN-BUTTON-LAYOUT=ON	If the option CTWIN-STYLE=ON is additionally set, the two button bars are displayed at the bottom of the screen.
AUTOMATIC-CHANGE-TO-START-DISPLAY=30	As of AIP 8.2.2.28: If this option is set, the display automatically changes to the main view after the configured time if no other interaction was performed in the meantime. The changing display is configured via the option <i>Show machine/OP</i> in the <i>Terminal configuration</i>, tab <i>MF functions</i>. - List: - Change from the detail views or function menus (operation, person, resource, etc.)

Entry	Comment
	○ Change to the main view with "tiles" • Icons: ○ Change from the detail views or function menus or from the main view with tiles ○ Change to the icon view of workplaces The configuration is specified in seconds. Do not specify less than 10 seconds. An automatic change to the start screen is not made if an input dialog is open.

8 Collection of Serial Numbers

General

Serial numbers are assigned to be able to differentiate between individual items of material. [Wikipedia](#) provides the following definition for serial numbers:

A **serial number** (also manufacturer's serial number or MSN) is a unique, alphanumeric code assigned for identification of a single unit. Although usually called a number, it may include letters, though ending with digits. Serial numbers are used to identify elements pertaining to series production also providing information about production conditions and thus enabling traceability of the used components, also for electronic products.

Serial numbers can be:

- generated/assigned at the end of the production process
- generated/assigned in earlier stages of the production process and monitored by the process
- integrated as used components in the production process and monitored by the process



[Merging serial numbers](#) represents the assembly of several components listed by serial number into one component part listed by serial number.

MES supports manufacturing businesses in many ways with the collection of serial numbers and enables complete traceability based on the data recorded in MES.

Definition of the supported variants

MES provides different variants to record serial numbers:

Variant	Description of the variant	Description	Documentation
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Variant	Description of the variant	Description	Documentation
1	Entry of serial numbers for OPs that are not subject to management in batches	Recording of serial numbers for operations that are not managed in batches enables the collection in relation to serial numbers if incorporated materials their batches and consumptions are not required to be recorded. The production order specifies the serial numbers to be processed as part of a production order. For each operation the user identifies the serial numbers and classifies them according to the produced quality. The system makes sure that this process can only be performed once for each serial number and operation.	here
2a	Entry of serial numbers for OPs that are subject to management in batches - manual input of the serial number	The user enters the serial number and the system is informed about it. In this case, the serial number is generated in another system. The recorded serial number is assigned a HYDRA batch number.	here
2b	Entry of serial numbers for OPs that are subject to management in batches - automatic assignment of serial numbers	In this case, the system generates the serial number in an unambiguous format applicable to the whole system. Serial numbers are generated according to the rules applying for the generation of batch numbers. In this case, the serial number matches the HYDRA batch number.	here

Variant	Description of the variant	Description	Documentation
2c	Entry of serial numbers for OPs that are subject to management in batches - automatic assignment of serial numbers	In this case, the system generates the serial number in an unambiguous format applicable to the whole system. Serial numbers are generated according to the rules applying for the generation of batch numbers. The serial number is assigned a HYDRA batch number.	Here
2d	Entry of serial numbers for OPs that are subject to management in batches - automatic transfer of serial numbers by using input batches.	In addition, to the conventional generation of output batches, the system particularly supports the generation of merged batches including individually assigned serial numbers. There is a direct connection between individual serial numbers and the generated output batch (merged batch), which logically embraces the serial numbers. Serial numbers are taken over as individual batches provided by logging an input batch/merged batch on. The serial number is the batch number. In addition, individual serial numbers are assigned to a superordinate merged batch.	This document

HYDRA batch number vs. serial number

In HYDRA the [HYDRA batch number](#) is unambiguous throughout the entire system. The serial number is kept throughout the complete process, provided that [serial numbers have not been merged](#). Consequently, it is differentiated between the internal and external HYDRA batch number to represent the collection of serial numbers:

HYDRA batch number

The HYDRA batch number is the ID by which the serial number is managed in HYDRA throughout the complete collection process.

The HYDRA batch number is used on labels to identify the material.

To ensure unambiguity within the system, an internal HYDRA batch number is assigned and managed for every HYDRA batch number when recording serial numbers.

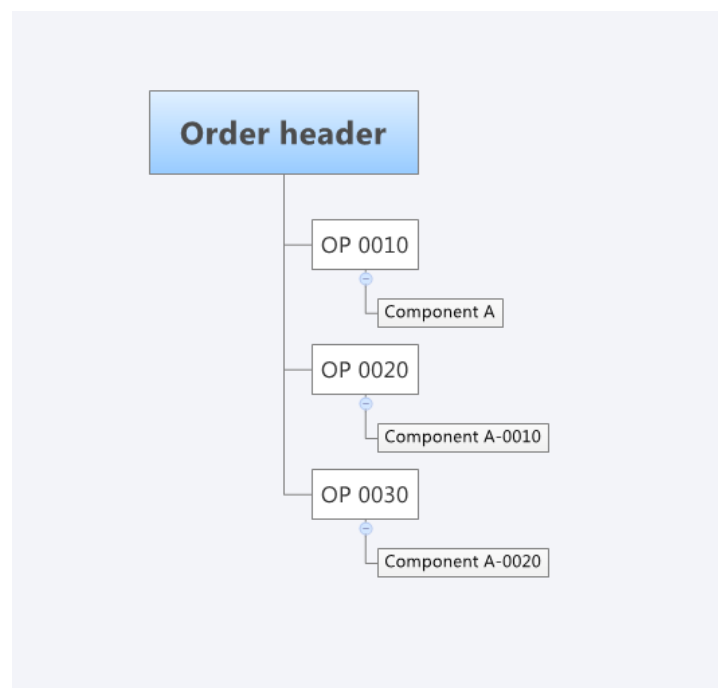
Internal HYDRA batch number

The internal HYDRA batch number represents a system-wide, unambiguous ID identifying batches and serial numbers as an alternative to the HYDRA batch number.

In contrast to the HYDRA batch number; this ID has to be changed from production step to production step to keep unambiguity.

This example explains the procedure:

A production order has the following structure:



The interface provided a merged batch with the serial number "BXC6GF7H". Upon posting this serial number, the HYDRA batch number and the internal HYDRA batch number change as follows:

Process step	Batch status	HYDRA batch number	Internal HYDRA batch number	Comment
Status <u>after</u> transferring the number to MES via the material interface and <u>prior</u> to logging the input batch on.	FREE	BXC6GF7H	BXC6GF7H	Initial status when the HYDRA batch number and the internal HYDRA batch number are identical.
Status after logging the merged batch on to OP 0010	RUNNING	BXC6GF7H	BXC6GF7H	No changes
Status after entering the serial number at OP 0010	Processed	BXC6GF7H	BXC6GF7H	The "old" batch is assigned the "processed" status.
	FREE	BXC6GF7H	PR1111X112	The "new" batch is assigned the "free" status, the same HYDRA batch number (serial number) and a new HYDRA internal batch number
Status after entering the serial number for the OP 0020	Processed	BXC6GF7H	BXC6GF7H	The "old" batch is assigned the "processed" status.
	Processed	BXC6GF7H	PR1111X112	The "old" batch is assigned the "processed" status.

Process step	Batch status	HYDRA batch number	Internal HYDRA batch number	Comment
	Free	BXC6GF7H	PR1111X113	The "new" batch is assigned the "free" status, the same HYDRA batch number (serial number) and a new HYDRA internal batch number
Status after entering the serial number for the OP 0030	Processed	BXC6GF7H	BXC6GF7H	The "old" batch is assigned the "processed" status.
	Processed	BXC6GF7H	PR1111X112	The "old" batch is assigned the "processed" status.
	Processed	BXC6GF7H	PR1111X113	The "old" batch is assigned the "processed" status.
	Free	BXC6GF7H	PR1111X114	The "new" batch is assigned the "free" status, the same HYDRA batch number (serial number) and a new HYDRA internal batch number

As regards a HYDRA batch number (serial number), there might exist 1-n HYDRA batches at different times, whereas only one of the batches may be "free" or "running" at a time.

Please note for the supply via interface

Merged batches and serial numbers can be transferred from other systems (e.g. ERP systems) to MES via the [material staging interface](#). Some particularities have to be taken into account when transferring the merged batch/combination of serial numbers:

- This order has to be observed when transferring serial numbers and their merged batches from Navision to HYDRA:
 1. Transfer of serial numbers
 2. Transfer of the merged batch record
- The below table shows how specific fields are assigned differently when being transferred to HYDRA.

Logical field	Interface field	Merged batch	Serial number
HYDRA internal batch number	HY_LOSNR	Merged batch number specified by ERP	Serial number specified by ERP
HYDRA batch number	DLL	Merged batch number specified by ERP	Serial number specified by ERP
PPS batch	PPS batch	Batch number specified by ERP	Batch number specified by ERP
Merged batch number	MCNR	BLANK	Merged batch number specified by ERP
Merged batch ID	SLOS	Fixed "J" (upper case "j")	BLANK
Merged batch type	SLOSTYP	Fixed "J" (upper case "j")	BLANK

Logical field	Interface field	Merged batch	Serial number
HU level	HULEVEL	1	0
Inventory type	BESTART	Fixed "H"	"S"
Number of individual batches	SLOS	Number of included serial numbers	0

Posting of serial numbers

If a merged batch with assigned serial numbers is finished, the system updates data and generates new documents in the form of log records and goods movements.

Transfer of data from the operation

This data is taken over from the running operation and transferred to the merged batch and the assigned serial numbers:

- Material and material name
- Material type

Transfer of entered data to serial numbers and the merged batch

The machine, producing operation, person performing the posting, material buffer and the transport unit are transferred to the merged batch and the serial numbers included in the merged batch.

Any comment that might be entered is transferred to the merged batch as additional information on the batch.

The status of the merged batch and serial numbers is set subject to the specified quality/class of the output batch:

Selected class	Assigned batch status	Reason accepted
Yield	Free	No
Scrap	Locked	Yes
Rework	Free	Yes

Selected class	Assigned batch status	Reason accepted
Open quantity	Free	Yes

Batch attributes collected in relation to the merged batch are transferred to the merged batch as batch attributes.

Batch attributes collected in relation to the serial number are transferred to serial numbers as batch attributes.

Document links collected in relation to the merged batch are transferred to the merged batch as document links.

Document links collected in relation to the serial number are transferred to serial numbers as document links.

Quantities of the merged batch

The generated merged batch includes the total quantities of the serial numbers assigned to the merged batch.

The number of assigned serial numbers is defined for the merged batch as number of individual batches.

Activities for serial numbers

New batches with a new, internal HYDRA batch number are generated for the serial numbers included in the merged batch (for further details on this procedure, please see section HYDRA batch number vs. serial number) and the preceding batches are assigned the "processed" status.

Generation of ADE log records

An H record is generated for the merged batch. This one includes the quantity(ies) of the merged batch, which in turn result(s) from the quantities of the assigned serial numbers. The batch number of the merged batch is stored at the H record.

Separate log records are not generated for serial numbers.

Generation of goods movements

The generation of goods movements can be controlled individually for consumptions as well as for the output material on the level of serial numbers and merged batches.

Consequently, it is possible to adjust the upload of these goods movements to the conditions of the relevant ERP system.

Details on how to configure goods movements can be found in the [configuration guidelines for serial number recording](#).

Tracing

Tracing information is updated for the merged batch and serial numbers in the system:

- The generated merged batch is connected with all integrated input batches logged on at the time of logging the merged batch on (by logging the OP on or by completing a merged batch beforehand) or that have been logged on since then.
- The serial numbers included in the merged batch are also connected with all integrated input batches logged on at the time of logging the merged batch on (by logging the OP on or by completing a merged batch beforehand) or that have been logged on since then.

[Establishing the connection](#) between serial numbers and input batches can be disabled if required.

9 Collection of Serial Numbers at AIP

Usage

You use the "enter serial number" dialog to enter and track serial numbers in the production process.

Prerequisite

Various configurations have to be made in the system to use the dialog. Further details can be found in the document dealing with the [Description of the configuration to use collection of serial numbers](#).

"Collection of serial numbers" dialog

The "enter serial number" dialog (A_SNR_A) consists of workflows providing extensive possibilities for data collection. By default, the dialog includes the following workflow steps:

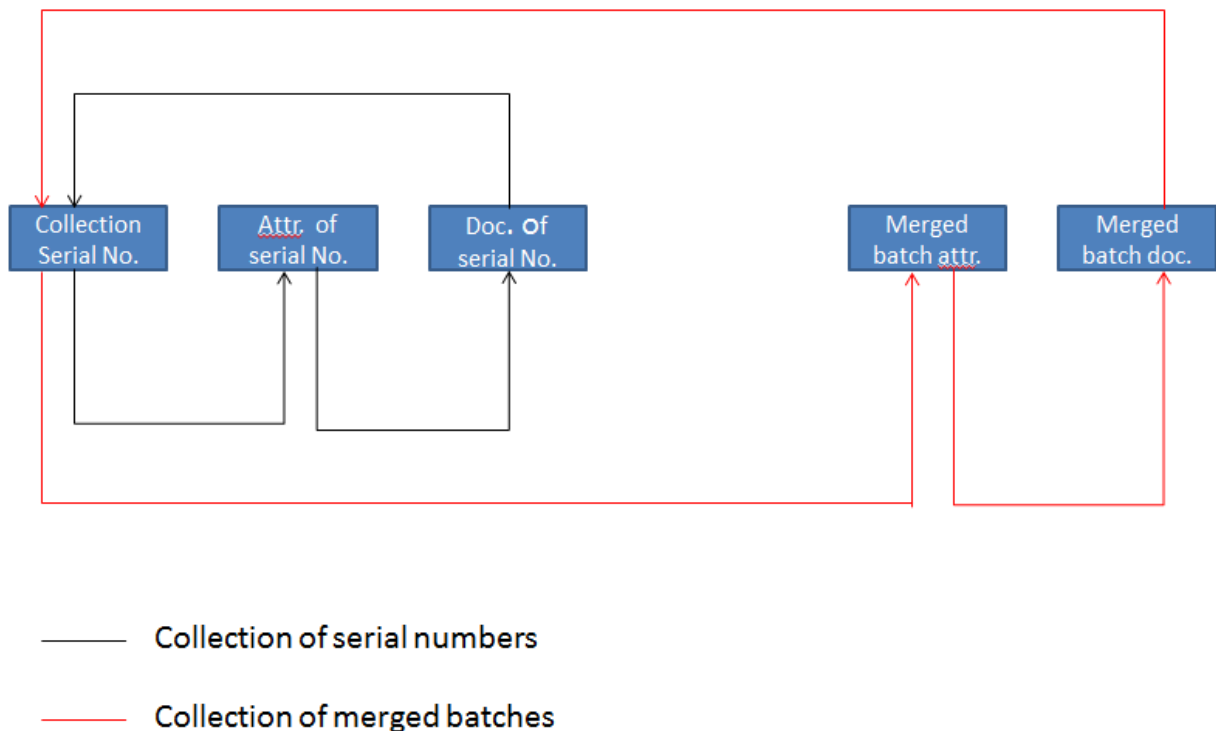
Workflow step/detailed dialog	Technical name	Usage
Collection of serial numbers	WF_A_SNR_A	This workflow step shows basic data of the serial number or the user can enter this data here.
Merged batch attributes	WF_SNR_ATTR	Entry of batch attributes for the merged batch
Merged batch documents	WF_SNR_DOC	Entry of document links for the merged batch
Serial number data	SNR-DATA	Detailed dialog to enter additional data specific to serial numbers. The detailed dialog can only be started for a single serial number.

Depending on the active workflow step, different function keys are provided.

Basically, data is collected in two steps:

- Data collection for serial numbers
Data is collected for serial numbers until a new merged batch can be completed.
Data collection restarts, once a merged batch has been completed.
- Data collection for merged batches
Basically, data is collected for the merged batch when the merged batch is completed; but it may also be started at an earlier point in time.

The following diagram illustrates the two acquisition cycles:



Workflow step: "collection of serial numbers"

The workflow step "collection of serial numbers" has been designed to enter data regarding one or several serial numbers.

The dialog shows the following data that cannot be modified:

Workplace

Current workplace as selected in the basic screen of the terminal.

Operation

Currently running operation as selected in the basic screen of the terminal.

Material

Material produced by the operation as laid down in the operation's article/item.

Merged batch

Merged batch number as defined by the system for the next merged batch to be completed.

The user may enter the following data:

Serial number

Subject to the type of serial number collection, the "serial number" field can be used to:

- enter serial numbers manually, instead of selecting them manually from the table of serial numbers
- scan serial numbers, instead of selecting them manually from the table of serial numbers
- If serial numbers have to be entered manually, they can be entered or scanned manually.

Staff badge number

Entry of the staff badge number is mandatory.

Quality

The user may define the quality. A reason has to be entered for the ratings "scrap", "open" and "rework".

Material buffer (*1)

The material buffer defined for the machine is suggested as the target material buffer.

Comment (*1)

A comment may be entered for the merged batch.

Transport unit (*1)

Suggests the transport unit that is identified as the default unit in the [assignment of material types to transport units](#) regarding the material type of the operation.

If nothing is assigned, "SYSTEM" is suggested.

Transfer (for each merged batch / all merged batches from now on)

For the fields labeled with (*1) it may be specified if they only apply for the merged batch that is currently being processed or for this and all future merged batches of this operation.

This definition applies until the operation is interrupted and/or logged off, even for different shifts.

The table of serial numbers shows the serial numbers of the registered input batch that have not yet been classified for this operation. The quality status (Q status) of serial numbers may be filtered, provided this status is set by quality data collection.

Single serial numbers can be selected by double clicking the table. Selected serial numbers are highlighted at the left-hand side of the table by an "X" with green background. There are also function keys for mass selections.

There is a number of function keys facilitating dialog handling, to input data for serial numbers or to complete a posting:

Designation	Usage
Close	Closes the dialog.

Designation	Usage
Complete	The currently generated merged batch is completed for the entered/assessed serial numbers. The merged batch number is displayed on the terminal. If the user confirms it, the dialog can be continued and remains opened to enter additional merged batches.
Select all	All serial numbers displayed in the list of serial numbers are selected.
Unselect all	All serial numbers displayed in the list of serial numbers are deselected.
Invert selection	Subject to the current state of selection, the system changes it: • Selected serial numbers are unselected • Unselected serial numbers are selected
Serial number data	Detailed dialog to enter data specific to serial numbers. The detailed dialog can only be started for a single serial number.
Next	Leads to the next workflow step

Detailed dialog "serial number data - batch attributes"

Batch attributes may be entered for the serial number in the "serial number attributes" workflow step of the "serial number data" detailed dialog.

The "attributes" function key allows entering attributes for a batch or for all serial numbers. The input dialog suggests all attributes that are assigned the "Capture attribute while generating batch" option in the [configuration of batch attributes](#) regarding the operation's material type.

The "go on" function key allows entering document links for serial numbers.



The list only shows the recorded batch attributes; batch attributes that might already be available are not shown.

Workflow step - merged batch attributes

Batch attributes may be entered for the serial number in the "merged batch attributes" workflow step.

The "attributes" function key allows entering attributes for the merged batch that is currently being processed or for this and all future merged batches of this operation. The input dialog suggests all attributes that are assigned the "Capture attribute while generating batch" option in the [configuration of batch attributes](#) regarding the operation's material type.

The "go on" function key allows entering document links for the serial number.



The list only shows the recorded document links; document links that might already be available are not shown.

10 Configuration for the Collection of Serial Numbers

Activation at machine/workplace

These configurations have to be made for the [machine / workplace](#) to enable the collection of serial numbers:

Parameter name	Value
Workplace configuration → Workplace master data → Workplace category	N Machine
Workplace configuration → Workplace master data → Workplace type	E Single workplace
Workplace configuration → MPL → Batch management	L Batch tracing (input/output batches)
Workplace configuration → MPL → Preceding material buffer	If required, define a preceding material buffer .
Workplace configuration → MPL → Subsequent material buffer	If required, define a subsequent material buffer .
Workplace configuration → MPL → Automat. generation of batch numbers	J Automatic generation of batch numbers for production batches (MPL) enabled

Maintain material types - for the operation

Maintain the [material types](#) to be defined for the operation and adapt them to your specific requirements of data collection.

Maintain material types – for components

Maintain the [material types](#) to be defined in the component list and adapt them to your specific requirements of data collection. Configure at least the following values for the component that is integrated as merged batch including assigned serial numbers in the operation:

Parameter name	Value

Parameter name	Value
Input batch processing → Inventory management	R - Yes, backflush (retrograde)
General → Options → transferred at interface	Enable this option if you expect consumption postings or final backflushes (notification of goods receipts) for this material type. Then another configuration can be used to control the transfer of merged batches and serial numbers separately.

Maintain reasons

Maintain [reasons for scrap and rework as well as reasons for open quantities](#), if you use these quality classes.

Maintain transport units

Create [transport units](#) in the system if you want to use them for data collection. Assign transport unit ans according material type.

Assign material types to transport units

If you want to use [transport units](#) and you created them in the system, you may assign them to the [material types](#) maintained in the system and define a default transport unit for each material type. This one will then be selected in advance in the input dialog.

Perform the assignment in the [assignment of TPU to material type](#).

Activation at the operation

These options have to be set for the operation.

Parameter name	Value
Batch management requirement	Yes
Serial number requirement	A Collection of serial numbers

These options are available to edit this information:

Manual maintenance in the operation

Edit the options manually for the [operation](#)

Maintenance of processing code (customizing)

Edit the options for the [processing code](#).

Maintenance of the template (customizing)

Edit the options of the [processing code](#) with value "A" - collection of serial numbers and assign the processing code to an [operation template](#).

Explicit specifications for the interface (recommended procedure)

Explicitly transfer the options for the operation at the interface.

Activation in components list

Maintain the input quantity in the [component list](#) by entering "1.0" units for components listed as merged batch including assigned serials numbers. Assign the relevant material types to the components.

Activation of the link between serial number and input batch

For tracing it is necessary to link the incorporated input batches with the produced serial numbers. To enable this processing, a configuration has to be enabled in [INI configuration](#). The entry itself is part of the default delivery and only needs to be enabled.

Parameter name	Value
INI name	MPL
Section	SERIALNUMBER
Key	CONNECT_SNR_CNR
Value	Y
Active	Enable the entry

Configure batch attributes for data collection when recording serial numbers

If you want to enter batch attributes manually when collecting serial numbers, create the [batch attributes](#) to be recorded in the system in relation to the [material type](#) of the operation. To do so, maintain at least these configurations:

Parameter name	Value
Options → Capture attribute while generating batch	Enable the option.
Options → Position	Specify the position - the system sorts the configured attributes in an ascending numeric order (bottom up).
Data type	Maintain the data type and length of the attribute to be recorded

Control the generation of goods movements

Define for incorporated merged batches and serials numbers as well as for produced merged batches and serial numbers if you require goods movements subject to uploads to be provided. To do so, configure the following settings in [advanced object configuration](#):



The goods movement option has to be enabled for the relevant material type to be able to use this configuration.

Parameter name	Value
Configuration for goods issues (consumptions)	
Object type	MPL
Object ID 1	SNR - serial number SAM - merged batch
Object ID 2	MATTYP
Object ID 3	Material type the entry applies for
Object ID 4	CMM_A
Parameter	CREATE_MOVEMENT
Parameter value	Y

Parameter name	Value
<i>Configuration for goods receipts (generated material)</i>	
Object type	MPL
Object ID 1	SNR - serial number SAM - merged batch
Object ID 2	MATTYP
Object ID 3	Material type the entry applies for
Object ID 4	CMM_E
Parameter	CREATE_MOVEMENT
Parameter value	Y

11 Merging Serial Numbers

Usage

Merging serial numbers allows for different material components listed by serial numbers to be combined specifically into one new component part. The serial number of the new component part may be the number of one of the incorporated serial numbers or assigned anew.

Process of merging serial numbers

Merging serial numbers is based on the management of serial numbers incorporated in [merged batches](#) registered as input batches. Incorporated serial numbers are assigned based on the serial numbers included in these merged batches.

This process also outputs a serial number. There are two strategies to get the serial number:

- An integrated serial number is continued keeping the ID
The indicator "superordinate serial number" at the component specifies which one of the incorporated components includes the serial number ID that is to be continued. These components and/or their serial numbers are designated as superordinate serial numbers.

For each operation only one component can be assigned the flag "superordinate serial number".

- A new serial number is assigned
If no component is identified as superordinate, the new serial number to be assigned can either be specified or the system assigns a number automatically.

If a merged batch with assigned serial numbers is finished, the system updates data and generates new documents in the form of log records and goods movements.

Posting of serial numbers

Transfer of data from the operation

This data is taken over from the running operation and transferred to the merged batch and the assigned serial numbers:

- Material and material name
- Material type

Transfer of entered data to serial numbers and the merged batch

The machine, producing operation, person performing the posting, material buffer and the transport unit are transferred to the merged batch and the serial numbers included in the merged batch.

Any comment that might be entered is transferred to the merged batch as additional information on the batch.

The status of the merged batch and serial numbers is set subject to the specified quality/class of the output batch:

Selected class	Assigned batch status	Reason accepted
Yield	Free	No
Scrap	Locked	Yes
Rework	Free	Yes
Open quantity	Free	Yes

Batch attributes collected in relation to the merged batch are transferred to the merged batch as batch attributes.

Batch attributes collected in relation to the serial number are transferred to serial numbers as batch attributes.

Document links collected in relation to the merged batch are transferred to the merged batch as document links.

Document links collected in relation to the serial number are transferred to serial numbers as document links.

Quantities of the merged batch

The generated merged batch includes the total quantities of the serial numbers assigned to the merged batch.

The number of assigned serial numbers is defined for the merged batch as number of individual batches.

Activities for serial numbers

New batches with a new, internal HYDRA batch number are generated for the serial numbers included in the merged batch (for further details on this procedure, please see section [HYDRA batch number vs. serial number](#)) and the preceding batches are assigned the "processed" status.

Generation of ADE log records

An H record is generated for the merged batch. This one includes the quantity(ies) of the merged batch, which in turn result(s) from the quantities of the assigned serial numbers. The batch number of the merged batch is stored at the H record.

Separate log records are not generated for serial numbers.

Generation of goods movements

The generation of goods movements can be controlled individually for consumptions as well as for the output material on the level of serial numbers and merged batches.

Consequently, it is possible to adjust the upload of these goods movements to the conditions of the relevant ERP system.

Details on how to configure goods movements can be found in [configuration instructions for the collection of serial numbers](#).

Tracing

Tracing information is updated for the merged batch and serial numbers in the system:

- The generated merged batch is connected with all integrated input batches logged on at the time of logging the merged batch on (by logging the OP on or by completing a merged batch beforehand) or that have been logged on since then.
- The serial numbers included in the merged batch are also connected with all integrated input batches logged on at the time of logging the merged batch on (by logging the OP on or by completing a merged batch beforehand) or that have been logged on since then.

[Establishing connections](#) between serial numbers and the input batch can be disabled if required.

12 Merging Serial Numbers on the AIP

Purpose

You use the dialog "Merge serial numbers" to group several materials that are listed by serial numbers. These materials are then listed under one number and can be used and planned more easily in the system. The system saves all serial numbers that are merged in one number.

Requirements

You must make different configurations in the system to use the dialog. For further details, refer to the document [Description of the configuration to use merged serial numbers](#).

Dialog *Merge serial numbers*

The "merge serial number" dialog (A_SNR_U) is a workflow dialog, which provides extensive possibilities for data collection. By default, the dialog includes the following workflow steps:

Workflow step / dialog	Technical name	Purpose
Merge serial numbers	WF_A_SNR_A	This workflow step shows basic data of the serial number or the user can enter this data here.
Merged batch attributes	WF_SNR_ATTR	Recording of batch attributes for the merged batch
Merged batch documents	WF_SNR_DOC	Recording of document links for the merged batch
Merge	A_SNR_MARRIED	Dialog to identify and assign the different ingoing serial numbers
Serial number data	SNR_DATA_MARRIED	Dialog to enter additional data specific to serial numbers. The dialog can only be started for a single serial number.

Depending on the active workflow step, different function keys are provided.

Basically, data is collected in two steps:

- Merge serial numbers

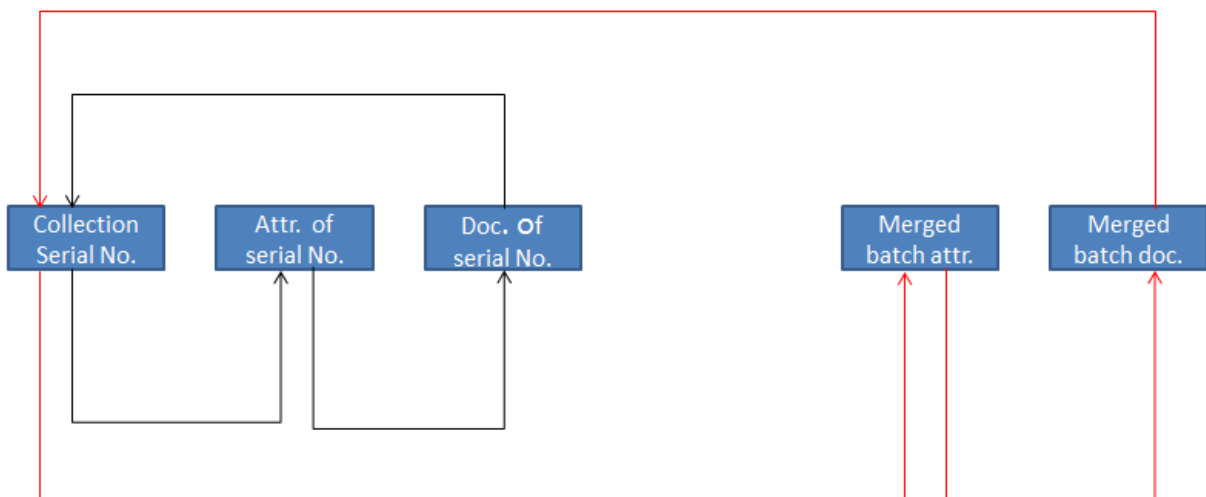
In the first step two or more serial numbers are merged. This process is completed and the "merge serial number" workflow step is opened again showing the superordinate serial number in the table.

Now either additional serial numbers can be added and merged or further data (batch attributes and document links) is recorded for an already merged serial number.

- Collection for merged batches

Basically, data is collected for the merged batch when the merged batch is completed; but data collection can also be started at an earlier point in time.

The following diagram illustrates the two cycles of data collection:



— Collection of serial numbers

— Collection of merged batches

Workflow step: merge serial number

You use the workflow step "Collection of serial numbers" to enter data for one or several serial numbers.

The dialog shows the following data that cannot be modified:

Workplace

Current workplace that is selected in the main view on the terminal.

Operation

Currently running operation that is selected in the main view on the terminal.

Material

Material produced by the operation that is specified for the article of the operation.

Merged batch

Number that the system assigns to the next merged batch that is set up.

The user can enter the following data:

Staff badge number

The input of the staff badge number is mandatory.

Quality

The user can specify the quality. You must enter a reason for the qualities "scrap", "open" and "rework".

Material buffer (*1)

The material buffer defined for the machine is suggested as the target material buffer.

Comment (*1)

You can enter a comment on the merged batch.

Transport unit (*1)

The system preassigns the transport unit that is identified as default unit for the material type of the operation in the [Assignment material type - transport unit](#).

If no assignment is available, "SYSTEM" is entered.

Transfer of data (for each merged batch / all merged batches from now on)

For the fields labeled with (*1), you can specify if the field specifications only apply for the merged batch that is currently being processed or for this and all future merged batches of this operation.

This definition applies until the operation is interrupted and/or logged off, also if the shift changes.

The serial number table shows the superordinate serial numbers that have already been merged and, therefore, are part of the currently running merged batch.

Single serial numbers can be selected in the table by clicking on the relevant row.

The button bar provides several function buttons. Using the buttons, you can record data for a serial number or complete a posting:

Designation (name)	Purpose

Designation (name)	Purpose
Close	Closes the dialog. Already merged serial numbers remain.
Complete	The currently generated merged batch is completed for the entered/merged serial numbers. The merged batch number is displayed on the terminal. If the user confirms it, the dialog remains open and the user can enter additional data for merged batches.
Merge	Dialog to select and assign serial numbers that are to be merged.
Serial number data	Dialog to enter data for a specific serial number. You can call the dialog for one specified serial number only. The number can also be the number of an already merged serial number from the grid of serial numbers.
Next	Continues with the next workflow step

Dialog "Merge"

In the dialog "Merge", the ingoing serial numbers (of the available merged batches) are identified, grouped and transferred to the superordinate serial number. You must call the dialog for each superordinate serial number and enter the data in the dialog.

You must enter the following data:

Superordinate serial number

A selection list opens where the available, superordinate serial numbers may be chosen. You can identify the superordinate serial number via the [identifier Superordinate SNR = Superordinate](#) in the component list.

Assigned serial number

A selection list opens where you can select the available, ingoing serial numbers. All serial numbers of the registered merged batches are offered that have not yet been assembled and that do not pertain to the superordinate serial number.

1 – n serial numbers can be selected and taken over.

Overview grid

The table shows the selected serial numbers that are to be merged. For each serial number it is obvious if it is a superordinate serial number or an ingoing serial number.

The button bar provides several function buttons:

Designation (name)	Purpose
Cancel	Exits the dialog without modifying, accepting or posting of data.
Accept	If the superordinate serial number or assigned serial number is entered manually in the input field, this function key takes it over in the overview grid.
Delete	The entire assignment is deleted. A new assignment process may be started directly after deletion.
Complete	Directly posts the assignment made.  The assignment is posted on the terminal and server, once it has been completed. Corrections can only be made via the Separating/Rebuilding Serial Numbers dialog.

Dialog "serial number data - batch attributes"

Batch attributes may be entered for the serial number in the "serial number attributes" workflow step of the "serial number data" dialog.

The "attributes" function key allows entering attributes for a batch or for all serial numbers. The input dialog lists all attributes where the option "Capture attribute while generating batch" is enabled for the material type in the [Configuration of the batch attributes](#).

You can use the function key "Continue" to record document links for the serial number.



The list only shows recorded batch attributes. Other available batch attributes are not shown.

Dialog "serial number data – document links"

You can specify document links for a serial number using the dialog "serial number data", workflow step "document links".

The user selects one of the following input options:

- For each serial number

If this option is enabled, the entered document links are only saved for this serial number. The entered document links are stored and suggested the next time a serial number is recorded.

- All serial numbers in this merged batch

If this option is enabled, the entered document links are saved for all serial numbers of this merged batch. The entered document links are stored and suggested the next time when the serial numbers for a merged batch are recorded. It is helpful to enable this option when you start to record document links. This way, all serial numbers in the merged batch include the document links and you need not specify the links for each serial number.

The list only shows the document links recorded in this process. If document links of the serial number are inherited from previous processes, these links are not shown.



You only use this dialog to record document links and not to show the links. You can display all document links that are available for a serial number or a batch/merged batch if you call the batch information (if configured accordingly in the document management).

You can only use this dialog to record document links (URL). Another function is currently not available.

Workflow step - merged batch attributes

You can use the workflow step "merged batch attributes" to record batch attributes for merged batches.

You can use the "attributes" function key to enter attributes for the merged batch that is currently being processed or for this and all future merged batches of this operation. The input dialog lists all attributes where the option "Capture attribute while generating batch" is enabled for the material type in the [Configuration of the batch attributes](#).

You can use the function key "Continue" to record document links for the merged batch.



The list only shows document links recorded, document links that might already be available are not shown.

Workflow step - merged batch documents

You can record document links for a merged batch in the workflow step "document links" of the dialog main view.

The user selects one of the following input options:

- For each merged batch

If this option is enabled, the entered document links are saved for this merged batch only. The entered document links are stored and suggested the next time a merged batch is recorded.

- All merged batches from now on

If this option is enabled, the entered document links are saved for the current merged batch. And the values are saved for all further merged batches (that are generated from now on)

The list only shows the document links recorded in this process. If document links are inherited from previous processes, these links are not shown.



You only use this dialog to record document links and not to show the links. You can display all document links that are available for a serial number or a batch/merged batch if you call the batch information (if configured accordingly in the document management).

You can only use this dialog to record document links (URL). Another function is currently not available.

13 Configuration of Merging Serial Numbers

Activation at machine/workplace

These configurations have to be made for the [machine / workplace](#) to enable the merge of serial numbers:

Parameter name	Value
Workplace configuration → Workplace master data → Workplace category	N Machine
Workplace configuration → Workplace master data → Workplace type	E Single workplace
Workplace configuration → MPL → Batch management	L Batch tracing (input/output batches)
Workplace configuration → MPL → Preceding material buffer	If required, define a preceding material buffer .
Workplace configuration → MPL → Subsequent material buffer	If required, define a subsequent material buffer .
Workplace configuration → MPL → Automat. generation of batch numbers	J Automatic generation of batch numbers for production batches (MPL) enabled

Maintain material types - for the operation

Maintain the [material types](#) to be defined for the operation and adapt them to your specific requirements of data collection.

Maintain material types – for components

Maintain the [material types](#) to be defined in the component list and adapt them to your specific requirements of data collection. Configure at least the following values for the component that is integrated as merged batch including assigned serial numbers in the operation:

Parameter name	Value

Parameter name	Value
Input batch processing → Inventory management	R - Yes, backflush (retrograde)
General → Options → transferred at interface	Enable this option if you expect consumption postings or final backflushes (notification of goods receipts) for this material type. Then another configuration can be used to control the transfer of merged batches and serial numbers separately.

Maintain reasons

Maintain [reasons for scrap and rework as well as reasons for open quantities](#), if you use these quality classes.

Maintain transport units

Create [transport units](#) in the system if you want to use them for data collection.

Assign material types to transport units

If you want to use [transport units](#) and you created them in the system, you may assign them to the [material types](#) maintained in the system and define a default transport unit for each material type. This one will then be selected in advance in the input dialog.

Perform the assignment in the [assignment of TPU to material type](#).

Activation at the operation

These options have to be set for the operation.

Parameter name	Value
Batch management requirement	Yes
Serial number requirement	U Merge serial numbers

These options are available to edit this information:

Manual maintenance for the operation

Edit the options manually for the [operation](#)

Maintenance of processing code (customizing)

Edit the options for the [processing code](#).

Maintenance of the template (customizing)

Edit the options of the [processing code](#) with value "U" – Merge serial numbers and assign the processing code to an [operation template](#).

Explicit specifications for the interface (recommended procedure)

Explicitly transfer the options for the operation at the interface.

Activation in components list

Maintain the input quantity in the [component list](#) by entering "1.0" units for components listed as merged batch including assigned serials numbers.

Verify whether or not the serial number of one of the incorporated components is to be continued. If this is the case, assign the "superordinate serial number" flag to this component.

Parameter name	Value
Component → Superordinate serial number	F Superordinate

Activation of the link between serial number and input batch

For tracing it is necessary to link the incorporated input batches with the produced serial numbers. To enable this processing, a configuration has to be enabled in [INI configuration](#). The entry itself is part of the default delivery and only needs to be enabled.

Parameter name	Value
INI name	MPL
Section	SERIALNUMBER
Key	CONNECT_SNR_CNR
Value	Y
Active	Enable the entry

Configure batch attributes for data collection when recording serial numbers

If you want to enter batch attributes manually when collecting serial numbers, create the [batch attributes](#) to be recorded in the system in relation to the [material type](#) of the operation. To do so, maintain at least these configurations:

Parameter name	Value
Options → Capture attribute while generating batch	Enable the option.
Options → Position	Specify the position - the system sorts the configured attributes in an ascending numeric order (bottom up).
Data type	Maintain the data type and length of the attribute to be recorded

Control the generation of goods movements

Define for incorporated merged batches and serials numbers as well as for produced merged batches and serial numbers if you require goods movements subject to uploads to be provided. To do so, configure the following settings in [advanced object configuration](#):



The goods movement option has to be enabled for the relevant material type to be able to use this configuration.

Parameter name	Value
Configuration for goods issues (consumptions)	
Object type	MPL
Object ID 1	SNR - serial number SAM - merged batch
Object ID 2	MATTYP

Parameter name	Value
Object ID 3	Material type the entry applies for
Object ID 4	CMM_A
Parameter	CREATE_MOVEMENT
Parameter value	Y
Configuration for goods receipts (generated material)	
Object type	MPL
Object ID 1	SNR - serial number SAM - merged batch
Object ID 2	MATTYP
Object ID 3	Material type the entry applies for
Object ID 4	CMM_E
Parameter	CREATE_MOVEMENT
Parameter value	Y

14 Separate/Rebuild Serial Numbers

Usage

This function enables material components listed by serial numbers and already mounted in a component to be exchanged specifically. Mounted material components are demounted and the new material components to be used are mounted. Mounting and demounting are documented in the system and recorded for the relevant serial number.

Procedure of separating/rebuilding serial numbers

In general, there might be two different situations requiring the exchange of components. At the moment, the system does not differentiate between them.

1. Components are demounted from a part while the process is running
An operation to "merge" serial number is running and it turns out that a serial number with a "damaged part" has been mounted. The serial number is recorded for an output batch/merged batch "as usual" and is ready to be demounted/rebuilt (batch status = free, class = yield, quality status = free/open).
2. Components are demounted in a subsequent or separate process step (e.g. rework)
The serial number was recorded for an output batch/merged batch and is ready to be demounted/rebuilt.
Specific identifiers (customer-specific) specify when a serial number gets to this process (e.g. batch status = blocked, class = rework/scrap, quality status = blocked). But by default these indicators are set: (batch status = free, class = yield, quality status = free/open).

In both cases, the function does not require operations or input components to be logged on. The dialog can be carried out at an independent workstation.

Posting of serial numbers

The below-mentioned postings have to be performed for the affected components listed by serial numbers, once they have been rebuilt:

Serial number of the complete, finished component part

This serial number is still assigned to the merged batch from which it was removed.

All relevant data (material number, material type, batch status, class, quality status) remains unchanged.

The connection to the demounted component is deleted. Now it is only included in the history. The connection to the newly mounted part is added accordingly.

Serial number data (attributes, document links) can be entered for the rebuilt part.

Serial number of the demounted component part

This serial number is no longer connected to the entire part from which it was demounted.

For traceability purposes, an entry is made in the batch history (indicating that this component was once mounted in this part).

The serial number of the removed component is not assigned to a new merged batch. It remains as single part (batch with quantity 1) on stock (class = scrap and batch status = blocked).

The merged batch which the demounted serial number derives from also remains unchanged (the inventory is not increased for this batch).

Serial number of the mounted component part

This serial number is taken from a provided merged batch the inventory of which is reduced accordingly.

As in this process no operation is logged on, no merged batch will be logged on as input batch. The inventory of the merged batch is reduced in the background after mounting.

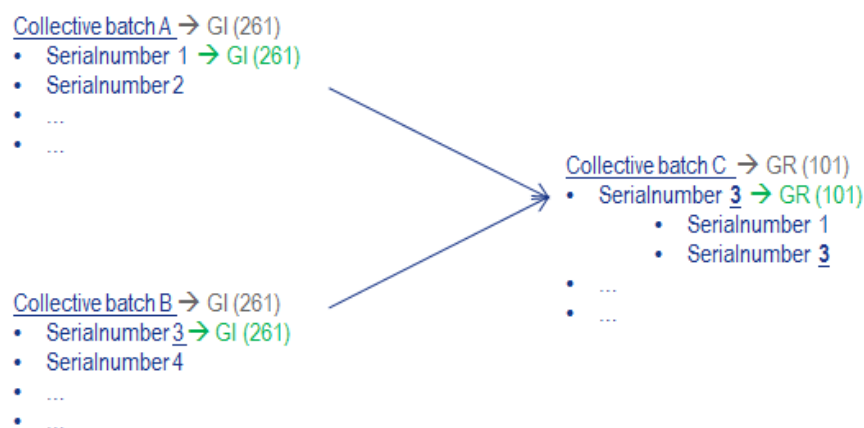
Batch data of the mounted component part is not changed. All relevant data (material number, material type, batch status, class, quality status) remains unchanged.

A connection to the consumed serial numbers/merged batches is established for traceability purposes.

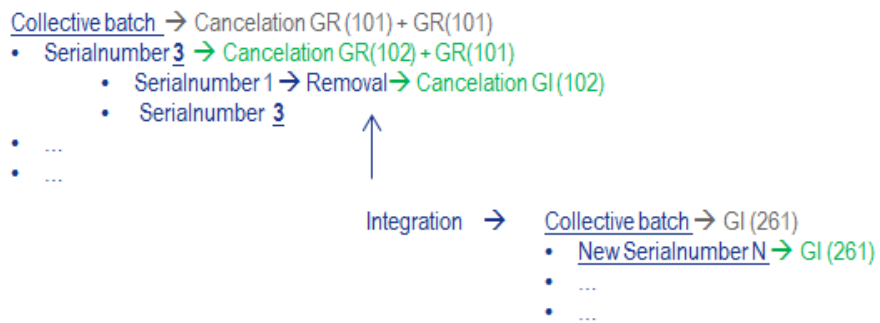
Generation of goods movements

These goods movements are performed for the mounted/demounted components, the complete component part and for the corresponding merged batches.

Goods movements when merging components listed by serial numbers:



Goods movements when separating components listed by serial numbers:



Subject to configuration, the generation of goods movements can be controlled individually for consumptions as well as for the output material on the level of serial numbers and merged batches. Consequently, it is possible to adjust the upload of these goods movements to the conditions of the relevant ERP system.

15 Separate/Rebuild Serial Numbers at AIP

Usage

The "separate/rebuild serial number" function is used to demount components listed by serial numbers from a component part and to mount a new component as a replacement during assembly processes. The serial number of the component part remains and is continued as a numeric value. All ingoing and outgoing serial numbers are to be identified.

Prerequisite

Various configurations have to be made in the system to use the dialog. Further details can be found in the document dealing with [description of the configuration to separate/rebuild serial numbers](#).

"Separate/rebuild serial numbers" dialog

The "separate serial number" dialog (A_SNR_D) consists of workflows providing extensive possibilities for data collection. By default, the dialog includes the following workflow steps:

Started by the workflow button "serial number data" → Dialog **SNR_DATA_D**

Including the dialogs:

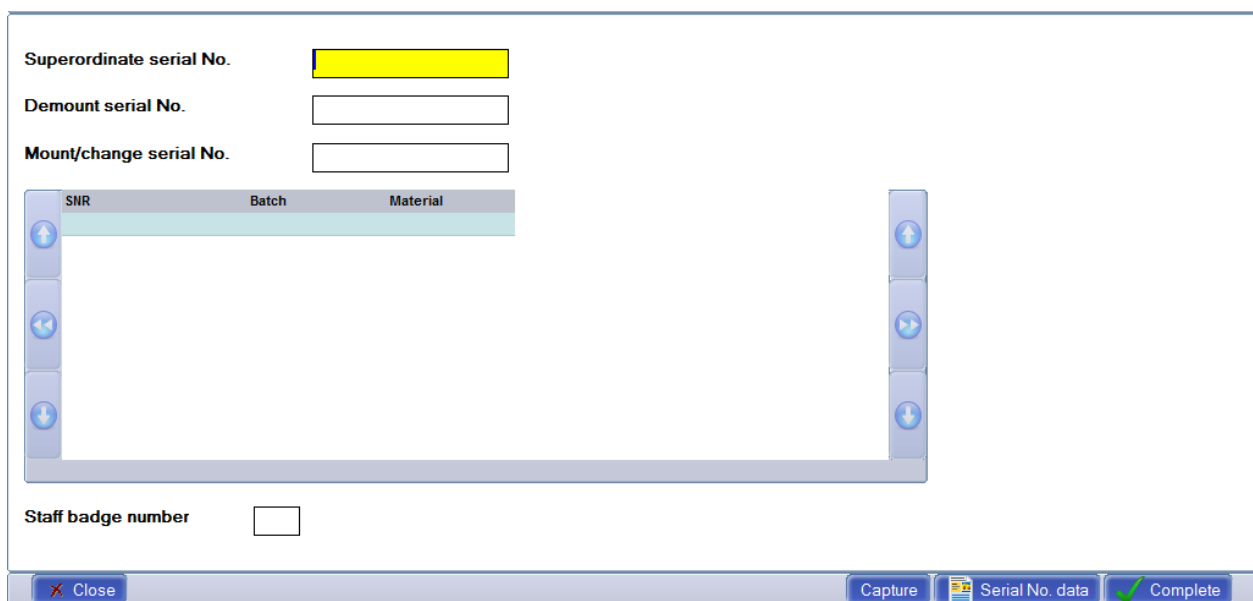
Workflow step/detailed dialog	Technical name	Usage
separate serial number	A_SNR_D	This workflow step shows basic data of the serial number or the user can enter this data here.
Serial number data	SNR_DATA_D	Detailed dialog to enter additional data specific to serial numbers. The detailed dialog can only be started for a single serial number.
Serial number data - attributes	WF_SNR_ATTR_D	Starts the input of attributes for a serial number
Serial number data - document links	WF_SNR_DOC_D	Starts the input of document links for a serial number

Depending on the active workflow step, different function keys are provided.

Basically, data is collected in two steps:

- **Separate/rebuild serial numbers**
At first demounting of a component and mounting of a replacement from/in a component part listed by serial number is documented. The reconstruction is recorded in the system, which leads back to the "separate serial number" workflow step.
The dialog may either be closed or additional data (batch attributes and document links) is entered for the component part listed by serial number.
- **Data collection for the serial number of the finished component part**
Basically, data is collected for the serial number when the serial number is completed.

"Separate/rebuild serial number" dialog



The following data is used in the dialog:

Superordinate serial number

The serial number of the component part from which a component is to be removed has to be entered (mandatory).

Demount serial number

The serial number of the component that is to be removed has to be entered (mandatory).

Mount serial number

The serial number of the component that is to be mounted has to be entered (mandatory).

Staff badge number

Entry of the staff badge number is mandatory.

The result list shows the components currently mounted in the serial number of the component part.

There is a number of function keys facilitating dialog handling, to input data for serial numbers or to complete a posting:

Designation	Usage
Close	Closes the dialog.
Complete	If the user confirms it, the dialog can be continued and remains opened to enter additional merged batches.
Serial number data	Detailed dialog to enter data specific to serial numbers. The detailed dialog can only be started for a single serial number and refers to already mounted serial numbers.

Procedure of the "separate/rebuild serial numbers" function

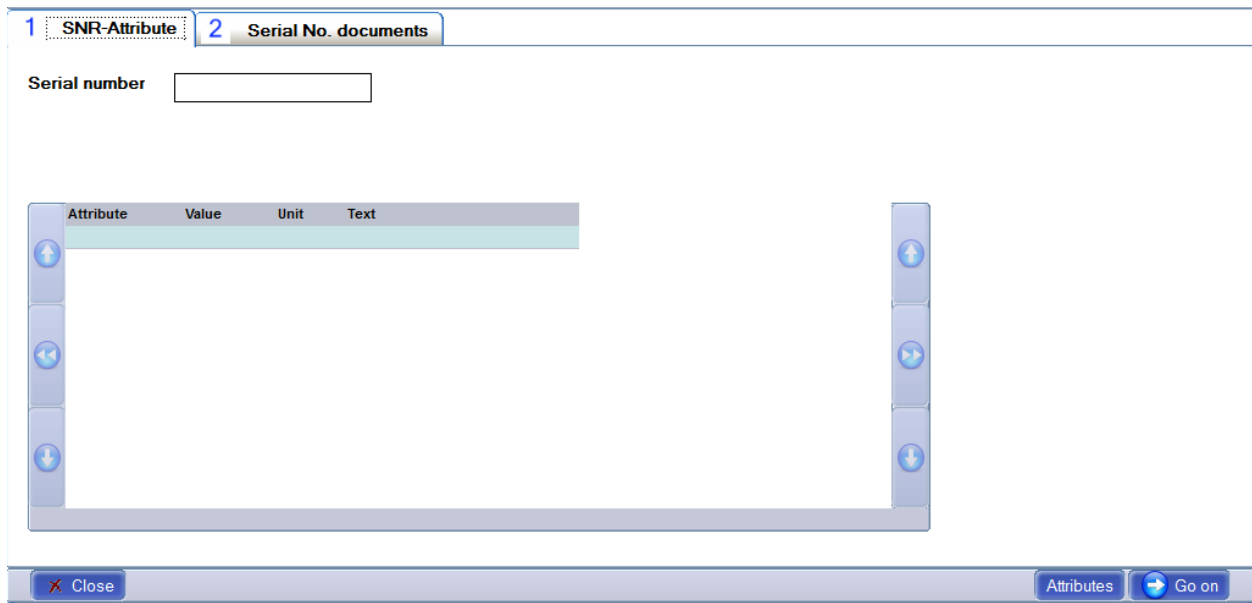
- The user opens the dialog "separate/rebuild serial numbers" by the function key in the basic terminal screen.
- The user enters the serial number (possibly by scanning) that is affected by separation/rebuilding.
- The result list shows all components mounted in this serial number (through the "merge serial numbers" function).
- The user enters the serial number of the component he/she wants to remove and/or takes it over (by double clicking).
- The user enters the new serial number he/she has mounted.
- Then the user presses the "capture" function key and the serial number is posted along with new component parts.
- The dialog remains opened until the user closes it explicitly. The entered "serial number" and result list remains and/or is still displayed.
- If several component parts need to be rebuilt, the dialog can still be used. Each exchanged component is entered separately.

Detailed dialog Serial number data - batch attributes

Batch attributes may be entered for the serial number in the "serial number attributes" workflow step of the "serial number data" detailed dialog.

The "attributes" function key allows entering attributes for a batch or for all serial numbers. The input dialog suggests all attributes that are assigned the "Capture attribute while generating batch" option in the [configuration of batch attributes](#) regarding the operation's material type.

The "go on" function key allows entering document links for serial numbers.



Detailed dialog "serial number data – document links (SNR_DATA_D)

The user may enter document links for each serial number (through [configuration](#)).

The procedure is as follows:

- The user presses the "serial number data" function key in the basic screen of the collection of serial numbers dialog.
- The user enters the document links. The entered/selected document links are displayed as assigned links in the result list.
- The user selects one of the following input options:
 - For each serial number

If this option is enabled, the entered document links are only saved for this serial number. The entered document links are stored and suggested the next time a serial number is recorded.
- Once the user has entered the document links for the serial number, he/she gets to the next workflow of the dialog or back to the basic screen by "go on".

Seriennummer trennen/umbuchen

1 SNR-Attribute
2 Serial No. documents

Serial number

Seriennummer-Dokumentenlink

Link/document

X Close

Add
Delete

↶ Back

✓ Complete

CTAIP V2.0.3.4 Server.scc7:4 TNR:131
04.07.14 07:53:58

16 Configuration for Separating / Reposting Serial Numbers

Dialog configuration

Define the function key "separate/repost serial number" by copying the layout *guil_anr.xml* to *guil_anr_ln.xml* and the key "BDE comments". Then the following entries must be changed in the configuration of the copied key:



If you do not use tiles for the AIP2, define the function key in the file **ctaipbut.ini**:

```
[ANR-LN-Page2]
...
1=A_SNR_D,L,SNR trennen, SNR_trennen.png
```

The below-mentioned sections are required in the layout configuration of **ctaiplay.ini**, irrespective of the used design:

```
[A_SNR_D_GRID.LST ]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite
GRID_ORDER=
```



```
DLL=C20,110,L,SNR
ALIAS LEER2=(DUMMY1)=C1,10,L
CNR=C20,0,L,CNR
ALIAS LEER3=(DUMMY2)=C1,40,L
SAPCNR=C20,110,L,Charge
ALIAS LEER4=(DUMMY3)=C1,10,L
ATK=C20,110,L,Material
```

```
[WF@DOC_DATA_D]
FILTER=
SECTION=DOC_DATA_D.LST
DATAFIELDS=
FILE=doc_data_d.lst
AUTOFILTERCOL=
MODE=DATALOCKUNTILSHOW=TRUE|
```

```
[DOC_DATA_D.LST]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite
GRID_ORDER=
```

```
LINK=C20,400,L,link/document
```

```
[WF@ATTR_DATA_D]
SECTION=ATTR_DATA_D.LST
DATAFIELDS=
FILE=attr_data_d.lst
AUTOFILTERCOL=
MODE=DATALOCKUNTILSHOW=TRUE|
```

```
[ATTR_DATA_D.LST]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite
GRID_ORDER=
```

```
ATTR=C20,100,L,Attribut
VALUE=C20,80,L,Wert
EINH=C20,60,L,Einheit
TEXT=C20,200,L,Text
```

System configuration

Configuration of batch attributes for data collection

If you want to enter batch attributes manually when collecting serial numbers, create the [Batch attributes](#) to be recorded in the system in relation to the [Material type](#) of the ready-mounted serial number. To do so, maintain at least these configurations:

Parameter name	Value
Options → Capture attribute while generating batch	Enable the option.
Options → Position	Specify the position - the system sorts the configured attributes in an ascending numeric order (top down).
Data type	Maintain the data type and length of the attribute to be recorded

Control the generation of goods movements

Define for incorporated merged batches, serials numbers and produced serial numbers if you require goods movements subject to uploads. To do so, configure the following in [Advanced object configuration](#):



The goods movement option must be enabled for the relevant material type in order to use this configuration.

Parameter name	Value
<i>Configuration for goods issues (consumptions)</i>	
Object type	MPL
Object ID 1	SNR - serial number SAM - merged batch
Object ID 2	MATTYP
Object ID 3	Material type the entry applies for
Object ID 4	CMM_A
Parameter	CREATE_MOVEMENT
Parameter value	Y
<i>Configuration for goods receipts (generated material)</i>	

Object type	MPL
Object ID 1	SNR - serial number SAM - merged batch
Object ID 2	MATTYP
Object ID 3	Material type the entry applies for
Object ID 4	CMM_E
Parameter	CREATE_MOVEMENT
Parameter value	Y

17 Variants of Batch Grouping

Summary

Batches are grouped and/or combined due to different situations:

- Not relevant to the process
It is assumed here that the identified material only switches positions and this movement is irrelevant to batch tracing (e.g. picking or transfer posting of batches to a group and provision for production).
- Relevant to the process
This function facilitates posting processes allowing to log on many input batches at once to a machine/OP, guide them through the process and to repost them afterwards to produced output batches for traceability purposes.

Variants

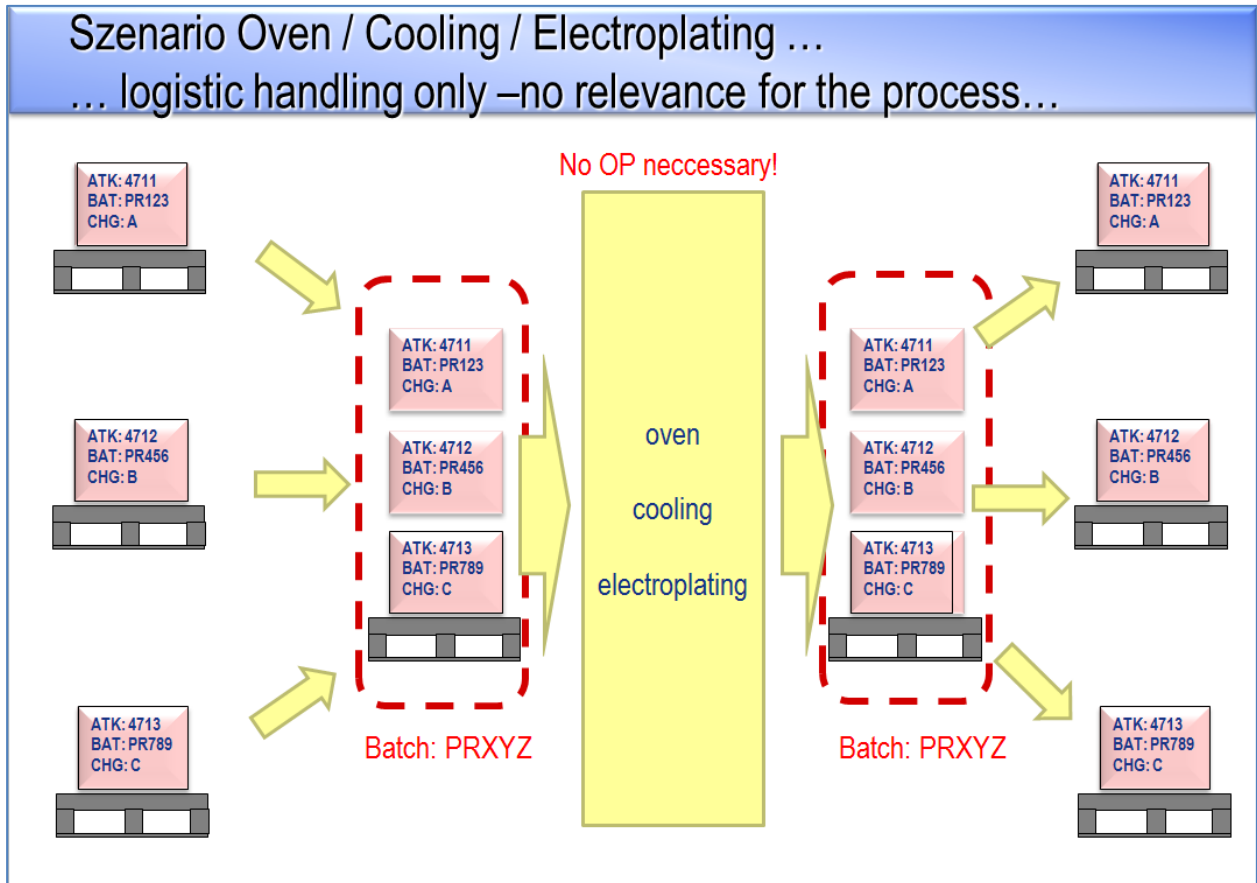
In general, there are the following two variants to group batches.

- Illustration: irrelevant to process --> generate/cancel batch group
- Illustration: relevant to process --> batch grouping

Generate/cancel group batch (not relevant to process)

General

The function is used every time when it is necessary to combine batches of different or identical materials, which is to be resolved/canceled at a later point in time providing the batches "unaffected".



Prerequisites, restrictions

- Original batches remain and are not "consumed" by grouping.
- During grouping, original batches cannot be processed or, for example, archived somewhere else.
- No OP is required - groupings may be performed at any point in time.
- Batches may still be removed unless the grouping has been created finally.
- A label can be printed for the generated grouping.
- Groupings may only be cancelled altogether. Individual batches cannot be removed.
- All group members are affected if the grouping is reposted (change of material buffer).
- Tracing does not consider the grouping.

AIP procedure

There are two steps:

- Generate group batch
- Cancel group batch

Results

The following results can be expected after generating the batch group:

- Batch characteristics
 - Individual batches still keep their material number, material type, material buffer, etc.
- Batch status
 - Once the "generate group batch" function has been executed, the single batches are in the "running" status. The group batch is assigned the "running" status.
- Goods movements
 - No goods movements are posted for the single batches or the group batch.
- Traceability
 - The group of batches is not relevant to tracing.
- History
 - The group of batches is documented in the history of individual batches.

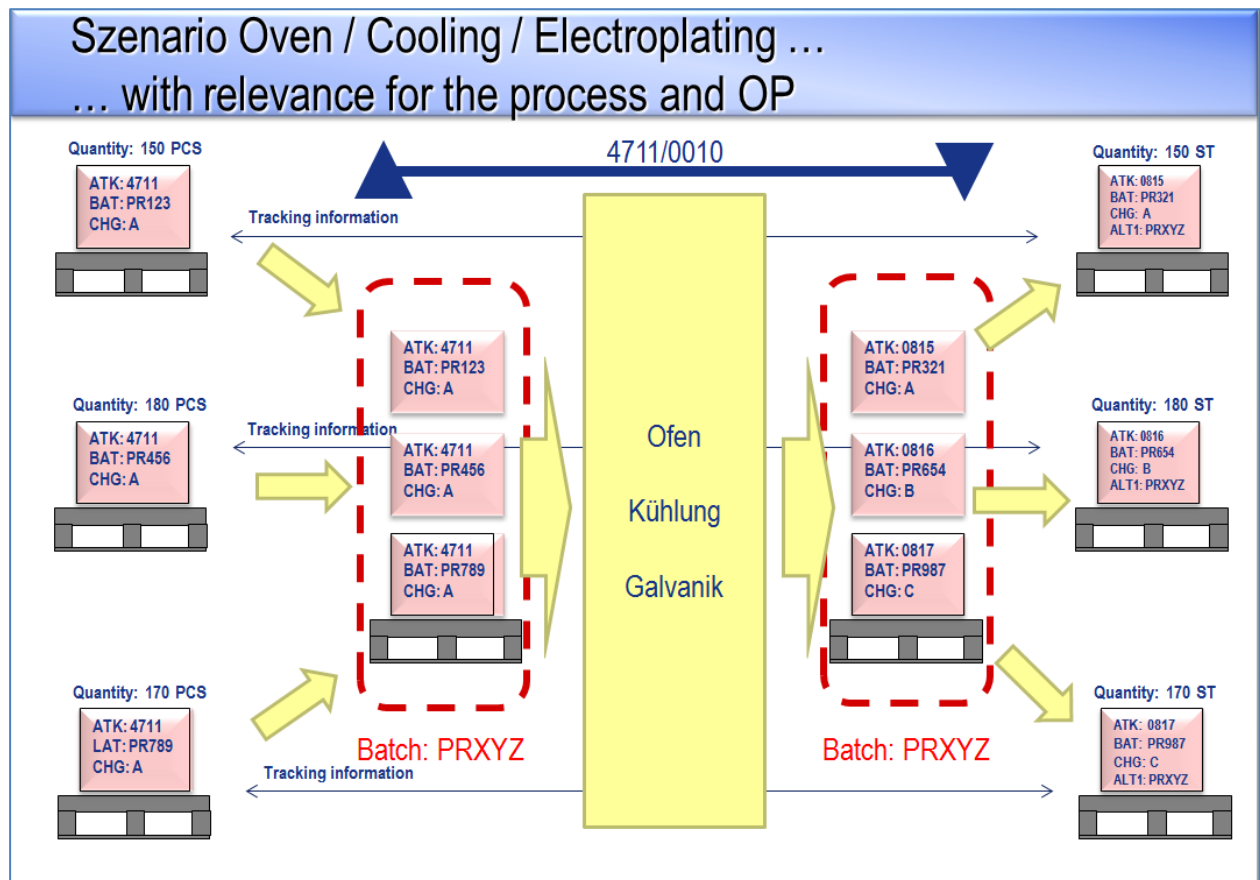
The following results can be expected after canceling the batch group:

- Batch characteristics
 - Individual batches still keep their material number, material type, material buffer, etc.
 - The group batch has taken over the following values from a "system batch" (copy template):
 - Material number
 - Material type
 - Material buffer
 - The quantity of the group batch is the total amount of all assigned individual batches and, therefore, is assigned the unit "PCE/pieces".
- Batch status
 - Once the "cancel group batch" function has been executed, the single batches are in the "free" status. The group batch is assigned the "processed" status. The "grouping" no longer exists.
- Goods movements
 - No goods movements are posted for the single batches or the group batch.
- Traceability
 - Cancelling group batches is not relevant to traceability.
- History
 - Cancelling the group batches is documented in the history of individual batches.

Batch grouping (relevant to process)

General

The function is used every time when batches of different or identical materials are used in a process and output batches are finally changed for all ingoing batches at a point in time.



Prerequisites, restrictions

- This function can only be used if grouping is directly performed "within the process". Groupings cannot be performed beforehand as part of a preceding picking process.
- The machine has not been configured as MPL machine.
- Input batches are "consumed" by grouping.
- During grouping, input batches cannot be processed or, for example, archived somewhere else.
- Operations are required: an OP appropriate for processing is searched via the order network or by reservations affecting all orders.

AIP procedure

The AIP procedure is described here.

Results/ processing

The "batch grouping" dialog (U_GROUPING) provides several functions.

- Add batch, start OP: CNRGRP.ADD
- Remove batch from list: CNRGRP.REMOVE:
- Complete group: CNRGRP.UNLOAD:

After executing the single functions, processing is performed in the dialog U_GROUPING.

Add batch/ start OP (CNRGRP.ADD)

This function adds the entered batch. To do so, the OP for the batch is logged on to the machine.

Inspections:

- General batch inspection:
 - The batch is available
 - The batch has the status "free"
- Automatic search for the operation matching the entered batch
 - OP is searched via the order network and batch assignment
 - The batch was generated in an OP, which is provided as piece of information for the batch. Due to the order network, the system knows the next OP to be processed within the order. The appropriate OP can be found by the order network.
 - If no OP is found via the network, it is searched using batch reservations.
 - An error message appears if no OP is found.
 - If an OP is found, it will only be started, unless it is already running at the machine.
- Further inspections:
 - Checking if the OP is planned for the machine and/or machine group.
 - Checking if the batch is reserved for another OP.
 - Checking if the batch can be logged on to the component.
 - An error message occurs if no component is found.
 - The component must be identical to the material number of the batch, otherwise an error message occurs.
- An error message appears if the admissible number of OPs running at the machine is exceeded.
- Start OP:

- The OP is started and set to the "running" status.
- The batch is reserved for the OP.

Remove batch (CNRGRP.REMOVE):

This function is used to cancel the reservation of the OP at the batch.

Inspections:

- The batch is available. Otherwise, an error message occurs.

Unload batch group (CNRGRP.UNLOAD):

This function is used to log on the reserved batches.

Inspections:

- Are there any running OPs including reserved batches at the machine?

Posting:

- Determination of all reserved batches
- Log output batch and input batch on.
 - At first generate a new batch number for the output batch
 - The new output batch is logged on with the reserved input batch.
- Log off output batch and input batch
 - The input batch is logged off
 - The output batch is logged off
 - The configured, alternative batch numbers and attributes are saved for the output batch.
 - The connection between input batch and output batch is saved for tracing purposes.
- This process is repeated for all batches included in the group.

18 Create/Cancel Group Batch

Summary

Within the production area, it is necessary to compress batches to a group with one ID and to provide this package/group within another area. It is assumed here that the identified material only switches positions and this movement is irrelevant to batch tracing (e.g. picking or transfer posting of batches in a group and provision for production or transport).

Usage

The function is used to combine batches of different or identical materials, which is to be resolved/canceled at a later point in time providing the batches "unaffected".

Prerequisite/configuration

The configuration for AIP is described [here](#).

The configuration for AIP2 is described [here](#).

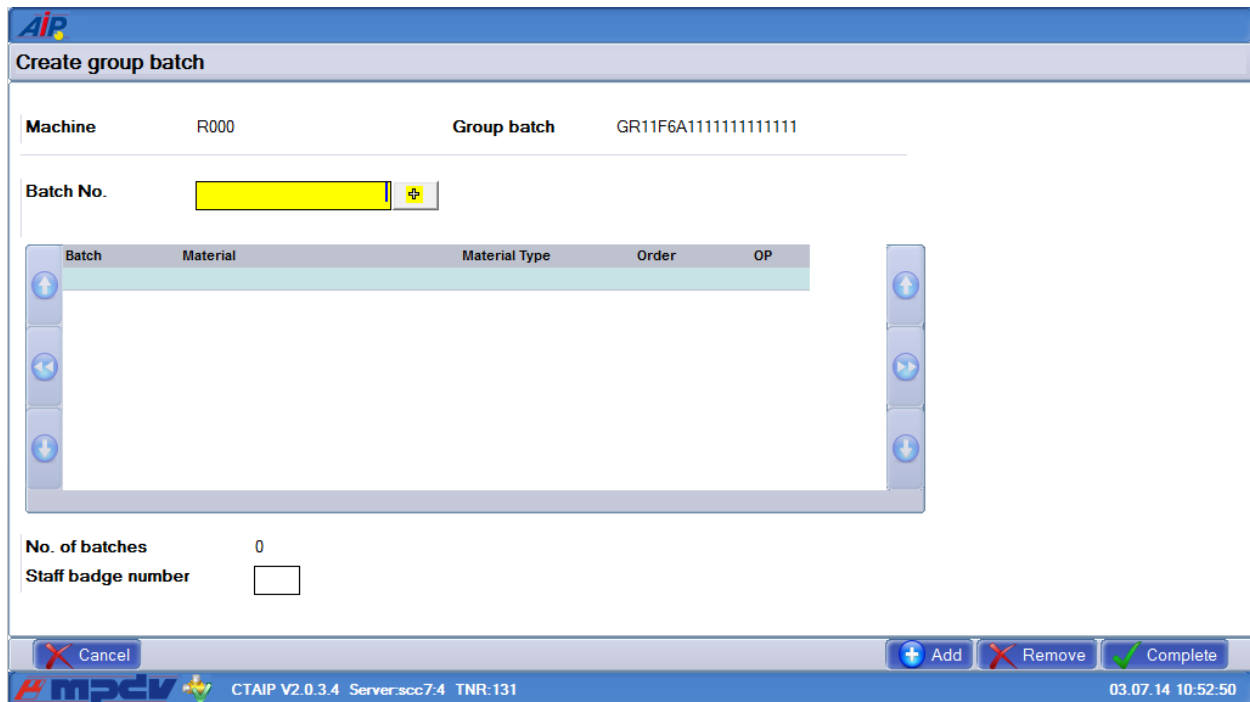
The logical process and posting are described [here](#).

Generate group batch



If the terminal is offline, only a limited number of posting functions is provided based on the available data. If the terminal is offline, errors are not displayed e.g. if posting failed on the server.

Dialog



Description of display fields:

- **Machine**
Used workplace/machine
- **Group batch**
Generated group batch that is currently being created.
- **Batch number**
Entered number of the batch that is to be recorded for the group batch.
- **Number of batches**
Number of individual batches included in the group batch.
- **Staff badge number**
The user's staff badge number.
- **Result list**
List of all individual batches included in the group batch with batch number, material number etc.

Function keys

- **Function key "cancel"**
The function key terminates the dialog and rejects data input.
- **Function key "add"**
An individual batch may be assigned to the group batch using this function key.
- **Function key "remove"**
A selected individual batch may be removed from the group batch using this function key.
- **Function key "complete"**

This function key completes the group batch, i.e. it is generated.

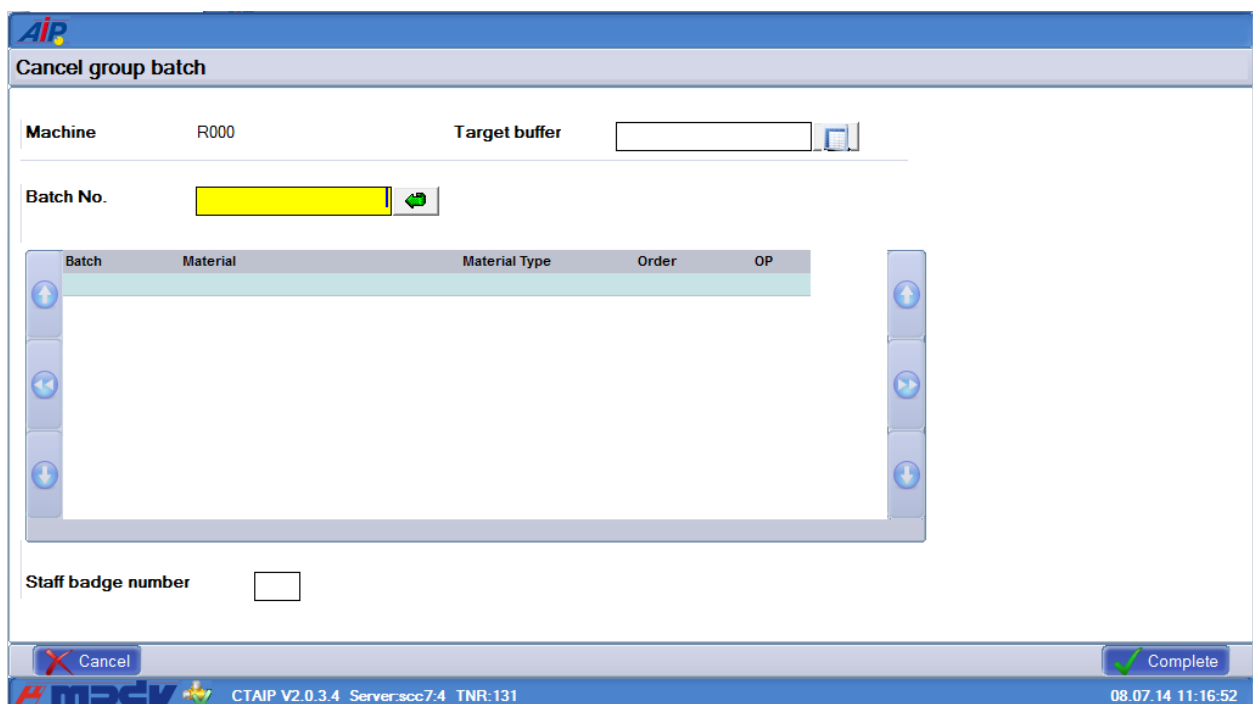
Procedure

For the user the procedure is as follows:

- The user opens the "create group batch" dialog in the basic screen of the terminal
- The user enters the batches (manually or by scanning) he/she wants to group
- The user may view the group batch number in the dialog
- The dialog's result list shows the entered batches that are to be grouped.
- Individual batches may still be removed or added, as long as the user has not confirmed the dialog by the "complete" key.

Cancel group batch

Dialog



Description of display fields:

- **Machine**
Used workplace/machine
- **Target buffer**
Material buffer to which the batch is transferred. If the field remains empty, no transfer posting takes place and the material buffer remains.
- **Batch number**

Entered number of the group batch that is to be canceled (re-posted).

- **Staff badge number**

The user's staff badge number.

- **Result list**

List of all individual batches included in the group batch with batch number, material number etc.

Function keys

- **Function key "cancel"**

The function key terminates the dialog and rejects data input.

- **Function key "complete"**

This function key cancels the group batch.

Procedure

For the user the procedure is as follows:

- The user opens the "cancel group batch" dialog in the basic screen of the terminal
- The user enters the group batch (manually or by scanning) he/she wants to cancel
- The dialog's result list shows the batches pertaining to the grouping.
- In the "material buffer" field the user may enter a new material buffer to which all individual batches of the group are to be re-posted.
 - If the user does not fill out the field, the material buffers of the individual batches remain as before.
- As long as the user has not closed the dialog by "cancel group", it remains open and/or can be cancelled. Data will be rejected.

19 Create / Cancel Group Batch

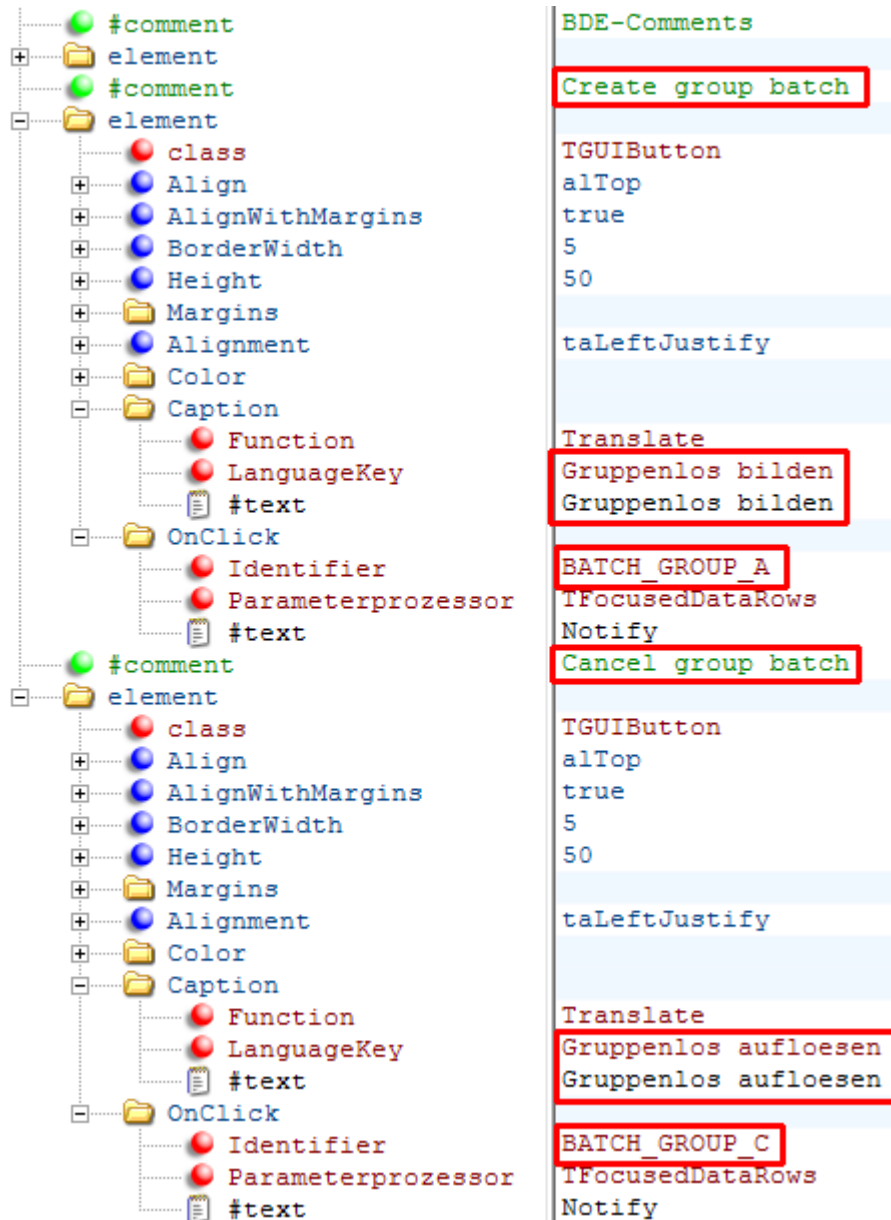
Usage

The function can be used to combine several batches into a group of batches at the AIP terminal and to cancel this grouping at a later point in time.

Dialog configuration

Dialog configuration (CNR.BATCH GROUP A)

Define the function keys "generate group of batches" and "cancel group of batches" by copying the layout *gui\l_anr.xml* to *gui\l_anr_ln.xml* and by copying twice the key "BDE comments". Then the following entries must be changed in the configuration of the copied keys:



If you do not use tiles for the AIP2, define the function keys in the file **ctaipbut.ini**:

[ANR-LN-Page4]

...

1=BATCH_GROUP_A,L,Gruppenlos bilden,Losgruppe_bilden.png

2=BATCH_GROUP_C,L,Gruppenlos auflösen,Losgruppe_aufloesen.png

The sections [BATCH_GROUP_A.LST] and [BATCH_GROUP_C.LST] are required in the layout configuration of *ctaiplay.ini*, irrespective of the used design:

```
[BATCH_GROUP_A.LST]

GRID_FONT=Arial

GRID_FONTSIZE=9

GRID_COLOR=clBlack

GRID_BACKGROUND=clWhite


CNR=C20,100,L,Batch

ATK=C20,100,L,Material

ATKBEZ=C40,140,Designation

HZTYP=C40,150,L,Material Type

RESAUNR=C12,100,L,Order

RESAGNR=C4,50,L,OP

; for set then position

;U_POSITION=C8,60,R,Position
```

```
[BATCH_GROUP_C.LST]

GRID_FONT=Arial

GRID_FONTSIZE=9

GRID_COLOR=clBlack

GRID_BACKGROUND=clWhite


CNR=C20,100,L,Batch

ATK=C20,100,L,Material

ATKBEZ=C40,140,Designation

HZTYP=C40,150,L,Material Type

RESAUNR=C12,100,L,Order

RESAGNR=C4,50,L,OP

; for set then position

;U_POSITION=C8,60,R,Position
```


System configuration

Further system configurations are not required in order to group/cancel batches.

20 Batch Grouping

Summary

In production it might be necessary to process batches at once in one process (e.g. heating furnace, conditioning system, washing machine). To this end, the user logs on the affected batches within a group to AIP.

Usage

To the user the OP is not important to record times and after processing, users do not want to post every single output batch in the system. Therefore, this function basically facilitates the posting process for the user.

Prerequisite/configuration

The configuration is described [here](#).

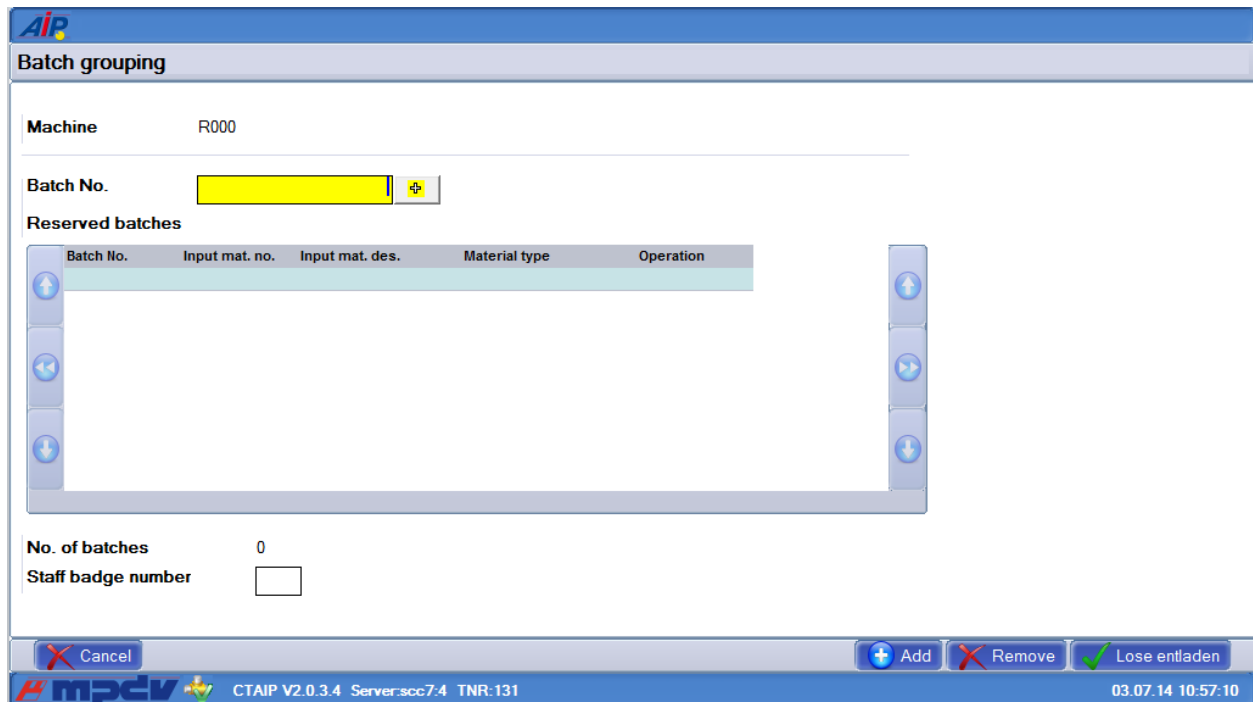
The logical process and posting are described [here](#).

Batch grouping



If the terminal is offline, only a limited number of posting functions is provided based on the available data. If the terminal is offline, errors are not displayed e.g. if posting failed on the server.

Dialog



Description of display fields:

- **Machine**
Used workplace/machine
- **Batch number**
Entered number of the batch that is to be recorded for the group.
- **Staff badge number**
The user's staff badge number.
- **Result list "reserved batches"**
List of all individual batches included in the group with batch number, material number etc.

Function keys

- **Function key "cancel"**
This function key terminates the dialog and rejects data input.
- **Function key "add"**
An individual batch may be added to the group using this function key.
- **Function key "remove"**
A selected individual batch may be removed from the group using this function key.
- **Function key "unload batch"**
This function key cancels the group and output batches are changed automatically for all input batches and operations registered in the background.

Procedure

For the user the procedure is as follows:

- The user opens the "batch grouping" dialog in the basic screen of the terminal
- The user enters the batch (manually/by scanning) he/she wants to add
- The user presses "add" and confirms the entered batch.
- The entered batch is assigned to the group. The batch is reserved for the OP.
- The user enters further batches he/she adds to the group.
- The user closes the dialog unless he/she wants to add further batches.
- To cancel the group and/or when completing the process, the user opens the dialog and presses the "unload batch" key. Consequently, the relevant registered operation is logged off and finished for every included input batch and an output batch is completed.

21 Grouping of Batches

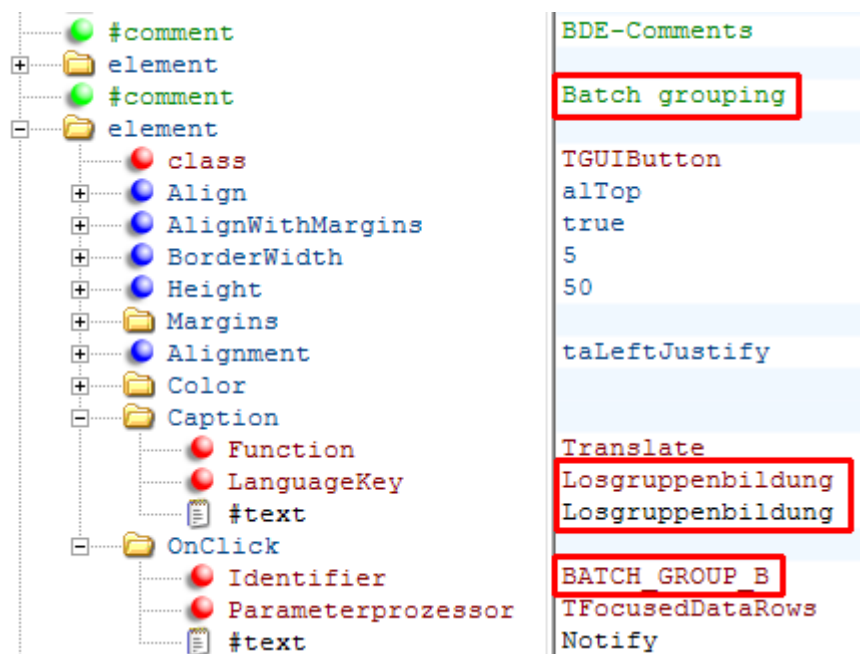
Usage

The function is used to apply several batches of a group in AIP processes.

Dialog configuration

Dialog configuration (BATCH GROUP B)

Define the function key "generate batch group" by copying the layout *gui\l_anr.xml* to *gui\l_anr_in.xml* and the key "BDE comments". Then the following entries must be changed in the configuration of the copied key:



If you do not use tiles for the AIP2, define the function key in the file *ctaipbut.ini*:

```
[ANR-LN-Page4]
```

```
...
```

```
3=BATCH_GROUP_B,L,Losgruppenbildung,Losgruppierung.png
```

The below-mentioned section is required in the layout configuration of *ctaipplay.ini*, irrespective of the used design:

```
[BATCH_GROUP_B.LST]
```

```
GRID_FONT=Arial
```

```
GRID_FONTSIZE=9
```

```
GRID_COLOR=clBlack
```

```
GRID_BACKGROUND=clWhite
```

```
CNR=C20,100,L,
```

```
ATK=C20,100,L,
```

```
ATKBEZ=C40,140,
```

```
HZTYP=C40,150,L,
```

```
RESANR=C12,100,L,Operation
```

System configuration

Further system configurations are not required in order to group batches.

An appropriate OP must exist in the system to be able to use the batches in the dialog and in order for an OP to be logged on in the background.



The relevant OP can be identified via:

- a special batch reservation (across all orders)
- the order network (sequence of OPs in an order)

22 Decision on Changing Input Batch

Usage

For traceability reasons, every time an input batch is changed, a new merged batch/output batch is to be produced for material listed by serial numbers.

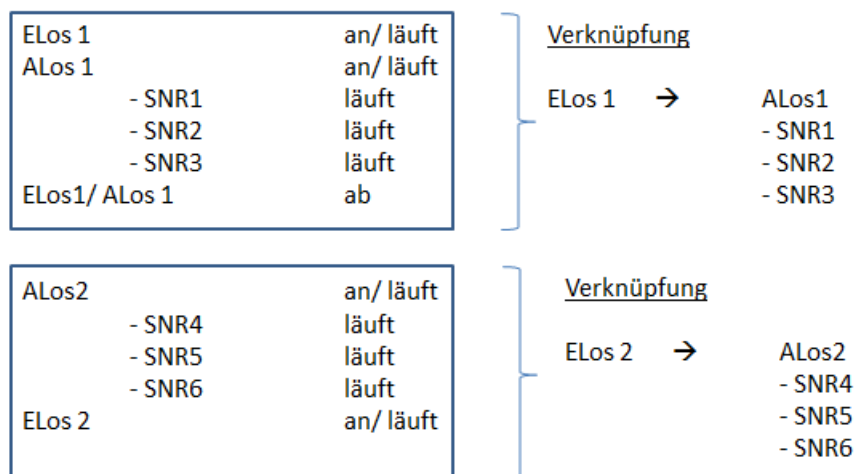
The function "decision on changing the input batch" is used every time when input batches are changed and it has to be ensured that the reported serial numbers deriving from different input batches are not mixed up in one output batch.

Example:

Two input batches from different lots are used for one material. Input batches are changed, once the first lot of the input batch has been consumed. The material/input batch of the first lot is logged off and the material of the second lot is logged on as input batch. There must be at least two different merged batches/output batches, as the serial numbers of two different input lots or input batches are assembled.

The "decision on changing the input batch" function and the relevant posting procedure for input batch changes make sure the right connections are generated between input and output batches and the user is forced to "pay attention" and to post all serial numbers before logging an input batch off.

Example:



Prerequisite/configuration

The warning message can be enabled and/or disabled. The configuration is described [here](#).

Results/inspections

The user is forced to make a decision when changing input batches:

- Log input batch off --> No
- Log input batch off --> Yes

The input batch is logged off by the system if the user decides to log off the input batch.

The dialog is executed and the input batch to be logged off is posted accordingly:

- Change the batch status (processed, free, blocked)
- Material movement (consumption --> goods issue (261))
- Connections between the output batch (merged batches and subordinate batches) and the input batch are completed in batch tracing. The system has already established a connection between the input batch to be logged off and the currently running output batch when the output batch was generated.

If this connection is not required, an option of the material type "output batch connection = J/N" makes sure the connection is deleted.

- The batch history is updated.

The user makes an entry (staff badge number) for the logged off input batch and the logged in input batch in the batch history. Consequently, one can trace back who decided to log on a new input batch and to ignore the serial numbers mounted with the old input batch. These serial numbers are only posted with the new input batch, although they have not been produced by it.



This function can be applied in the "merge serial numbers" dialog and to "change output batches". The text has to be adjusted specific to the project.

23 Decision on Changing Input Batch

Summary

For traceability reasons, every time an input batch is changed, a new merged batch/output batch is to be produced for material kept with serial numbers. The function "Decision on changing the input batch" is provided for this purpose.

Usage

This function is used when input batches are changed and it has to be ensured that the reported serial numbers deriving from different input batches are not mixed up in one output batch.

Decision on changing the input batch - AIP warning



If the terminal is offline, only a limited number of posting functions is provided based on the available data. If the terminal is offline, errors are not displayed e.g. if posting failed on the server.

Dialog

Step 1: Change input batch

POS Input mat. no.	Input mat. des.	Custom. batch no.	Remainde Unit
1 453-9970	453-9970	SAM100	5 ST

Step 2: Decision on changing the input batch

The user receives a message at the terminal asking whether the batch to be changed is actually to be logged off or if there are still remaining serial numbers to be entered. The dialog text can be customized.

Procedure

For the user the procedure is as follows:

- The user opens the "input batch change" dialog by pressing "input batch change" in the basic screen of the terminal.
- Step 1: Change input batch
 - The user selects the input batch to be logged off and/or enters the new input batch to be logged on and uses the "post batch" key.
- Step 2: Decision on changing the input batch

The user receives a message at the terminal asking whether the batch to be changed is actually to be logged off or if there are still remaining serial numbers to be entered.
- Step 3: User decision
 - The user wants to enter further serial numbers for the running input batch and clicks "No". The user has to enter his/her staff badge number.
 - The user decides to confirm the message by clicking "yes". Consequently, the input batch is logged off.

24 Configuration of Decision on Changing Input Batch

Usage

The function "decision on changing the input batch" is used when input batches are changed and it has to be ensured that the reported serial numbers deriving from different input batches are not mixed up in one output batch.

For this reason, the "decision on changing the input batch" alert may be configured as an intermediate dialog when changing input batches.

System configuration

Material type:

These options have to be set for the material type:

- Input batch processing > decision on changing the input batch = ON
If this option is enabled, the alert is shown in the "decision on changing the input batch" dialog.
- Output batch processing > delete batch assignment = ON
Using this option deletes the most recent assignment between input batch and current output batch. This deletion is necessary as the assignment is established immediately after the input batch has been logged on.

25 Splitting Batches

Summary

The "split batch" function can be used to divide an existing batch into several individual batches. The following "batch types" may be split:

- "simple" batch (split off quantities with new batch numbers)
- merged batch (split off subordinate batches/serial numbers into new merged batches)

Prerequisite

In general, all batches to be split off have to be available.

The "split batch" function can only be applied to batches meeting the below-mentioned conditions:

- only batches in the batch status "free" may be split
- only batches assigned the batch class "yield" may be split

Variants

In general, there are the following two variants to split batches.

- [Splitting batches at AIP](#)
AIP provides a function to split the entered batches/merged batches into several batches/merged batches. A label may be printed each for the batches/merged batches split off.
- [Splitting batches at MOC](#)
The MOC batch data overview provides a function to split the entered batches/merged batches into several new batches/merged batches.

Result

The below-mentioned results can be expected after splitting batches at AIP or MOC:

Splitting "simple" batches:

- the batches split off include the entered quantity
- the batches split off are assigned the status "free"
- If no remaining quantity is available, the old batch is in the "free" or "processed" status.
- The old batch has the batch class "yield"

- the batch number of the old batch:
 - is a new batch number
 - is the original batch number

Splitting merged batches:

- the merged batches split off include the selected subordinate batches/serial numbers and the relevant quantity.
- the merged batches split off are assigned the status "free"
- If no remaining quantity is available, the old merged batch is in the "free" or "processed" status.
- the old merged batch is assigned the "yield" batch class and includes all subordinate batches/serial numbers that have not been split off and/or no subordinate batches/serial numbers if all of them were split off.
- the batch number of the old merged batch:
 - is a new batch number
 - is the original batch number

26 AIP Batch Splitting

Summary

Some production processes (e.g. goods issue) require a batch (e.g. with large quantities) to be split into several batches (e.g. with smaller quantities) to be able to provide materials in smaller bins/containers at machines/workplaces. With this function the terminal provides an opportunity to separate batches and to print a label for the new batches.

Usage

These split variants are possible:

- "Simple" batches
When splitting "simple" batches, the user separates the required quantities from the original batch and generates new batches for these split quantities.
- Merged batches
When splitting merged batches, the dialog shows the individual sub-batches/serial numbers assigned to the merged batch. The user selects the sub-batches/serial numbers he/she wants to split off to generate a new merged batch.

Prerequisite

The relevant configuration has to be made to be able to use this function.

Terminal functions

The function is triggered manually by using the key of the "split batch" dialog (BATCH_SPLIT).



If the terminal is offline, only a limited number of posting functions is provided based on the available data. If the terminal is offline, errors are not displayed e.g. if posting failed on the server.

Dialog/start

The dialog opens by input of the batch number/merged batch number. The batch information on the batch/merged batch is shown.

Batch information	
Batch	1902FASS09
Material	4711
Designation	22402
Type	MAT
Class	G
Operation	C140000510000006
Machine	4202_00
Material buffer	MELT_OUT
Remaining qty.	10,0 KG
Prod.date	10.12.2013 14:54:26
Status	F
Note	
Merged batch	N

Cancel
 Splitten

Splitting "simple" batches

AIP

Split batch

Batch No. 1001234
Class G
Remaining qty. 61,0 ST

Quantity ST
Reason

☒ Yield
☐ Scrap
☐ Rework
☐ Open quantity

Quantity	Class	Reason
10,000	G	0
10,000	A	1
5,000	H	0
4,000	O	0
10,000	G	0

☐ Repost remaining quantity to new batch
☒ Reduce batch to remaining quantity

Cancel
 Add
 Remove
 Split

CTAIP V2.0.3.4 Server:scg7:4 TNR:131
03.07.14 11:40:58

Description of display fields:

- **Batch number**
Entered number of the batch that is to be split.
- **Class**
Batch class of the batch to be split.
- **Remaining quantity**
Remaining quantity of the batch to be split. At the beginning, it is the original quantity.

- **Quantity**
Quantity to be split off. The user enters this quantity.
- **Class**
Batch class of the quantity to be split off. This can be:
 - Yield
 - Scrap
 - Rework
 - Open quantity
- **Reason**
Reasons for the selected batch class.
- **Repost remaining quantity to new batch**
The remaining quantity of the original batch is reposted to a new batch.
- **Reduce batch to the remaining quantity**
The original batch is reduced to the remaining quantity.
- **Result list**
The result list shows all batches split off including the respective quantity, batch class and reason.

Function keys

- **Function key "add"**
The result list shows the entered quantity as separate entry (later the batch split off).
- **Function key "remove"**
The concerned row of the result list is selected and can be removed. The envisaged quantity is not split off and assigned again to the original batch.
- **Function key "split"**
When using the function key, all entries of the result list are split off including the relevant quantity from the entered batch.
- **Function key "cancel"**
This function key deletes all entries and the dialog is closed.

Procedure:

- The user opens the "split batch" dialog in the basic screen of the terminal
- The user selects the batch he/she wants to split off.
- Batch data of the entered batch is displayed.
- The user enters the split quantity, batch class and the reason, if necessary.
- The "add" function key adds the quantity and/or batch to be split off to the result list and displays it.
- The user may split off additional batch quantities

- The result list shows all quantities/batches to be split off
- The user decides if the remaining quantity of the original batch is to be reposted to a new batch or if it is to remain with the original batch.
- The user closes the dialog by the "split" function key and the split quantities will be posted.

Splitting merged batches

Description of display fields:

- **Batch number**
Entered number of the merged batch that is to be split.
- **Class**
Batch class of the merged batch to be split.
- **Remaining quantity**
Remaining quantity of the merged batch to be split. At the beginning, it is the original quantity.
- **Class**
Batch class of the merged batches to be split off. This can be:
 - Yield
 - Scrap
 - Rework
 - Open quantity
- **Reason**
Reasons for the selected batch class.
- **Display list**
The display list shows all sub-batches/serial numbers including the relevant quantity and article number assigned to the entered merged batch. Individual sub-batches/serial numbers are selected and split off into a new merged batch.

Function keys

- **Function key "reload"**
The display list is updated.
- **Function key "reverse"**
The selected sub-batches/serial numbers are unselected.
- **Function key "split"**
The selected sub-batches/serial numbers are assigned to a new merged batch and removed from the existing merged batch.
- **Function key "cancel"**

The dialog is closed by this function key.

Procedure:

- The user opens the "split batch" dialog in the basic screen of the terminal.
- The user selects the batch he/she wants to split off.
- Batch data of the entered batch is displayed:
 - Material
 - Quantity
 - Class
 - Reason
 - Batch status
- If it is a "merged batch" (indicator at the batch), the result list shows the serial numbers/subordinate batches assigned to this merged batch.
- The user chooses (multiple selection) the serial numbers/sub-batches he/she wants to split off into a new merged batch.
- The user selects the class for the new merged batch (and thus for the newly assigned serial numbers/sub-batches)
- The function key "split" generates the new merged batch with the split off serial numbers. The list of serial numbers is reduced and additional merged batches may be generated and, as a result, serial numbers can be split off.
- The remaining serial numbers/sub-batches remain with the original merged batch (existing number).
- Then the user closes the dialog.

Result

Result when splitting "simple" batches:

- A new batch is generated for the quantity split off.
- The batch quality depends on the selection made
- Subject to the selected quality (yield/scrap), the batch status of the new batch is --> released/blocked
- The quantity of the old batch is reduced by the quantities split off. If the quantity is zero, the batch status of the old batch is "processed".
- Other data of split batches, such as the material number, designation, material type, storage location, material buffer, PPS batch is taken over from the original batch.
- The split event is displayed in the batch history (old batch and new batches).
- The graphic/tabular overview shows which batches have been split off from a batch.

Result when splitting a merged batch:

- The new merged batch is generated and includes all sub-batches/serial numbers selected for splitting
- The quality is identical for the merged batch and sub-batches/serial numbers (as defined)
- The quantity included in the new merged batch is the total quantity of all sub-batches/serial numbers
- Subject to the selected quality (yield/scrap), the batch status of the new merged batch and the sub-batches/serial numbers is --> released/blocked
- The old merged batch is no longer connected to the sub-batches/serial numbers split off
- The quantity of the original merged batch has been reduced by the quantity of the sub-batches/serial numbers split off. If the quantity is zero and/or if there are no longer sub-batches/serial numbers, the batch status of the old batch is "processed".
- Other data of sub-batches/serial numbers and the new merged batch, such as the material number, designation, material type, storage location, material buffer, PPS batch are taken over from the original batch.
- The split event is displayed in the batch history (old merged batch and new merged batches + serial number).
- The graphic/tabular overview shows which merged batches have been split off from a merged batch.



This dialog does not allow entering attributes/document links for individual serial numbers/sub-batches pertaining to the merged batch.

27 Splitting Batches in the AIP

Usage

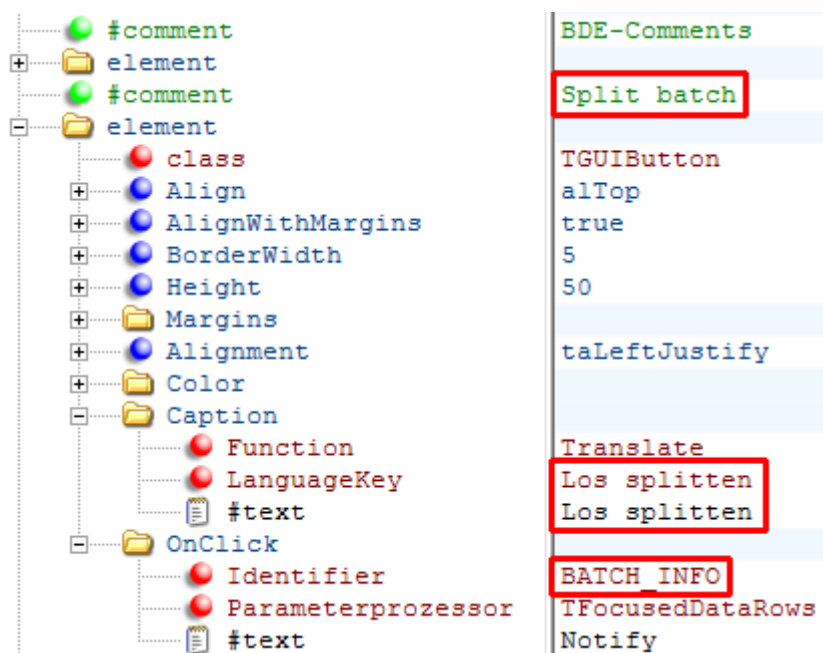
You use the function to split up a batch into several batches in the AIP terminal. Subject to the batch type, batches may be split in the following ways:

- Splitting off quantities into individual batches
- Splitting off individual batches from existing merged batches into a new merged batch

Dialog configuration

Dialog configuration (CNR.SPLIT)

Define the function key "split batch" by copying the layout *guiL_anr.xml* to *guiL_anr_ln.xml* and the key "BDE comments". Then the following entries must be changed in the configuration of the copied key:



If you do not use tiles for the AIP2, define the function key in the file **ctaipbut.ini**:

```
[ANR-LN-Page3]
```

```
...
```

```
3=BATCH_INFO,L,Los splitten,Los_splitten.png
```

The sections [BATCH_S_SPLIT] and [BATCH_SPLIT] are required for the layout configuration of **ctaisplay.ini**, irrespective of the used design:

```
[BATCH_S_SPLIT]

GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite

GRID_CELLPAINT=ON
EXAMINE_CELLBKCOLOR=SELECT,SELECT,X-clLime

SELECT=C1,50,Z,*
DLL=C20,150,Z,batch
ATK=C20,150,L,article
SGR:REST=N12.0,100,R,quantity
EINH=C3,60,Z,unit
;CKL=C1,150,Z,class

[BATCH_SPLIT]

GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite

GRID_CELLPAINT=ON
EXAMINE_CELLBKCOLOR=CKL,CKL,G-clLime|A-clRed|N-clBlue|O-clYellow

EGR=N12.3,250,R,quantity
CKL=C1,150,Z,class
EGG=C4,100,Z,reason
```

System configuration

Further system configurations are not required to split batches.

28 Merging Batches

Summary

The "merge batches" function can be used to merge existing batches into one common batch. The following "batch types" may be combined:

- "simple" batch and "simple" batch (adding up quantities)
- merged batch and merged batch (grouping of assigned subordinate batches/ serial numbers)

Prerequisite

In general, all batches to be merged have to be available.

The batch merge function can only be applied to batches meeting the below-mentioned conditions:

- Only batches with the same material (material number) can be merged
- Only batches with the same material type can be merged
- All batches must have a positive remaining quantity (batch class = yield, remaining quantity > 0). Consequently, scrap batches cannot be merged.

The following combinations are neither supported nor allowed:

- combination of "simple" batches and merged batches
- combination of an individual batch/serial number and a merged batch
- Combination of an individual batch/serial number and an individual batch/serial number

Variants

In general, there are the following two variants to merge batches.

- [Merge batches at AIP](#)
AIP provides a function to merge the entered batches/merged batches into a new batch/merged batch. A label can be printed for the merged batch.
- [Merge batches at MOC](#)
The MOC batch data overview provides a function to merge the entered batches/merged batches into a new batch/merged batch.

Result

The below-mentioned results can be expected after merging batches at AIP or MOC:

Merging "simple" batches:

- The merged batches have a remaining quantity of 0
- The merged batches are assigned the status "processed"
- The new batch has the status "free"
- The new batch has the batch class "yield"
- The batch number of the new batch:
 - is a new batch number
 - is a batch number pertaining to the merged batches

Grouping merged batches:

- the grouped merged batches have a remaining quantity of 0 and do no longer include subordinate batches/serial numbers
- the combined merged batches are assigned the status "processed"
- the new merged batch has the status "free"
- the new merged batch is assigned the "yield" batch class and includes all subordinate batches/serial numbers of the combined merged batches
- The batch number of the new merged batch:
 - is a new batch number
 - is a batch number pertaining to the combined merged batches

29 AIP Batch Merge

Summary

In production partly consumed containers with the same material are often combined to one container. However, the system considers the materials to be located in different batches. With this function the terminal provides an opportunity to merge batches and to print a label for the new batch.

Usage

At the moment, it is impossible to:

- combine "normal" batches with merged batches and/or serial numbers
- combine merged batches and serial numbers or to add several serial numbers to one merged batch. The merged batch number including the serial number always has to be entered.
- combine individual serial numbers into one merged batch. Their merged batch number always has to be entered. It is also possible to use the standard function "collection of serial numbers" A_SNR with the relevant configuration.

Prerequisite

The relevant configuration has to be made to be able to use this function.

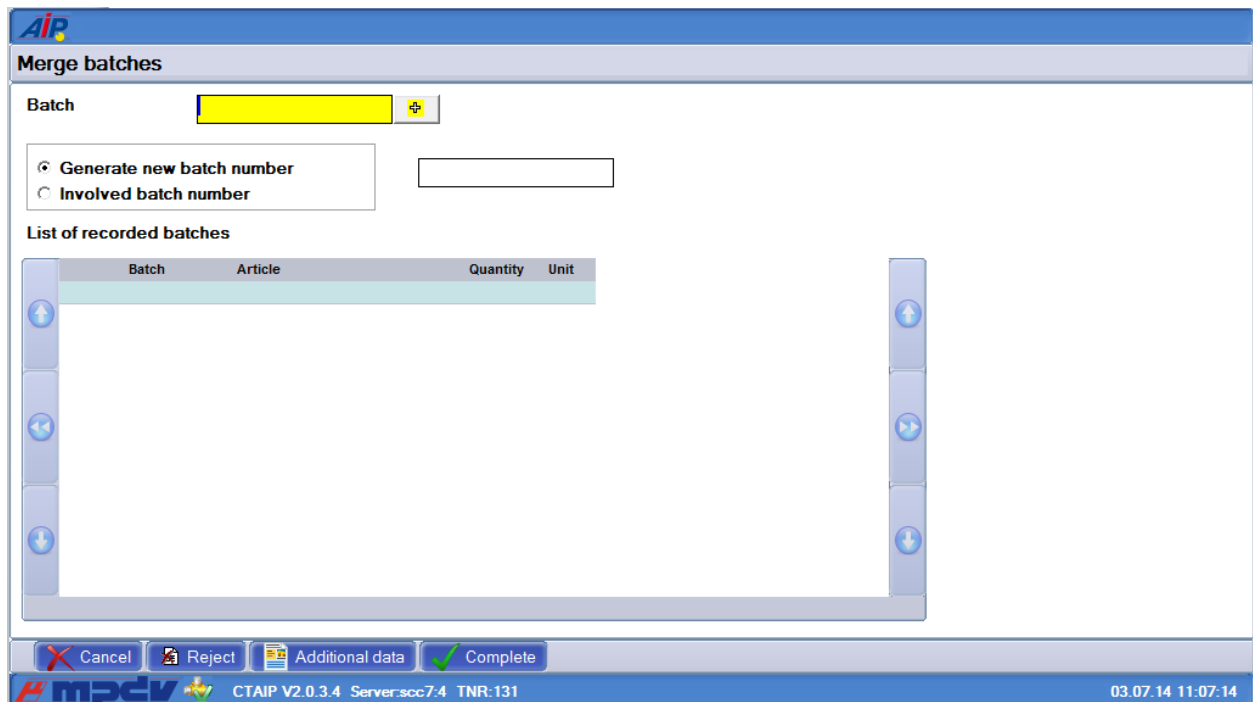
Terminal functions

The function is triggered manually by using the key of the "merge batches" dialog (BATCH_MERGE).



If the terminal is offline, only a limited number of posting functions is provided based on the available data. If the terminal is offline, errors are not displayed e.g. if posting failed on the server.

Dialog



Description of display fields:

- **Batch**
Batch number to be entered by the user (manually/by scanning)
- **Generate new batch number / involved batch number**
Specifies whether the new batch is to be generated with a new batch number or if an involved batch number is used for the new batch. If an involved batch number is used, this one has to be entered.
- **List of recorded batches**
Shows the entered batches that are to be merged.

Procedure for the "merge batches" function:

- The user opens the "merge batches" dialog by the relevant function key in the basic screen of the terminal
- In the "batch" field the user enters the affected batches (manually or by scanning) he/she wants to merge. Data may be entered manually or by scanning. If batches are entered manually, the input will be confirmed after each batch by . This empties the field for the next input.
- The user specifies whether the new batch is to be generated with a new batch number or if an involved batch number is used for the new batch. If an involved batch number is used, this one has to be entered.
- The "list of recorded batches" shows the batches the user has already entered and wants to merge.

- The user applies the "complete" key. Then the "batch" field and the "list of recorded batches" are emptied. The user receives an intermediate message with the generated batch number for the merged batch. Additional batches may be merged.
- The dialog remains opened until the user closes it explicitly by pressing "cancel". Then batches are merged into a new batch.
- A label can be printed for the merged (new) batch.

Function keys

- **Function key "cancel"**
This function key rejects the entered data and/or empties the result list. New data may be entered. The dialog remains open.
- **Function key "complete"**
Entered batches are merged into one batch. By using this function key, the dialog is emptied (i.e. no data included) and the new batch (batch number) is shown on the terminal. A label is printed subject to configuration.
- **Function key "reject"**
Entered batches are not merged into one batch. The data input is rejected.
- **Function key "additional data"**
Subject to configuration, this function leads to the next workflow to record attributes and/or document links.

Result

- **By the "cancel" function**
The dialog remains open. All entered data are rejected. The input data are not posted. Data input can be restarted.
- **By the "complete" function**
This function key closes the dialog. Entered batches are merged into one batch and data are posted accordingly. The amount of merged batches is reduced accordingly and set to the "processed" batch status. The new batch includes the merged amount. If merged batches are combined, the serial numbers will be assigned to the new merged batch. Label printing may be triggered by this.
- **By the "enter" function**
Entered data are recorded for each batch but not yet posted.
This function key records the individual batches once they have been entered and displays them in the result list. The dialog remains open.
- **By the "additional data" function**

Similar to the output batch change, it is possible enter attributes and document links for each combined batch/merged batch. If they are configured, relevant workflows are integrated for data collection. Attributes and document links from previous batches are not taken over.

This dialog does not allow entering attributes/document links for individual serial numbers/sub-batches pertaining to the merged batch.



In addition, the event is also shown in the batch history (old batches and new batch and old merged batches and new merged batch and serial number).

The graphic/tabular overview shows which batches have been merged into one batch.

30 Configurations for Merging Batches in the AIP

Usage

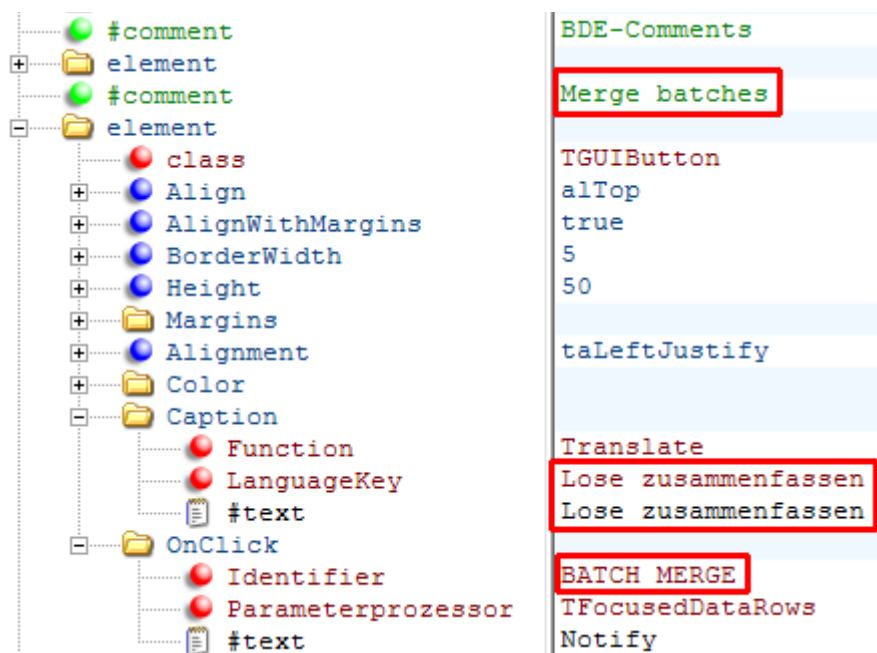
You use the function to merge several batches into one batch at the AIP terminal. Subject to the batch type, batches may be merged in the following ways:

- Merging simple batches and/or their quantities to one batch
- Merging individual batches belonging to an already existing merged batch to one new merged batch

Dialog configuration

Dialog configuration (CNR.SUMMARIZE)

Define the function key "pool batches" by copying the layout *guil_anr.xml* to *guil_anr_ln.xml* and the key "BDE comments". Then the following entries must be changed in the configuration of the copied key:



If you do not use tiles for the AIP2, define the function key in the file **ctaipbut.ini**:

```
[ANR-LN-Page3]
```

```
...
```

```
2=BATCH_MERGE,L,Lose zusammenfassen,Lose_zusammenfassen.png
```

The sections [BATCH_MERGE] and [DOC_BATCH_MERGE.LST] and [ATTR_BATCH_MERGE.LST] are required for the layout configuration of **ctaiplay.ini**, irrespective of the used design:

```
[BATCH_MERGE]

GRID_FONT=Arial

GRID_FONTSIZE=9

GRID_COLOR=clBlack

GRID_BACKGROUND=clWhite


EXAMINE_SCANEXPR1=SLOS=J

EXAMINE_SCANCOLOR1=clBlue


DLL=C20,150,Z,batch

ATK=C20,150,L,article

SGR:REST=N12.0,100,R,quantity

EINH=C3,60,Z,unit


[DOC_BATCH_MERGE.LST]

GRID_FONT=Arial

GRID_FONTSIZE=9

GRID_COLOR=clBlack

GRID_BACKGROUND=clWhite

LINK=C20,400,L,link/document


[ATTR_BATCH_MERGE.LST]

GRID_FONT=Arial

GRID_FONTSIZE=9

GRID_COLOR=clBlack

GRID_BACKGROUND=clWhite

;GRID_ORDER=DAT=- | ZEI=- | SNR


ATTR=C20,100,L,attribute

VALUE=C20,80,L,value
```

`EINH=C20,60,L,unit`

`TEXT=C20,200,L,text`

System configuration

Further system configurations are not required in order to merge batches.

31 Advanced Batch Information

Summary

Various batch information needs to be made available to manufacturing processes.

Usage

The function is used to display the information stored to the batch. This refers to:

- General batch information
- Batch quantities
- Batch attributes
- Alternative batch numbers



The dialog is structured in workflows. These workflows can be started by clicking on the individual tabs or by clicking "next" in each individual workflow.

Prerequisite

Batch attributes have to be configured for the batch to be able to display them.

Alternative batch numbers can only be displayed if they are used in specific projects.

Show general batch information

Dialog



If the terminal is offline, only a limited number of posting functions is provided based on the available data. If the terminal is offline, errors are not displayed e.g. if posting failed on the server.

Batch information

1 Batch information
2 Quantities
3 Attributes
4 ...

Batch No. 1001234

Material

XX

Designation

Diffusor Rohr

Type

ROHR

Class

G

Operation

Machine

50500

Material buffer

MP_V_1168

Prod.date

04.03.2013 00:00:00

Status

F

Q-Status

O

M-Status

U

Note

beschichtet

X Cancel
Go on

CTAIP V2.0.3.4 Server:scv7.4 TNR:131
03.07.14 12:52:40

Description of display fields:

- **Batch number**
Entered number of the batch for which information is to be displayed.
- **Material**
Material number of the batch
- **Designation**
Material name of the batch
- **Type**
Material type of the batch
- **Class**
Batch class (yield, scrap, rework, open quantity)
- **Operation**
Operation that has produced the batch or which it is logged on to.
- **Machine**
Machine/workplace that has produced the batch or which it is logged on to.
- **Material buffer**
Material buffer that includes the batch.
- **Manufacturing date**
Manufacturing date of the batch
- **Status**
Batch status
- **Q-status**

Quality status of the batch.

- **M-status**

Material status of the batch.

Function keys

- **Function key "cancel"**

This function key terminates the dialog and rejects data input.

- **Function key "go on"**

This function key leads the user to the next configured workflow of the dialog (quantities, attributes, alternative batch numbers).

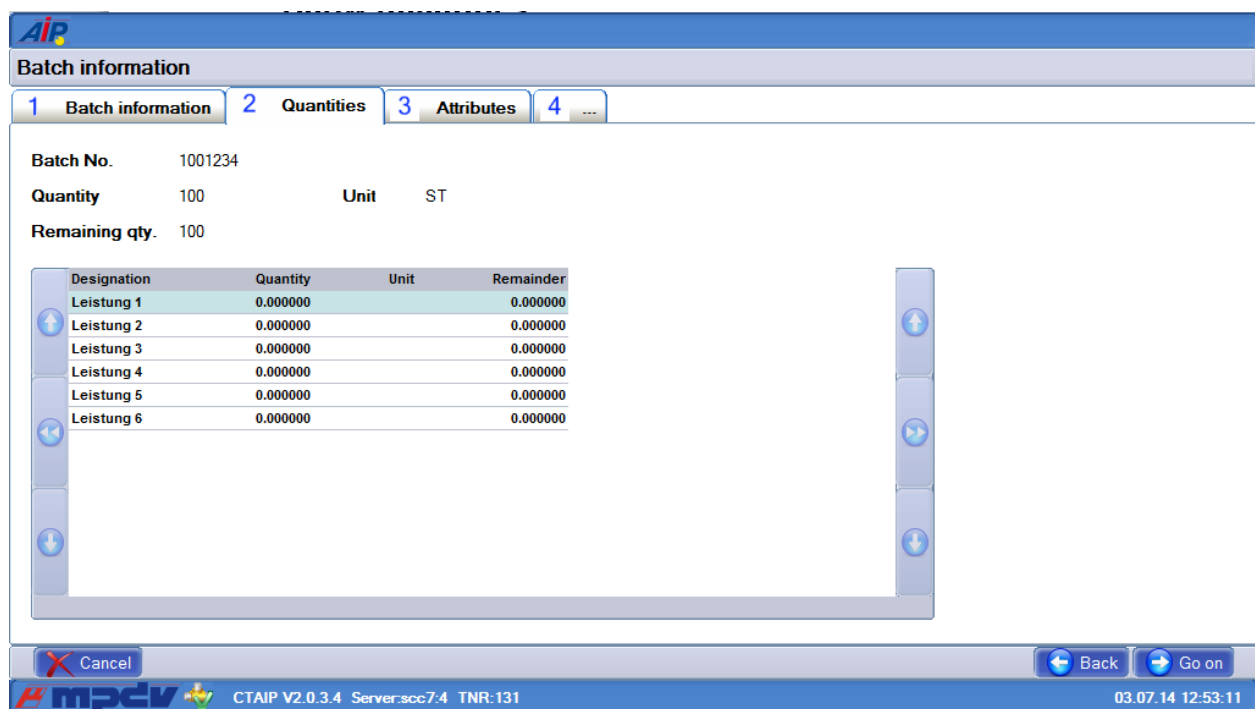
Procedure

For the user the procedure is as follows:

- The user opens the "batch information" dialog in the basic screen of the terminal.
- The user enters the batch (manually or by scanning) for which he/she wants to view batch information.
- The user applies the "batch information" workflow.

Show quantities

Dialog



Batch information

1 Batch information 2 Quantities 3 Attributes 4 ...

Batch No. 1001234

Quantity 100 Unit ST

Remaining qty. 100

Designation	Quantity	Unit	Remainder
Leistung 1	0.000000		0.000000
Leistung 2	0.000000		0.000000
Leistung 3	0.000000		0.000000
Leistung 4	0.000000		0.000000
Leistung 5	0.000000		0.000000
Leistung 6	0.000000		0.000000

Cancel Back Go on

mpciv CTAIP V2.0.3.4 Server:scs7:4 TNR:131 03.07.14 12:53:11

Description of display fields:

- **Batch number**
Entered number of the batch for which information is to be displayed.
- **Quantity**
Initial quantity of the batch.
- **Remaining quantity**
Current remaining quantity of the batch.
- **Unit**
Batch unit
- **Activity (1-6)**
Activity fields of the batch (to convert into other units)

Function keys

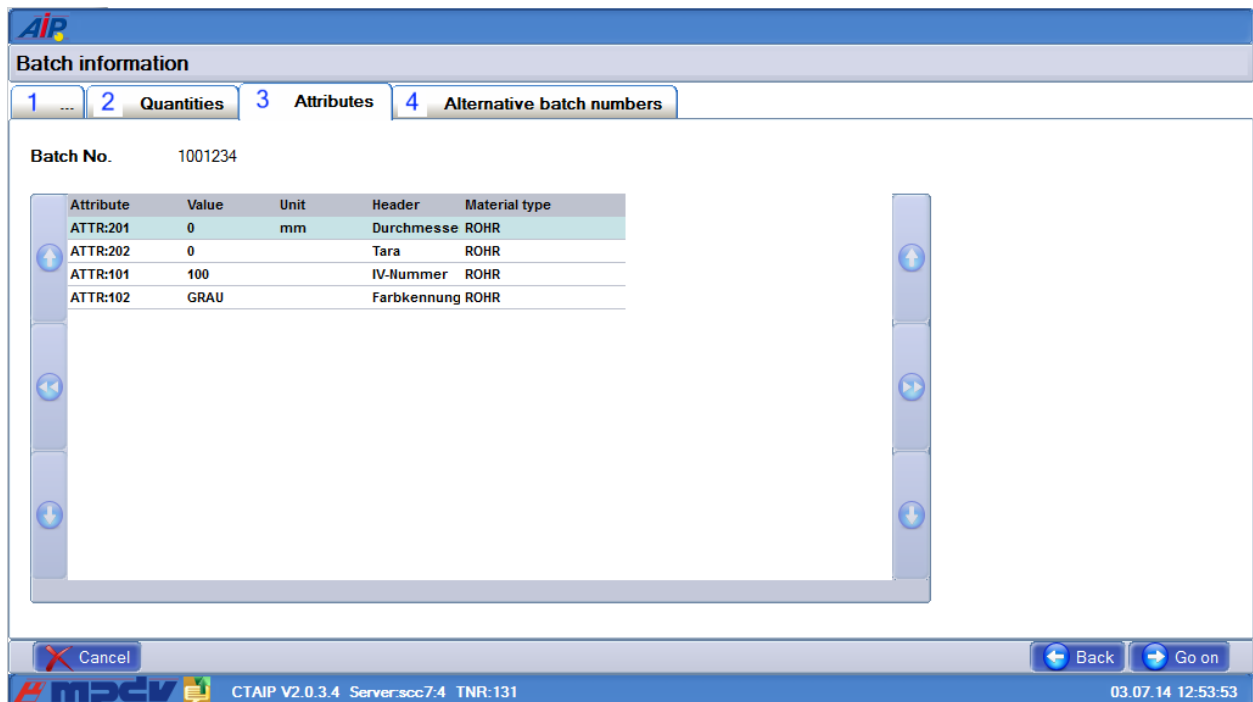
- **Function key "cancel"**
This function key terminates the dialog and rejects data input.
- **Function key "go on"**
This function key leads the user to the next configured workflow of the dialog (attributes, alternative batch numbers).
- **Function key "back"**
This function key leads the user back to the previously configured workflow of the dialog (batch information, quantities, attributes, alternative batch numbers).

Procedure

For the user the procedure is as follows:

- The user opens the "batch information" dialog in the basic screen of the terminal.
- The user enters the batch (manually or by scanning) for which he/she wants to view batch information.
- The user applies the "quantities" workflow.

Show batch attributes**Dialog**



Attribute	Value	Unit	Header	Material type
ATTR:201	0	mm	Durchmesse	ROHR
ATTR:202	0		Tara	ROHR
ATTR:101	100		IV-Nummer	ROHR
ATTR:102	GRAU		Farbkennung	ROHR

Description of display fields:

- **Batch number**
Entered number of the batch for which information is to be displayed.
- **Result list: attributes**
Shows the configured batch attributes for the material type of the batch including value and relevant captions. The displayed columns are configurable.

Function keys

- **Function key "cancel"**
This function key terminates the dialog and rejects data input.
- **Function key "go on"**
This function key leads the user to the next configured workflow of the dialog (attributes, alternative batch numbers).
- **Function key "back"**
This function key leads the user back to the previously configured workflow of the dialog (batch information, quantities, attributes, alternative batch numbers).

Procedure

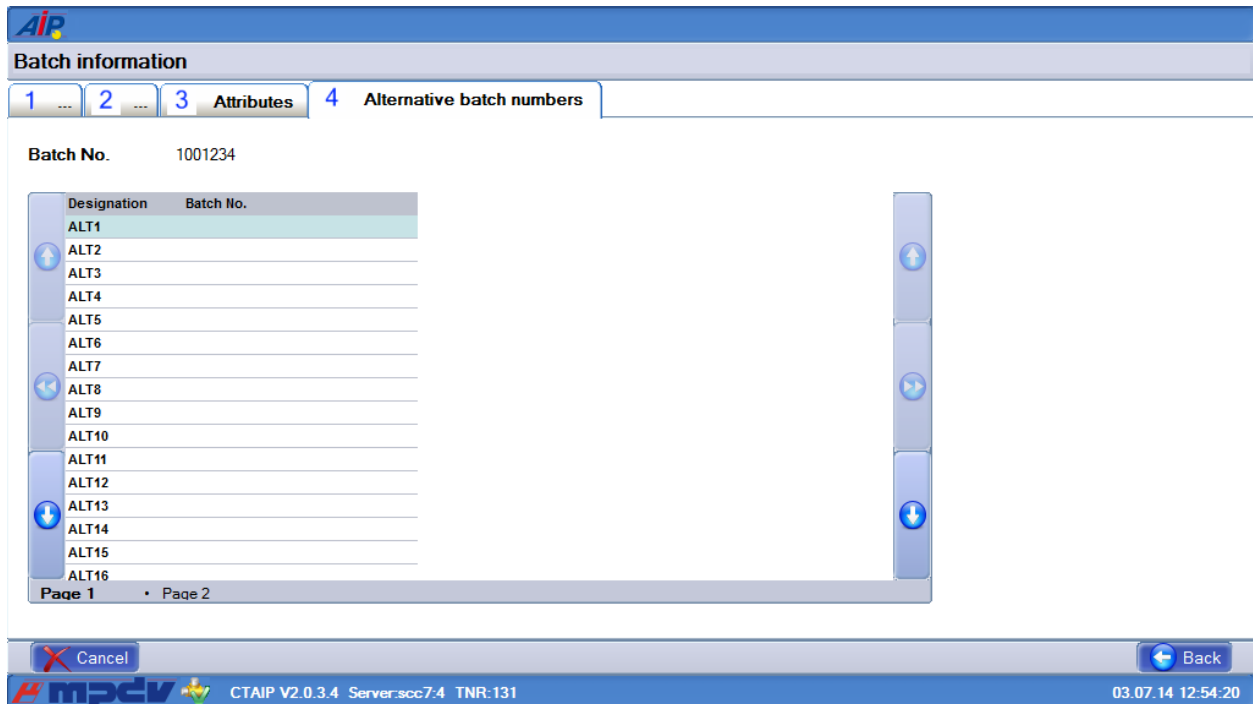
For the user the procedure is as follows:

- The user opens the "batch information" dialog in the basic screen of the terminal.
- The user enters the batch (manually or by scanning) for which he/she wants to view batch information.

- The user applies the "attributes" workflow.

Show alternative batch numbers

Dialog



Batch information

1 ... 2 ... 3 Attributes 4 Alternative batch numbers

Batch No. 1001234

Designation	Batch No.
ALT1	
ALT2	
ALT3	
ALT4	
ALT5	
ALT6	
ALT7	
ALT8	
ALT9	
ALT10	
ALT11	
ALT12	
ALT13	
ALT14	
ALT15	
ALT16	

Page 1 • Page 2

Cancel Back

CTAIP V2.0.3.4 Server:sc7:4 TNR:131 03.07.14 12:54:20

Description of display fields:

- **Batch number**
Entered number of the batch for which information is to be displayed.
- **Result list: alternative batch numbers**
Shows the alternative batch numbers for the material type of the batch including relevant value. Alternative batch numbers are optional and entered for specific projects.

Function keys

- **Function key "cancel"**
This function key terminates the dialog and rejects data input.
- **Function key "back"**
This function key leads the user back to the previously configured workflow of the dialog (batch information, quantities, attributes, alternative batch numbers).

Procedure

For the user the procedure is as follows:

- The user opens the "batch information" dialog in the basic screen of the terminal.

- The user enters the batch (manually or by scanning) for which he/she wants to view batch information.
- The user applies the "alternative batch numbers" workflow.

32 Additional Batch Information

Usage

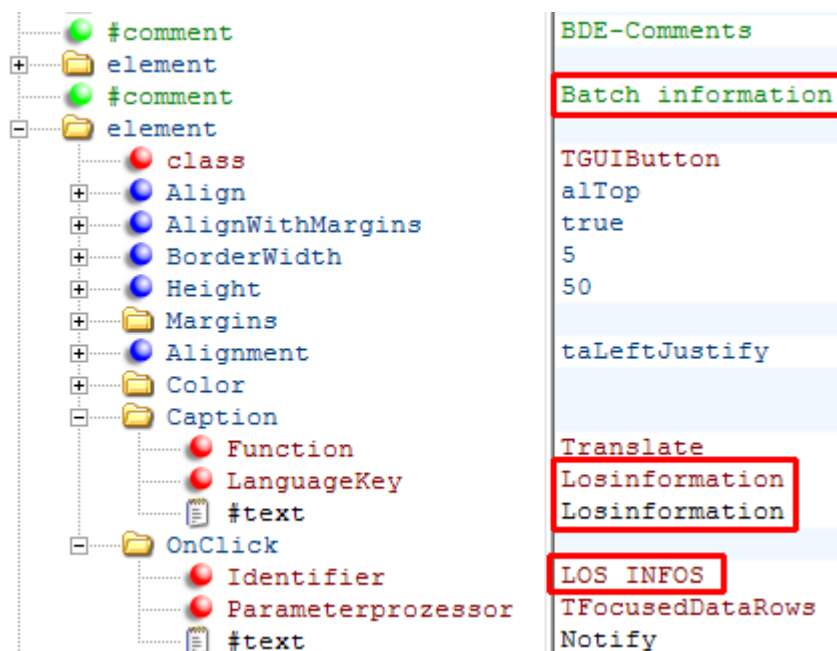
You use the function to view information about a batch in the AIP terminal. Additional batch information include:

- General batch information
- Quantities
- Batch attributes
- Alternative batch numbers

Dialog configuration

Dialog configuration (LOS_INFOS)

Define the function key "batch information" by copying the layout *guil_anr.xml* to *guil_anr_ln.xml* and the key "BDE comments". Then the following entries must be changed in the configuration of the copied key:



If you do not use tiles for the AIP2, define the function key in the file **ctaipbut.ini**:

```
[ANR-LN-Page3]
```

```
1=LOS_INFOS,L,Losinformation,Shipping Box Closed Information.png
```

The sections [LOS_INFOS_ATTR.LST] and [LOS_INFOS_QUA.LST] and [LOS_INFOS_ALTER.LST] are required for the layout configuration of *ctaisplay.ini*, irrespective of the used design.

```
[LOS_INFOS_ATTR.LST]
```

```
GRID_FONT=Arial
```

```
GRID_FONTSIZE=9
```

```
GRID_COLOR=clBlack
```

```
GRID_BACKGROUND=clWhite
```

```
KENNUNG=C20,100,L,attribute
```

```
ATTR_VAL=C20,80,L,Value
```

```
EINH=C20,80,L,unit
```

```
TXT=C20,80,L,
```

```
MATTYP=C40,140,
```

```
[LOS_INFOS_QUA.LST]
```

```
GRID_FONT=Arial
```

```
GRID_FONTSIZE=9
```

```
GRID_COLOR=clBlack
```

```
GRID_BACKGROUND=clWhite
```

```
KENN=C20,100,L,
```

```
ALIAS LEER1=(DUMMY1)=C1,30,L
```

```
MENGE=C20,80,R,
```

```
EINH=C4,40,R,
```

```
ALIAS LEER2=(DUMMY1)=C1,30,L
```

```
REST=C20,80,R,
```

```
[LOS_INFOS_ALTER.LST]
```

```
GRID_FONT=Arial
```

```
GRID_FONTSIZE=9
```

```
GRID_COLOR=clBlack
```

```
GRID_BACKGROUND=clWhite
```

```
ALT=C20,100,L,
```

```
CNR=C20,200,L,
```

System configuration

Further system configurations are not required in order to display additional batch information.

33 Weighing Components

Usage

For operations that are not subject to batch management, it is possible to record batches in relation to discrete material components. In this special case, batches may be entered in relation to the charge via a special terminal function.

This entry function replaces the collection of quantities for the operation (e.g. partial upload) and records material consumption in relation to material components. This consumption is also posted as material movement in the system.

Prerequisites

- The function can weigh more than one charge. In this case, the operation's secondary quantity has to include the number of charges. The additional data fields of the components and operation regarding the weighing function (e.g. target quantity per charge, tolerances, etc.) also need to be taken into account.
- Every time an operation is changed, the target quantity for each charge is recalculated (formula: $\text{target quantity per charge} = \text{primary target quantity} / \text{secondary target quantity}$). It is neither possible to set the value for the target quantity per charge manually nor to display it on MOC. If the secondary target quantity is not set, the value 1 is assumed by default and only one charge is processed in the weighing operation.
- In this case, the machine does not allow to enter automatic quantities additionally.
- All components have to be managed by the "weight" unit (kilogram).



The entry function cannot be used offline.

Configuration

The configuration document describes the configurations required to use and enable the entry function at the terminal.

Posting

Data for label printing

The schema **a_vbrkomp** is provided in initial data for label printing.

This data can be printed on the label during weighing (dialog action KEW_RECORD):

Acronym	Type	Length	Description
DLG.MNR	C	10	Machine
DLG.ANR	C	40	Operation
DLG.ATK	C	40	Material number
DLG.SLP	C	40	BOM item
DLG.CHARGE	C	20	Batch
DLG.CST	C	1	Batch status F/S (free/blocked)
DLG.SGR:GUT	DEC		Target quantity
DLG.SGE:GUT	C	3	Unit of the target quantity
DLG.EGR:GUT	DEC		Actual quantity
DLG.EGE:GUT	C	3	Unit of the actual quantity
DLG.EGR:MENGE	DEC		Input quantity
DLG.EGE:MENGE	C	3	Unit of the input quantity
DLG.DAT	DATE		Date of the posting
DLG.ZEI	ZEIT		Time of the posting
DLG.KNR	C	10	The reporting person's badge number

This data can be printed on the label when completing the charge (dialog action KEW_ABSCHLUSS):

Acronym	Type	Length	Description
DLG.MNR	C	10	Machine
DLG.ANR	C	40	Operation
DLG.ATK	C	40	Material number
DLG.ATKBEZ	C	40	Material designation
DLG.CHARGE	C	20	Batch
DLG.EGR:GUT	DEC		Quantity of the batch
DLG.EGE:GUT	C	3	Unit
DLG.DAT	DATE		Date of the posting
DLG.ZEI	ZEIT		Time of the posting
DLG.KNR	C	10	The reporting person's badge number

Scale interfacing

The weight can also be entered in the weighing dialog by means of a connected scale. Upon opening the KOMP_WIEG dialog, the value is requested from the scale via the PCC driver interface and entered in the "input quantity" field.

This is an example for entering the scale value in the INI file of the driver:

<WAAGENTREIBER>.INI
V:WAAGE:NETTO=Nettogewicht_Waage

If an OPC interfacing is used, changed scale values can be sent automatically by the OPC server. For this purpose, the below-mentioned parameter has to be entered in the file "OPCMPDV.INI"
SETVALEVENTS=V:WAAGE:NETTO

<WAAGENTREIBER>.INI
SETVALEVENTS=V:WAAGE:NETTO ← additional entry
V:WAAGE:NETTO=Nettogewicht_Waage

34 Weighing of Components

Summary

Some production areas require components to be weighed using scales and to provide them for further production processes.

Usage

The function is always used when the user prepares charges using the relevant material quantities. To save time, users often weigh several charges at once.

The logical process and posting are described [here](#).

Prerequisite/configuration

The configuration is described [here](#).

Weighing components

Components for a charge are weighed by the AIP dialog A_VBRKOMP.



If the terminal is offline, only a limited number of posting functions is provided based on the available data. If the terminal is offline, errors are not displayed e.g. if posting failed on the server.

The "weighing" function at the terminal provides a list of required input materials (discrete material components). The component-related data collection can be opened from this list.

Once the operation to be weighed has been logged on, the "weighing components" dialog can be started by the "weigh" function key from the toolbar of the basic screen of the terminal. After opening the dialog, the terminal automatically generates a new batch for the order. This is presented as batch in the following dialog.

Please note: The dialog only opens if "discrete" material components (consumption type = D) exist for the running operation.

The actual weighing process of the relevant components is performed by the "weigh charge" function in a detailed dialog (see below).

Dialog

AIP
<A_VBRKOMP>

Workplace W2030
Operation M30002000010
Batch PR4RH6B11K

Material 4529-42
Designation

Status
☒ Free
☐ blocked

Components

POS	Article	Target qty.	Actual qty.	Remaining qty.	UOM	Designation	Tolerance	Dev
1	9901	169,000	0,000	0,000	L	Ethylalkohol	0,000	
2	3011	3380,000	0,000	0,000	G	Phosphorsäure	0,000	
3	4011	152,100	0,000	0,000	G	Glyzerin	0,000	
4	8511	16731,000	0,000	0,000	L	Destilliertes Wasser (Aquadest)	0,000	

Staff Number

Cancel
Weigh comp.
Quantity adjustm.
Reject charge
Complete charge

04.07.14 10:37:22

Workplace

Weighing workplace to which the operation to be weighed is logged on.

Operation

Weighing order/operation

Batch

Batch number for the charge that is currently to be produced.

Material

Material number for the charge that is currently to be produced.

Status

Batch status for the charge that is currently to be produced after weighing. This can be:

- Free
- Locked

Staff badge number

The user's staff badge number.

Components list

The component list shows all components of the currently registered weighing operation that are relevant for the charge to be produced. These details for the component are displayed to perform weighing:

- Article number (material number of the component from the component list)
- Target quantity
- Actual quantity (quantity entered upon weighing, at first 0,000)
- Remaining quantity (computed remaining quantity after weighing, at first 0,000)
- Designation (component name from the component list)
- Tolerance (calculated, admissible tolerance of the component/component list)
- Deviation (calculated, admissible deviation of the component/component list)

Procedure:

The user selects the first component that is to be weighed and starts weighing.

- The relevant component row is highlighted in **green** if the component is within the variance tolerance (at the component, percentage).
- Materials falling short of the variance tolerance are highlighted in **red**.
- Materials the weighing result of which exceeds the variance tolerance but is still within the tolerances of quantity adjustment (of the component) are highlighted in **blue**.
- The row is shown in black font, provided that the component has not yet been weighed.
- If the weight value is beyond the variance tolerance but within the tolerance of quantity adjustment, posting will be accepted but the user has to adjust quantities for the other components (automatically in the system). The input dialog cannot be closed if this modification is not made.

Quantities have to be entered for all input materials included in the component list.

Functions

The paragraphs that follow describe the functions provided in the input dialog to weigh components:

"Weigh charge"

The "weigh charge" function opens the dialog to weigh a selected individual component (KOMP_WIEG).

AIP

Weigh charge

<KOMP_WIEG>

Workplace

W2030

Operation

M30002000010

Batch

PR4RH6P113

Component

4529-42

BOM item

1

Component batch

PR4RD1D113

Default quantity

169,000

L

Actual qty.

150,000

KG

Balance

0,000

KG

Input quantity

0,000

KG

Allowed deviation

0,000

%

Actual deviation

100,000

%

Staff Number

9999

Cancel

Weigh charge

mpciv

18.07.14 09:09:15

The current status of target and actual quantity is presented after selecting a component that is to be weighed. The user has to enter a relevant batch for consumption posting in the "component batch" input field. The batch has to be available in HYDRA with the appropriate material and the status "free" and it must have a remaining quantity ≥ 0 . The dialog can be closed by entering the weighing quantity and the staff badge number (optional).

Successful weighing has the following additional effects on the system:

- Indicators are recorded for the component
- The weighed quantity is added to the actual quantity of the component
- Material movements are generated for consumption (261)
- A batch assignment is generated for the batch ⇔ component batch.



- The article of the batch always has to match the article of the component.
- A component may also be weighed several times.
- The amount (stock) of component batches is only reduced if the relevant material type is assigned the "retrograde inventory collection" flag.

Once the weight has been entered, the displayed actual quantity and remaining quantity are updated.

The displayed component requirements are updated in the table once the component has been weighed successfully.

<A_VBRKOMP>

Workplace W2030

Operation M30002000010

Batch PR4RH6B11K

Material 4529-42

Designation

Status

☒ Free
 ☐ blocked

Components

POS	Article	Target qty.	Actual qty.	Remaining qty.	UOM	Designation	Tolerance	Dev
1	9901	169,000	150,000	19,000	L	Ethylalkohol	0,000	
2	3011	3380,000	150,000	3230,000	G	Phosphorsäure	0,000	
3	4011	152,100	0,000	0,000	G	Glyzerin	0,000	
4	8511	16731,000	0,000	0,000	L	Destilliertes Wasser (Aquadest)	0,000	

Staff Number 9999

Cancel

Weigh comp.

Quantity adjustm.

Reject charge

Complete charge

04.07.14 10:39:51

A warning message the user may skip is output if the value of the actual deviation is greater than the one of the target deviation.

"Reject charge" function

Due to the weighing process, it might be the case that an entire batch cannot be used. The "reject charge" function can be used to identify the current charge/batch as "scrap". The material used for this charge is uploaded to ERP and the generated scrap quantity (identified as charge scrap by SYSTEM reason 910) is posted to the order. The batch is generated with the "locked" status and the "scrap" batch class (goods movement 531).

A confirmation prompt the user has to affirm comes up with this function.

Then the original target quantity for each charge is restored and the components are reset. Consequently, the weighing process can be restarted for a new batch.

"Adjust quantity" function

Once the first component has been weighed, the user can apply the "adjust quantity" function to automatically adjust the input quantities of the other components in proportion to the weighing result of the first component and its default quantity. This function can only be used once for each charge.

The function adjusts the target quantities of all components to the weighed quantity of one component. Quantities are adjusted by changing the primary target quantity for each charge of the operation.

However, quantities may only be adjusted if the component is weighed within its tolerances.

"Complete charge" function

Using this function, the charge/batch is completed (generated) and the dialog is closed. However, the charge/batch can only be completed, once all components have been weighed and are within their specified tolerances.

Completed successfully, the charge/batch is generated and transferred as goods movement 101 to ERP. Optionally, the charge/batch is assigned a minimum shelf-life (from the material type of the OP). In addition, the PPS batch from the order is determined and stored as PPS batch in the batch of the charge (optionally).

The quantity of the generated batch results from the input quantity recorded as consumption for each component.

The user may also directly block the batch/charge by selecting a status from the component requirements dialog. By default, the batch is always assigned the status "free".

35 Configuration of Weighing Components

Usage

You use the "pass batch attributes on" function if attributes of input batches are to be transferred to the generated output batch when changing output batches.

Dialog configuration

The input function is controlled by the dynamic dialogs A_VBRKOMP and KOMP_WIEG

Activation of the posting function at the terminal

Specific posting functions are enabled at AIP by an entry in the file **ctaipbut.ini**.

This is an example for the entry in ctwinbut.ini:

CTAIPBUT.INI
F1=A_VBRKOMP, weigh



The dynamic dialogs A_VBRKOMP and KOMP_WIEG must be available.

System configuration

Operation data

The following additional fields have be filled out for the operation using the PPS interface:

- No batch management requirement
- Target quantity per charge (calculated form primary target quantity / secondary quantity)(ab.soll_menge_ansatz)
- Number of charges (secondary quantity in pieces) - default = 1
- Batch

Data included in component list

These parameters have to be filled out for discrete material components using the PPS interface:

- Tolerance (in percent) → mlst_hy.mengen_tol
- Deviation (absolute value) → mlst_hy.mengen_abweichung

- Input quantity and unit of input quantity
- Component type must be D - discrete

36 Palletizing and Packing on the AIP

Log packing OP on

Use the input function (dialog A_AN_HU or A_P_AN_HU) to log on a packing OP requiring batch management to the terminal.

When logging the OP on, an operation is selected from the sequencing list and logged on to the machine. The logon is performed without input material for OPs requiring batch management and posted using the PDM command A_AN or A_P_AN.

If the machine is configured so that the person has to be logged on with the OP, the A_P_AN command is used and the dialog has to include an input field for the badge number.

Assign batches

The below input function (dialog CE_AN_HU) is used on the terminal to assign batches to the current TPU (running output batch) for a running OP requiring batch management:

Assign batches <CE_AN_HU>

Machine: MPDV-PAC Transp. unit: HUPBF9011D
Order: PLATE000320000850

Batch No.

Batches in transport unit

Batch No.	Material	Weight	Einh	Designation
- PRPBF90116	PLATE_WIP07	1.00	KG	Plate finishing Article

No. of batches: 1 Total qty.: 1
Staff badge number:

Exit Assign batch Remove batch Complete TPU

Figure: "Assign batches" dialog - CE_AN_HU

Enter a batch number that is known in HYDRA and click the button *Assign batch* to logically assign the batch to the current TPU. Once assigned successfully, the batch is set to the "processed" status and displayed in the list, which includes all batches assigned to the TPU.

Note: You can assign only batches of the same material.

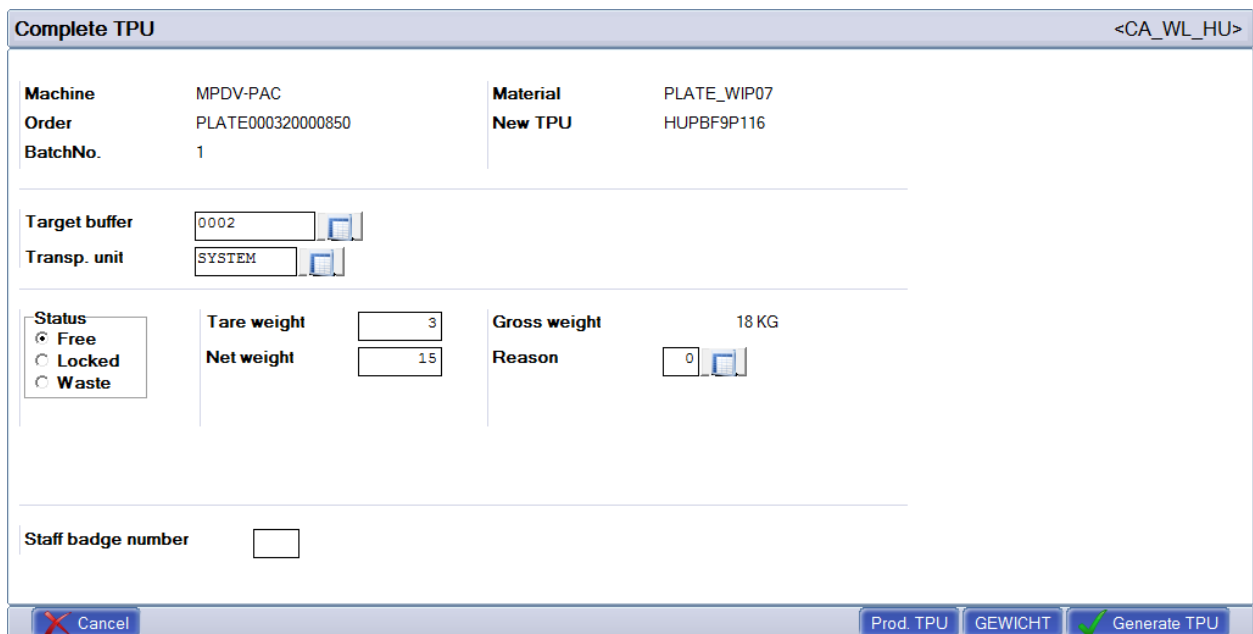
Use the button *Remove batch* to cancel the assignment of a selected batch (double-click on the relevant row). Then the batch changes back to the "free" status.

As additional status information on the current TPU, the dialog shows the number of assigned batches and their total quantity in kg.

You configure the contents of the "batches in transport unit" table in the section [C_PAL_ZUORD.LST] of the file ctaipplay.ini.

Complete TPU

You use this input function (dialog CA_WL_HU) to complete the currently running output batch (TPU) on the terminal.



Machine MPDV-PAC		Material PLATE_WIP07	
Order PLATE000320000850		New TPU HUPBF9P116	
BatchNo. 1			
Target buffer 0002			
Transp. unit SYSTEM			
Status ☒ Free ☐ Locked ☐ Waste	Tare weight 3 Net weight 15	Gross weight 18 KG Reason 0	
Staff badge number			

Buttons: Cancel, Prod. TPU, GEWICHT, Generate TPU

Figure: "Complete TPU" dialog – CA_WL_HU

Using the function *Complete TPU*, you can complete a TPU and enter the weight and the status yield, locked, scrap. When you complete the TPU, the relevant quantity (net weight) is booked to the running output batch and completed and a goods receipt (movement type 101) is posted. At the same time, the next output batch is logged on to the OP.

You can collect the following data with this posting:

Target buffer

Material buffer where the TPU batch is posted. By default, this is the machine's output buffer.

Gross weight/tare weight/net weight

By default, the total (in kg) of the assigned single batches is entered as net weight. If entered manually, it is calculated as follows:

$$\text{Gross weight} = \text{Tare weight} + \text{Net weight}$$

The values "gross weight" and "tare weight" are additionally saved as batch attributes for the TPU batch:

Gross weight - batch attribute 301 (ATTR:301)

Tare weight - batch attribute 302 (ATTR:302)

Transp. unit (transport unit)

A transport unit existing in the system can be assigned to the TPU batch.

Note: the weight stored for the transport unit is currently not taken over as tare weight to the input dialog.

Status (free/locked/waste)

By using the status selection, you can assign a status to the generated TPU batch: status "free" (yield material), status "locked" (class: yield material) or status "waste" (class: scrap).

In case of scrap, the quantity is posted as scrap for the operation and a scrap reason is entered.

Number of batches

Shows the number of batches currently assigned to the TPU. The value is saved as batch attribute 201 (ATTR:201).

Once the input has been confirmed by clicking OK, you can optionally enter further batch attributes if the attributes have been defined for the material type of the operation.

Once the posting has been completed successfully, the TPU batch is generated as "merged batch" and the relevant quantity and the goods movement are posted for the operation.

Log OP off / Interrupt OP

The packing OP can be interrupted or logged off on the terminal using this input function (dialog A_UN_HU or A_AB_HU).

Once selected, the user can interrupt or finish the packing OP:

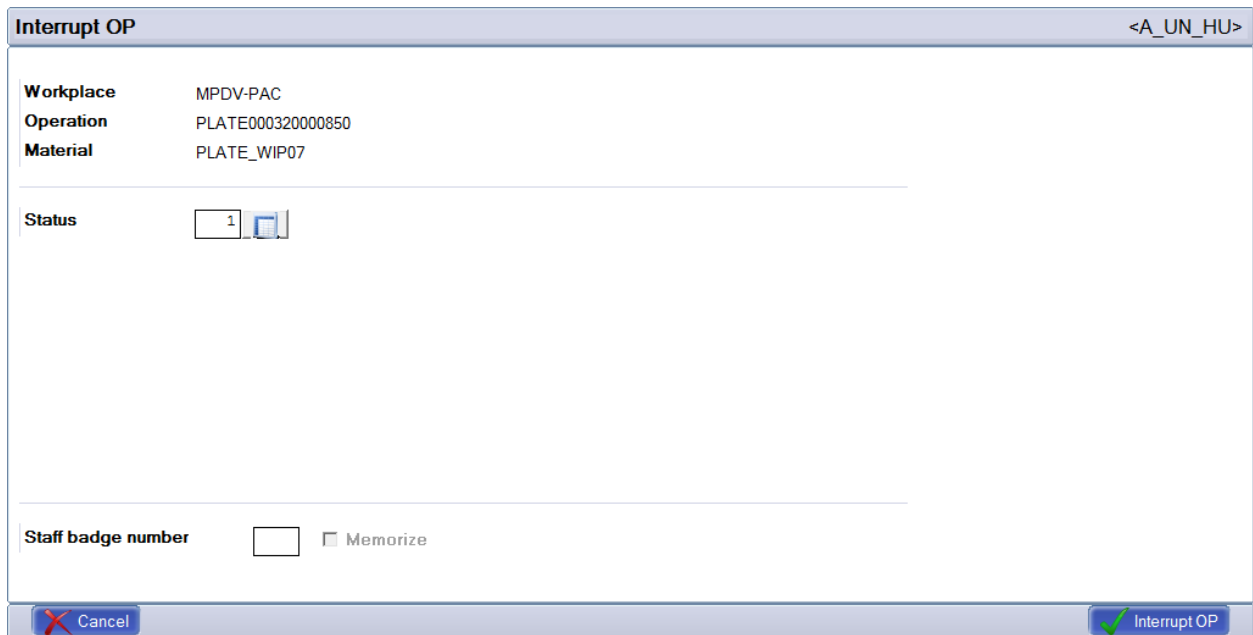


Figure: "Interrupt packing OP" dialog – A_UN_HU

The running packing OP is interrupted/finished after confirming by clicking OK.

Notes / restrictions:

- The last active output batch of the packing OP is completed without quantity and with status "deleted".
- If batches are still assigned to the TPU when you try to log the OP off, the logoff is rejected with error message.
- The ADE checks for overdelivery/underdelivery of the operation are not active when you use these posting functions.
- Quantities cannot be posted using this function.

Unpack / Repack

Requirements

Service pack 13 must be installed and activated. The document [Activating MPL TRT Dialogs \(SP13\)](#) describes how to use the dialog on the shop floor client.



For new customers, the new/updated MPL or TRT dialogs are directly available once the service pack 13 has been released.

General

Using this dialog, the operator can enter the packed handling unit and unpack selected batches. The operator can also unpack all batches and therefore cancel the handling unit.

Terminal procedure - Unpack individual batches

You start the function "Unpack TPU" (Dialog: CE_DEL_HU) in the main view by clicking the button *Unpack TPU*.

The user enters the handling unit or merged batch number. The number can be entered manually or by scanning the barcode. If you enter the number manually, you must click the green arrow. If you scan a barcode, the system automatically performs the transfer and requests the detail data of the batch.

The list displays the data of the entered handling unit. You can double-click specific batches to select the batches that you want to unpack. The selected batches are labeled in the list with an "x". Double-click a selected batch to deselect the batch. The display returns to normal ("-").

Use the button *Unpack selected ones* to unpack selected batches.

A confirmation prompt opens asking whether you want to execute the dialog:

- Click *No* to cancel execution. The user returns to the dialog. The selected batches remain.
- Click *OK* to execute the dialog.

Then a message confirms that the unpacking process has been completed successfully. If you click *OK*, you get back to the dialog for further processing.

If you click *Exit*, the dialog closes. Selected unpacked batches remain unchanged in the handling unit.

Terminal procedure - Unpack all batches

You start the function "Unpack TPU" (Dialog: CE_DEL_HU) in the main view by clicking the button *Unpack TPU*.

The user enters the handling unit or merged batch number. The number can be entered manually or by scanning the barcode. If you enter the number manually, you must click the green arrow. If you scan a barcode, the system automatically performs the transfer and requests the detail data of the batch.

The list displays the data of the entered handling unit. You can double-click specific batches to select the batches that you want to unpack. The selected batches are labeled in the list with an "x". Double-click a selected batch to deselect the batch. The display returns to normal ("-").

The function key *Unpack all* unpacks all batches contained in the handling unit.

Then a message confirms that the unpacking process has been completed successfully. If you click *OK*, the dialog is closed because there are no more batches.

Terminal procedure - Add new batch

You start the function "Unpack TPU" (Dialog: CE_DEL_HU) in the main view by clicking the button *Unpack TPU*.

The user enters the handling unit or merged batch number. The number can be entered manually or by scanning the barcode. If you enter the number manually, you must click the green arrow. If you scan a barcode, the system automatically performs the transfer and requests the detail data of the batch.

The list displays the data of the entered handling unit.

If you click the button *Add*, you can add batches to the handling unit. A dialog opens where you can enter the batch number that is added, either manually or by scanning. After confirmation, the batch is added to the handling unit.

Posting procedure - Unpack individual batches

If you want to remove child batches from a handling unit, the system checks the following:

- The child batches must exist, but can be archived.
- The handling unit must be available but not running or processed.

The following actions are performed for each unpacked child batch:

- The batch status of the child batch is set to the status of the handling unit.
- The remaining quantity of the child batch is calculated from the total quantity of the goods movements when the child batch is packed.
- An entry in the [Batch history](#) is generated.
- The connection between handling unit and child batch is removed
- The batch assignment between child batch and handling unit is removed.

Actions performed for the handling unit:

- The remaining quantity of the handling unit is reduced by the sum of all remaining quantities of all unpacked batches.

- If the remaining quantity of the handling unit is zero, the batch is set to status "A" (processed)
- A goods movement is generated specifying the reduced quantity.

Posting procedure - Unpack all batches

See above "Unpack individual batches". The same processing is applied. All batches are unpacked and the handling unit with the remaining quantity = 0 is set to status "A" (processed).

Posting procedure - Add new batch

If you want to add child batches to an existing handling unit, the system checks the following:

- The handling unit must be available but not running or processed.
- The batch to be added must have the status "F" (free) and the same article number as the handling unit batch.

Note for the added child batch:

- All batch data of the handling unit is transferred to the child batch.
- The batch class of the child batch is identical to the batch class of the handling unit.
- The batch status is "A" (processed).
- The child batch is linked to the handling unit.
- An event „CE_AN_PA“ is added for the child batch in the batch history.
- The batch assignment between child batch and handling unit is generated.
- A goods movement (goods issue) is generated for the child batch.

Actions performed for the handling unit:

- The remaining quantity and the quantity of the handling unit increase by the remaining quantity of the added child batch.
- The system generates a goods movement (goods receipt) with the new quantity of the handling unit.

37 Activating Palletizing/ Packaging/ Assembling

Requirements

Ensure that:

- The flag "Batch management requirement" is set at the operation
- The operation material type does not have the flag "Consumption balance"
- The primary quantity unit for the operation is "KG"

Ensure that at the [workplace](#):

- The flag "Batch administration" is set
- The machine type "Packing location" is selected
- Machine monitoring is not activated

Procedure

Assign the [workplace to a terminal](#). Reboot the terminal.

Results

The following specific dialogs are available at the terminal for the configured packing location:

<i>Dialog</i>	<i>Description</i>
A_AN_HU	Log packing OP on (no input batches)
A_P_AN_HU	Log packing OP on (no input batches, with person)
CE_AN_HU	Assign batch to running OP/ TPU
CA_WL_HU	Complete TPU (output batch change)
A_UN_HU	Interrupt packing OP
A_AB_HU	Finish packing OP
A_AUT_HU	Log off/ interrupt OP selection
C_VLOS_HU	Display preceding TPU



The TPU batch numbers created are permanently generated at the terminal with the prefix HU.
The prefix is not currently configurable.



There is no off-line entry capability

38 Consumption Balance

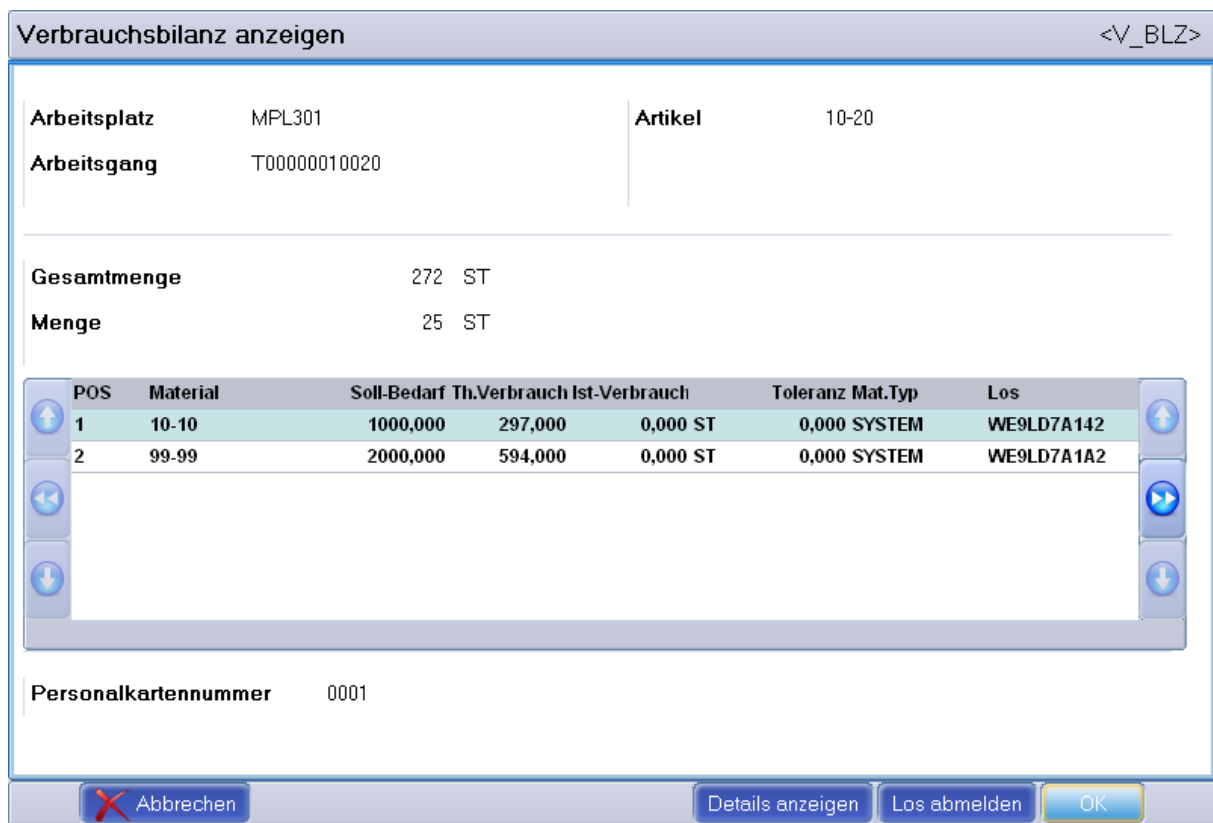
Summary

When logging an OP off, the “consumption balance” (V_BLZ dialog) can be displayed, which can be configured via the machine and the material type of the operation.

The consumption balance is shown if this option is active at the machine and material type.

The consumption balance shows the material consumption based on batches and the user is able to log still running batches off.

Dialog



Verbrauchsbilanz anzeigen <V_BLZ>

Arbeitsplatz: MPL301 Artikel: 10-20
Arbeitsgang: T00000010020

Gesamtmenge: 272 ST
Menge: 25 ST

POS	Material	Soll-Bedarf	Th.Verbrauch	Ist-Verbrauch	Toleranz	Mat.Typ	Los
1	10-10	1000,000	297,000	0,000 ST	0,000	SYSTEM	WE9LD7A142
2	99-99	2000,000	594,000	0,000 ST	0,000	SYSTEM	WE9LD7A1A2

Personalkartenummer: 0001

Abbrechen Details anzeigen Los abmelden OK

Figure: Consumption balance – V_BLZ

Display of consumption postings

The “show details” function allows for consumption quantities, which have been collected so far, to be displayed.

Detail Verbrauchsbilanz anzeigen
<V_BLZ_DTL>

Arbeitsgang T00000010020

Artikel 10-10

Artikelbezeichnung

Personalkartennummer 0001

Errechnete Menge 297 ST

Gemeldete Menge 90 ST

POS	Losnummer	Verbrauch	Anmeldedat	Anmeldezeit	Abmeldedat	Abmeldezeit	STA
1	WE9LD7A141	90,000 ST	09.08.2010	16:53:58	10.08.2010	09:22:58	F

Figure: Display of consumption postings – V_BLZ_DTL

Logging input batches off

By way of the “log batch off” function, the user can choose a currently running batch from the list and log it off by entering consumption.

Eingangslos abmelden				<V_BLZ_CEAB>	
Los	WE9LD7A142		Material	10-10	
Stücklistenposition	1		Materialkurztext		
			Materialtyp	SYSTEM	
Status ☐ ABGEARBEITET ☐ GESPERRT ☒ mit Restmenge			Verbrauch	297	ST
			Loshinweis	mg 100000	
Personalkartennummer			0001		
Abbrechen			Eingangslos abmelden		

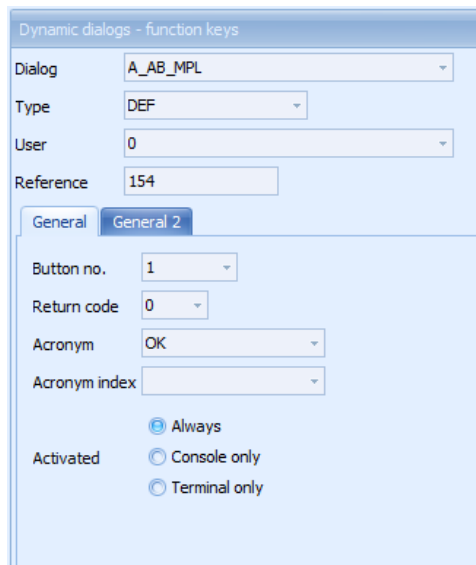
Figure: Log batch off – V_BLZ_CEAB

39 Configuration of Consumption Balance

Dialog configuration:

A special configuration (function = "DLG=V_BLZ;BREAK-ON-CANCEL") has to be defined for the OK button of the logoff dialog (e.g. A_AB_MPL) to make sure that the consumption balance is started when the OP is logged off.

Sample configuration of the OK button for starting the consumption balance:



Dynamic dialogs - function keys

Dialog: A_AB_MPL

Type: DEF

User: 0

Reference: 154

General General 2

Button no.: 1

Return code: 0

Acronym: OK

Acronym index:

Activated:

- ☒ Always
- ☐ Console only
- ☐ Terminal only

Dynamic dialogs - function keys

Dialog: A_AB_MPL

Type: DEF

User: 0

Reference: 154

General General 2

Key:

Text: OK

X pos.: 505

Y pos.: 12

Width: 105

Height: 30

Information: Uebernehmen

Symbol: UPD.BMP

Function: DLG=V_BLZ;BREAK-ON-CANCEL

License:

User-def. 1: AUTO

☐ Blocked

User-def. 2: FORM-VALIDATION

Figure: Example for the OK button configuration including consumption balance

40 AIP - on-screen keyboard decimal symbol configuration

Overview

Menu	All AIP input screens
Transaction code	n/a
Function authorization	n/a

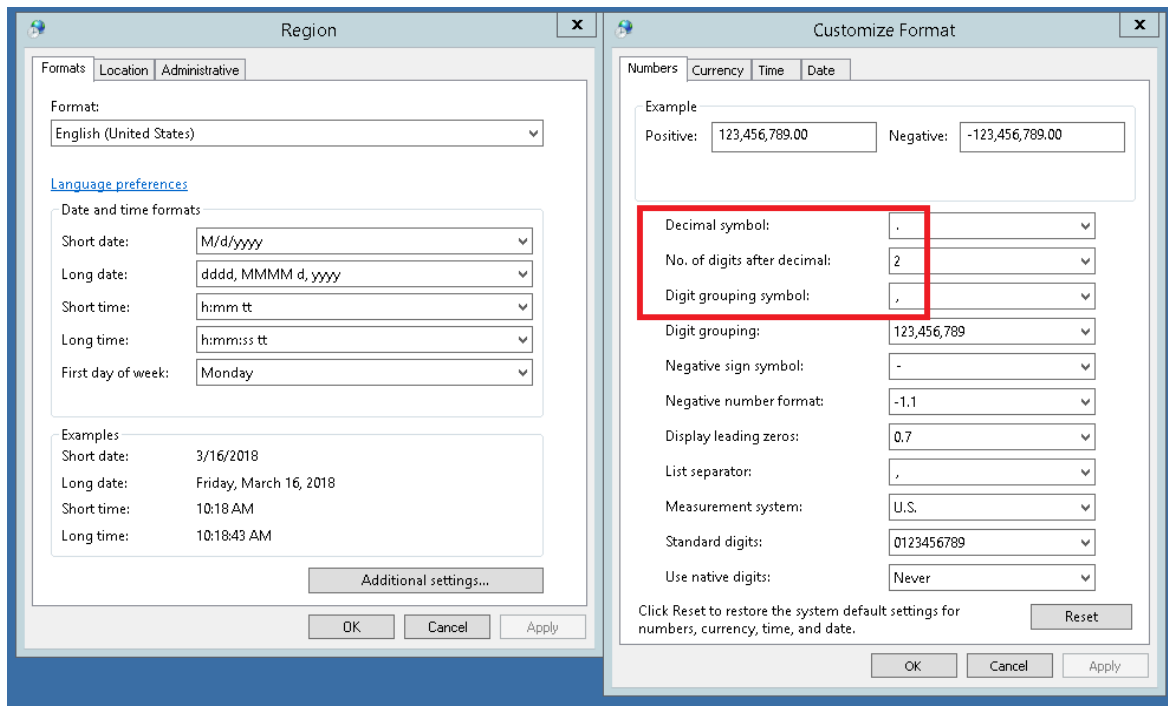
You cannot use the numeric on-screen keyboard (virtual keyboard) to enter decimals, if you configure different decimal symbols in the keyboard settings and the regional settings of the operating system.

The screenshot shows the AIP application interface. At the top, it says 'Change target cycle for operation 01010100000200010 0010 Turning'. Below this, there are two main sections: 'Operation 0010 0010 Turning' and 'Workplace 01011004 Turning'. The 'Operation' section shows fields for Order (010101000002), Article (0101H34729-0010), Target qty. (50 PCE), Yield (70 PCE), Scrap (0 PCE), and Completion (140%). The 'Workplace' section shows Current status (PRODUCTION), Part. (2), Current target cycle (5,000 [Sec./cycle]), and Act. cycle (0,000 [Sec./cycle]). At the bottom, there is a numeric keypad with a red box around the decimal separator field, which currently shows a comma (','). Another red box highlights the 'Change targ. cycle' button.

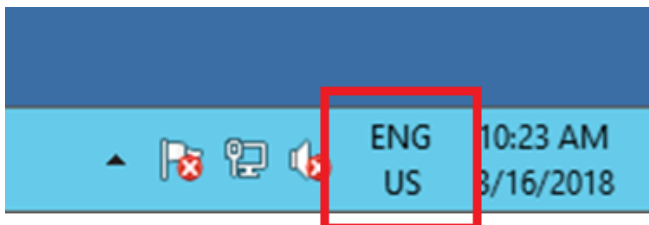
Solution: Keyboard settings and regional settings should match. Example: if you configure a comma "," as the decimal symbol in the regional settings, you also have to select a comma as the decimal separator in the keyboard settings.

Purpose

Every time you start the AIP, the application identifies which decimal symbol is displayed and expected in the input field. The AIP application gets this information from the Windows system setting "region".



Every time you start the on-screen keyboard, the keyboard identifies which decimal symbol is displayed on the numeric keypad. The on-screen keyboard gets this information from the currently selected keyboard settings:



Integration

If you set both configurations correctly (see above), you can use the numeric on-screen keyboard to enter decimals (see the following screenshot):

If the configurations do not match, for example:

- you select the "comma" in the "region" tab of the Windows system settings and
 - you select the "decimal point" in the keyboard settings
- you cannot use the on-screen keyboard to enter the decimal symbol:



Change target cycle for operation 01010100000200010 0010 Turning

Operation 0010 0010 Turning

Order	010101000002	Target qty.	50 PCE
Article	0101H34729-0010	Yield	70 PCE
Handle		Scrap	0 PCE
		Completion	140%

Workplace 01011004 Turning

Current status	PRODUCTION	Part.	2
for	2017-02-22 10:21:43	Current target cycle	5.000 [Sec./cycle]
Duration	9288:23:12	Act. cycle	0.000 [Sec./cycle]

New target cycle

5.000

Sec./cycle]

Staff badge number

Cancel

Change targ. cycle

18-03-16 10:45:09